

Five OEIS Asymptotic Conjectures for Distinct-Partition Norms

Proofs, all-orders logarithmic expansions,
exact-saddle corrections, and inverse
and extreme-value laws

Research report prepared for Vladimir Reshetnikov
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Merged with an independent manuscript on A022629
(batch 73O1: base source 40, member source 48)
and with two further independent manuscripts
(batch 75: sources 75-01 and 75-06),
and with a fifth (batch 77: source 77-51),
and with three more (batch 100: sources 100-82, 100-89, 100-93)

Central family

$$A_\alpha(q) = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad a_\alpha(n) = [q^n] A_\alpha(q), \quad \alpha > 0.$$

The cases $\alpha = 1, 2, 3, 4, 5$ are [A022629](#), [A092484](#), [A265840](#), [A265841](#), and [A265842](#). All five consulted entries label their leading logarithmic asymptotic as a conjecture. This report proves those formulas and develops a single refinement mechanism for every fixed positive α .

Scope and verification. The results below have detailed English proofs and accompanying exact and numerical checks. They are not claimed to be Lean-verified or peer-reviewed. The literature comparison is targeted, not an exhaustive priority certification. The complete exponentially improved coefficient transseries is not claimed: the report distinguishes its proved logarithmic block, exact-saddle expansion, and Fourier-resonance scale. Source 48, merged here, states the same scope for $\alpha = 1$ and real $\alpha > 0$: no Lean verification, no worldwide priority, and no convergence of the inverse-logarithmic series.

Abstract

For a partition λ into distinct parts, let $N(\lambda)$ be the product of its parts. We analyze the fixed positive norm moments $a_\alpha(n) = \sum_{\lambda \vdash n \text{ distinct}} N(\lambda)^\alpha$. A direct argument proves the logarithmic growth formulas still marked conjectural in five OEIS entries. With $K = \sqrt{2n}$, $r = \log K$, and $c = \pi^2/(6\alpha^2)$, we then prove, to every fixed order,

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \frac{c}{r} + \frac{c}{r^2} + \frac{c - c^2/2}{r^3} + \cdots \right).$$

An explicit inverse-function construction generates all coefficients; seven are printed. A separate local-limit argument gives multiplicatively accurate exact-saddle asymptotics and all fixed Edgeworth orders. This separation matters: exponentiating a fixed truncation of the logarithmic expansion does not give a relative asymptotic equivalent. We invert the dominant block using the principal Lambert function and obtain asymptotics for the first index attaining a large value. Under the norm-biased measure, the occupation boundary has a logistic profile, while the largest part has a Gumbel limit and is asymptotic to $(1 + 1/\alpha)\sqrt{2n}$. Finally, we prove that the first Fourier resonances have logarithmic modulus of order $-K/(\alpha^3(\log K)^3)$, explaining a striking finite-size limitation visible in the exact computations. Reproducible scripts, source-status notes, and further research questions accompany the proofs.

This report merges two manuscripts that proved these theorems independently: source 40 (the base text, five OEIS sequences, real $\alpha > 0$) and source 48 (A022629 and the same real family, written $\prod(1 + k^s q^k)$). Every shared theorem is printed once and credited to both, with identical coefficients. Source 48 adds a second, elementary proof of the upper bound, a second derivation of all coefficients by an explicit differential operator, explicit centring of the Gumbel law in $\log \sqrt{2n}$, the radius of convergence of $\log A_\alpha$ at the origin, the logarithm of the largest single weight, the entropy factor $\log(a_\alpha(n)/\max N^\alpha) \sim \pi^2 K/(6\alpha r)$, exact data to $n = 6400$ and its own research questions.

Two further manuscripts of batch 75, sources 75-01 and 75-06, prove the same theorems a third and a fourth time, with identical coefficients; those duplicates are recorded in one table, not reprinted. Their new material forms Sections 16–18: sharp signed cumulant scales, a saddle remainder $O((\alpha Ms)^{-J-1})$ with first correction $\sim -3/(8\alpha Ms)$, localization of the integer threshold to $1 + o(1)$, eventual strict log-concavity and fixed-degree Jensen hyperbolicity (source 75-01); and the convolution powers $\prod(1 + k^\alpha q^k)^\mu$ (A297321, A022630, A022631) with μ -independent coefficients, two further inverse coefficients, and an exact rational check of P_8 (source 75-06).

A fifth manuscript, source 77-51 of batch 77, proves the core theorems once more (recorded in one table) and adds the slot-multiplicity family $\prod(1 + k^\alpha q^k)^{k^\beta}$, $\beta \geq 1$ an integer, with the logarithmic equivalent conjectured for A266891 (Section 20).

Three more manuscripts of one series, sources 100-82, 100-89 and 100-93 of batch 100, follow (Sections 21 and 22). The first two prove the core theorems a sixth and a seventh time, with identical coefficients, and add P_9 , P_{10} , Δ_9 , Δ_{10} , a smooth lattice-minus-integral difference with explicit Taylor coefficients, an all-orders relative expansion with the exact integral, and an all-orders exact-range inverse. The third treats powers growing like $\alpha \asymp (2n)^{1/6}/\log n$: the Fourier resonances of Section 11 then enter the leading order, the coefficient carries a Jacobi theta multiplier, and no single-Gaussian equivalent holds uniformly.

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1 The problem, its significance, and the source audit

1.1 Five conjectures in one natural family

Let $\mathcal{D}_n$ be the set of partitions of n into distinct positive parts. The empty partition has norm 1. For fixed $\alpha > 0$, define

$$N(\lambda) = \prod_{k \in \lambda} k, \quad a_\alpha(n) = \sum_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha. \quad (1.1)$$

Choosing whether each part occurs gives the locally uniformly convergent product

$$A_\alpha(q) = \sum_{n \geq 0} a_\alpha(n) q^n = \prod_{k \geq 1} (1 + k^\alpha q^k), \quad |q| < 1. \quad (1.2)$$

For integral α the coefficients are integers. One may interpret k^α as the number of colors available for a part of size k , with at most one part of each *size*, regardless of color. This is not the product $\prod_k (1 + q^k)^{k^\alpha}$, which allows several differently colored copies of the same size.

[Batch 77, 2 October 2026.] Source 77-51 (Section 20.3) treats a third product between these two, $\prod_k (1 + k^\alpha q^k)^{k^\beta}$ with integer $\beta \geq 0$: k^β slots of size k , each with internal weight k^α .

The OEIS entries consulted on October 1, 2026 have the following status [1, 2, 3, 4, 5].

α	Sequence	Date of the conjectural formula
1	A022629	May 8, 2018
2	A092484	December 27, 2020
3	A265840	December 27, 2020
4	A265841	December 27, 2020
5	A265842	December 27, 2020

All five formulas are attributed there to Václav Kotešovec and have the unified form

$$\log a_\alpha(n) \sim \alpha \sqrt{\frac{n}{2}} (\log(2n) - 2). \quad (1.3)$$

The full array of nonnegative integral moments is [A292189](#); its columns 0, 1, $\dots$, 5 are identified in that entry [6]. The zeroth column is the ordinary distinct-partition function, but $\alpha = 0$ is a different asymptotic regime and is excluded from our fixed-positive-parameter theorems.

Source 48, which treats A022629 and the real family, makes two further points [1]. Its use of “currently” refers to the entry as retrieved on October 1, 2026, not to an exhaustive determination of the publication history. And a logarithmic equivalent does not determine a multiplicative equivalent: even the constant -2 inside the parentheses of (1.3) is not specified by the relation $\sim$, since changing it leaves the same equivalence. The additive statement that the corresponding $-\sqrt{2n}$ term is correct is sharper, and Proposition 3.2 proves it. For non-integral α , (1.2) is a positive weighted enumeration, and the product converges normally on compact subsets of $|q| < 1$ (source 48).

These sequences are not arbitrary numerical constructions. They simultaneously encode weighted enumeration, moments of a fundamental multiplicative partition statistic, and a conditioned model of independent binary occupancies. Their common structure makes a family theorem more informative than five separate fits to numerical data.

1.2 What is established here, and what is not a priority claim

The conjectural labels in the consulted entries are verifiable facts; the claim that no proof has ever appeared anywhere would be much stronger and is not made. This report supplies self-contained proofs of the displayed formulas, followed by explicit refinements and probabilistic consequences. Novelty of the refinements should be assessed against the broader literature before publication.

Schneider and Sills survey partition norms, including extremal norms over distinct partitions and their connection with triangular numbers [7]. Bridges and Craig study norm distributions and moments for *unrestricted* partitions [8]; repeated parts change the analytic problem substantially. Rana, Kaur, and Kumar give recurrences and related norm identities for distinct partitions [9]. None of those results is used as an unproved asymptotic input here.

A particularly relevant recent source is Gikunda’s March 2026 dissertation [10]. Its Chapter 4 uses $\prod_k (1 + u^{\log k} q^k)^{b_k}$ and proves log-norm central limit theorems. For $b_k = 1$ and $u = e^\alpha$ this is our generating family. However, the coefficient estimate in Proposition 4.2.5 and the surrounding analysis use a *shrinking* tilt $\log u = O(t^\delta)$ as the saddle parameter t tends to zero. Our tilt $\alpha > 0$ stays fixed. Thus that estimate does not directly supply the fixed-tilt results below. This is a distinction between asymptotic regimes, not a claim that the underlying family has not previously been studied.

Source 48 places the method in the classical saddle-point and probabilistic tradition of asymptotic enumeration; standard background is in Flajolet and Sedgewick [13]. Granovsky and Stark [14] study a broad product framework and, in their Section 3, the related products $\prod (1 + k^{-q} z^k)$ with $q > 0$. Their main theorem assumes parameters $0 < a_k \leq 1$, so substituting $a_k = k^\alpha > 1$ is not a direct application of that theorem. This observation is not a claim that no other general theorem can cover the present case (source 48).

[**Merge note, batch 77.**] Source 77-51 (Section 20.5) repeats the Gikunda and Granovsky–Stark distinctions in its own words and adds Naranjo and Ramírez [27], who discuss A022629 explicitly in a combinatorial treatment of coloured partitions.

1.3 Relation to the ProveIt repository

The repository snapshot consulted was 1f1981f682b2878bde51a6ad40c22777f362fc05. Its `Analysis/Transseries/README.md` describes the transseries and inversion volumes, accompanying verification programs, and the separate Lean library for asymptotic scales and power-logarithmic monomials [11]. The present report follows that program by separating three precision levels: a logarithmic asymptotic block, a relative exact-saddle expansion, and flat Fourier contributions. Its Lambert- W inversion is given explicitly rather than left as a formal instruction to “reverse the series.”

No repository theorem is assumed here without proof. In particular, the existence of general transseries infrastructure is not treated as a formal certificate for our analytic remainder estimates. The new material is an independent, reproducible research contribution suitable for integration after review; it does not modify the repository.

Source 48 read the repository’s `Analysis/Transseries` material, which consolidates polynomial–logarithmic asymptotics, Lambert-core inversion, error transport, and staircase inversion [12]; those are the organizing ideas it uses. No repository theorem is imported as an unexamined analytic hypothesis: the needed estimates are proved below. A targeted repository search for A022629 returned no match during its preparation. That search and the literature checks are a useful duplication screen, not a proof of originality (source 48).

[**Merge note, batch 73O1.**] The repository still contains no other treatment of these sequences; this report is the first. Its methodological relative is the partition-number chapter of the volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion`, which treats $p(n)$ (A000041); nothing is imported from it.

[**Merge note, batch 75.**] Sources 75-01 and 75-06 (Section 15) inspected the ProveIt commit 29aca108e, which held the archives of sources 40 and 48 unopened and predates their placement. Source 75-01’s statement that a repository search for A022629 found nothing was true of that snapshot only; this report is the repository’s treatment, and both sources are merged into it.

[**Merge note, batch 77.**] Source 77-51 (Section 20) names no ProveIt commit and records no repository search; it was written without knowledge of this report.

[**Merge note, batch 100.**] Sources 100-82 and 100-89 (Section 21) name no ProveIt commit and record no repository search. Source 100-93 (Section 22) examined the snapshot 6591a56b3 (1 October 2026, 18:55 PDT), eighteen minutes before this report’s first placement 9df4ba51a; its bounded source comparison saw neither this report nor the lattice-peak report cited in Section 22.8.

1.4 The two sources and this merge

This report is built from two manuscripts of batch 73O1 of ProveIt’s incoming-reports intake, written independently on October 1, 2026.

- **Source 40** (the base text), *Five OEIS Asymptotic Conjectures for Distinct-Partition Norms*: the family $\prod(1 + k^\alpha q^k)$ for real $\alpha > 0$, the five OEIS sequences $\alpha = 1, \dots, 5$, all fixed orders with seven printed coefficients and an eighth checked numerically, the exact-saddle expansion, Lambert- W inversion, the logistic boundary and Gumbel maximum, the fixed-resonance theorem, and finite-size diagnostics to $n = 5000$. Its section order is kept.
- **Source 48**, *The A022629 Conjecture: All-Orders Logarithmic Asymptotics, Saddle-Point Expansions, Inverse Growth, and the Largest Part of a Weighted Partition*: the same theorems for A022629 and for $\prod(1 + k^s q^k)$, real $s > 0$, with six printed coefficients. Its s is source 40’s α ; with that identification its coefficients $c_j(A)$, $A = \pi^2/(6s^2)$, are exactly source 40’s $P_j(c)$, $c = \pi^2/(6\alpha^2)$, and its inverse coefficients and first Edgeworth correction are identical.

Every result, proof, remark, question and non-claim of both manuscripts is printed. A theorem proved by both is printed once, in source 40’s wording, and credited “sources 40 and 48”. Where source 48’s proof is genuinely different, it follows as a “Second proof (source 48)”. Material only source 48 has is marked (source 48): Propositions 3.4 and 3.5, Lemma 3.6, Corollary 8.1, the operator route of Section 7.3, the explicit forms in Sections 9 and 10.3, the computations of Section 12.5, its questions, its OEIS wording, and Appendix C. Text added in the merge is marked [**Merge note, batch 73O1.**]. Appendix D records the provenance and every choice the merge made.

[**Merge note, batch 75.**] Batch 75 added two further independent manuscripts, sources 75-01 and 75-06. Their shared results are recorded once, in Table 2 of Section 15, and their new material is Sections 16–19 (marked (source 75-01) and (source 75-06)). Text added in that merge is marked [**Merge note, batch 75.**], and dated notes on earlier text [*Batch 75, 2 October 2026.*]. Earlier sections are otherwise unchanged, so their numbering is that of the batch-73O1 report.

[**Merge note, batch 77.**] Batch 77 added a fifth independent manuscript, source 77-51. Its shared results are recorded once, in Table 6; its new material is Section 20 (marked (source 77-51)). Text added in that merge is marked [**Merge note, batch 77.**], and dated notes on earlier text [*Batch 77, 2 October 2026.*]. Every earlier section, statement, equation and table keeps its number.

[**Merge note, batch 100.**] Batch 100 added three manuscripts of one series, sources 100-82, 100-89 and 100-93 (called by their bundle report numbers). The shared results of the first two are recorded once, in Table 8; their new material is Section 21, and the third is Section 22 (marked (source 100-82), (source 100-89), (source 100-93)). Text added in that merge is marked [**Merge note, batch 100.**], dated notes on earlier text [*Batch 100, 5 October 2026.*], and statements and proofs supplied by the write rather than by a source [write]. Every earlier section, statement, equation and table keeps its number.

1.5 Notation across the two sources

The two manuscripts use the same letters for different objects. This report uses source 40's notation throughout; source 48's material is translated as follows.

Notion	This report (source 40)	Source 48
power in the weight generating function, coefficients	α $A_\alpha(q), a_\alpha(n)$	s (renamed) $F_s(q), a_s(n); F(q), a(n)$ for A022629
$\sqrt{2n}, \log \sqrt{2n}$	K, r	x, λ
$\pi^2/(6\alpha^2)$; coefficients	$c; P_j(c)$	$A; c_j(A)$
log of the transition site	$s = -W_{-1}(-t/\alpha)$	L
transition site e^s	M	r
$\log A_\alpha(e^{-t})$	$F(t)$	$S_s(t)$
mean, variance	$-F'(t), B = F''(t)$	$m(t), V(t)$
Bernoulli variables, total	$X_k, S = \sum kX_k$	B_k, X_t
log-expansion series	$\mathcal{H}_\alpha(r)$	$\mathcal{H}_s(\lambda)$
boundary operator and series	$\mathcal{T}, \mathcal{C}_\alpha, \mathcal{B}_\alpha$	$\mathcal{T}, \mathcal{C}_s, \mathcal{B}_s$
truncation order (log, Edgeworth)	R, J	K, M
block length (Lemma 5.1)	h	H
largest single weight	$\Pi_\alpha(n)$	$Q_s(n)$
largest part	L_n	M_n
log threshold; Lambert data	$Y; w, R_0, K_0$	$y; w, \ell, x_0$ (and $Y = y/s$)
explicit Gumbel centring	$\beta_\alpha(r)$	$d_s(\lambda)$
small exponent in proofs	σ, β	α

Readings to avoid.

- Source 48's s is the *power* ($= \alpha$); source 40's s , used here, is the *logarithmic transition coordinate* $-W_{-1}(-t/\alpha)$, which source 48 calls L .
- Source 48's $r = e^L$ is the transition *site* (here M); source 40's r , used here, is $\log \sqrt{2n}$ (source 48's λ).
- Source 48's K is a truncation order or a cutoff index; here K is always $\sqrt{2n}$. Source 48's M is an Edgeworth order; here M is always e^s .

- Source 48's $Q_s(n)$ is the largest single weight (here $\Pi_\alpha(n)$), not source 40's mean-relation series $Q_\alpha(s)$. Source 48's Gumbel centring $d_s(\lambda)$ (here $\beta_\alpha(r)$) is not source 40's boundary series $d_\alpha(s)$.
- B is the variance; $\mathcal{B}_\alpha$ is source 48's mean-boundary series.
- Source 48's Edgeworth sum $\mathcal{E}_M$ is cumulative (all weights below $2M$); source 40's $\mathcal{E}_\ell$ is the single weight 2ℓ . Here $1 + \sum_{\ell < J} \mathcal{E}_\ell$ is source 48's $\mathcal{E}_J$.

[Merge note, batch 75.] Sources 75-01 and 75-06 are translated as follows. The layer count μ , the sharp parameter $\varepsilon_\sharp$, the Gaussian inverse n_* , the inverse coefficients Δ_j and the Jensen and Hermite symbols are new in this report.

Notion	This report	Source 75-01	Source 75-06
power in the weight	α	s (renamed)	r (renamed)
number of layers	μ	—	s (renamed)
generating function, coefficients	$A_\alpha(q)^\mu, a_{\alpha,\mu}(n); a_\alpha(n)$	$F_s(q), a_s(n)$	$F_{r,s}(q), a_{r,s}(n)$
$\sqrt{2n}, \log \sqrt{2n}$	$K, r; K_\mu = \sqrt{2n/\mu},$ r_μ	N, ℓ	$m = \sqrt{2n/s}, L$
$\pi^2/(6\alpha^2)$; coefficients	$c; P_j(c)$	$P/6, P = \pi^2/s^2;$ $c_j(P)$	$u/6, u = \pi^2/r^2;$ $b_j(u)$
log transition coordinate, site	$s, M = e^s$	L, M	v, K
log generating function	$F(t), \mu F(t)$	$\Phi_s(t)$	$\mathcal{L}(t)$
Bernoulli cumulant polynomials	$b_j(p)$	$P_j(p)$	$B_j(p)$
Edgeworth terms	$\mathcal{E}_\ell, \mathcal{E}_\ell^{(\mu)}$	E_r	Q_j
small parameters	$\varepsilon = s/M;$ $\varepsilon_\sharp = 1/(\alpha Ms)$	$\varepsilon = 1/(sML)$	$\varepsilon = v/K$
truncation orders (log, Edgeworth)	R, J	K, J	M, J
boundary-inverse derivatives	$U_m(s)$	$u_L^{(m)}(0), a = L - 1$	$Y_v^{(m)}(0)$
boundary and mean series	$d_\alpha(s), Q_\alpha(s)$	$h_s(L) - \frac{L}{2} + 1,$ $2B_s(L)$	$c(v), 1 + d(v)$
log threshold; Lambert data	$Y; w, R_0, K_0, n_0$	$\log A; w, z_0, N_0, n_0$	$\log Y; w, L_0, m_0, n_0$
threshold index	$N_\alpha(Y), N_{\alpha,\mu}(Y)$	$\mathcal{N}_s(e^Y)$	$N_{r,s}(e^Y)$
inverse coefficients	$\Delta_j(c)$	coefficients of $\mathcal{N}_s/n_0$	$d_j(u)$
Gaussian inverse	$\Psi_0, n_*(Y)$	$\Psi_0, x_0(A)$	—
Jensen, Hermite, local data	$\mathcal{J}_\alpha^{d,n}, \mathbf{H}_d; \delta_n, \tau_n, \gamma_k$	$J_s^{d,n}, H_d; \delta_n, A_n,$ g_k	—
largest part; centring	$L_n; b$	—	$M_n; b_n$
smoothing kernel	$g(v)$	$R(v)$	$h(U)$
divisor sum in (12.2)	$C_\alpha(m)$	$b_s(m)$	$\beta_r(m)$

Readings to avoid (batch 75).

- Source 75-01's s and source 75-06's r are the *power* α ; source 75-06's s is the *number of layers* μ . Neither is this report's $s = -W_{-1}(-t/\alpha)$, which source 75-01 calls L and source 75-06 calls v .

- Source 75-06's K is the transition site M , not $\sqrt{2n}$; its L is $\log \sqrt{2n/\mu}$ (here r_μ). Source 75-01's K is a truncation order and its L is s .
- Source 75-01's $P_j(p)$ are cumulant polynomials (here $b_j(p)$), not the coefficients $P_j(c)$; source 75-06's $b_j(u)$ are the coefficients ($b_j(6c) = P_j(c)$), not the cumulant polynomials $b_j(p)$.
- Source 75-06's $c(v)$ is $d_\alpha(s)$ and its $d(v)$ is $Q_\alpha(s) - 1$; its $d_j(u)$ are the inverse coefficients $\Delta_j(c)$ (at $u = 6c$), unrelated to $d_\alpha(s)$.
- Source 75-01's $\varepsilon = 1/(\alpha Ms)$ is written $\varepsilon_\#$ here; $\varepsilon = s/M$ keeps the meaning of (4.8), and $\varepsilon_\# = \varepsilon/(\alpha s^2)$.
- Source 75-01's $x_0(A)$ is the Gaussian inverse, written $n_*(\log A)$ here; it is not source 48's x_0 ($= K_0$).
- Source 75-01's Hermite H_d is written H_d ; H remains the transition width. Its E_r and source 75-06's Q_j are the Edgeworth terms $\mathcal{E}_\ell$, not $Q_\alpha(s)$.
- Source 75-06's $U(x) = tx - r \log x$ is minus the log odds; it is not the Lagrange coefficient $U_m(s)$.

[Merge note, batch 77.] Source 77-51 (Section 20) is translated as follows. The family notation $A_\alpha^{[\beta]}$, $a_\alpha^{[\beta]}$, $K^{[\beta]}$, $r^{[\beta]}$, $K_0^{[\beta]}$, $R_0^{[\beta]}$ is new in this report.

Notion	This report	Source 77-51
power in the weight; slot exponent	$\alpha; \beta$	$\alpha; \beta$, and $d = \beta + 1$
slot family, coefficients $\sqrt{2n}$, $\log \sqrt{2n}$	$A_\alpha^{[\beta]}(q)$, $a_\alpha^{[\beta]}(n)$ K, r ; $K^{[\beta]} = ((\beta + 2)n)^{1/(\beta+2)}$, $r^{[\beta]}$	$F_{\alpha,\beta}(q)$, $a_n^{(\alpha,\beta)}$ m, L ; m_0, L_0
$\pi^2/(6\alpha^2)$ at $\alpha = 1$; π^2	c ; $6c$	C ; P
transition site; its logarithm	$M = e^s$; s	M ; $\ell = \log M$
small parameter	$\varepsilon = s/M$; $\mathcal{W} = M^{\beta+1}/s$	$w = M/\ell$; $\mathcal{W}$
log generating function, mean, variance	$F(t)$, $-F'(t)$, B ; $F^{[\beta]}$, $B^{[\beta]}$	$f(t)$, $\mu(t)$, $V(t)$
cumulant order	j	r
Edgeworth sum through order R	$1 + \sum_{\ell \leq R} \mathcal{E}_\ell$	E_R
logarithmic and inverse coefficients ($\alpha = 1$)	$P_j(c)$; $\Delta_j(c)$	d_j ; b_j
boundary-inverse derivatives; smoothing series	$U_j(s)$; Lemma 7.1	$R_j(x)$; $T(L)$
log threshold; threshold index	Y ; $N_\alpha(Y)$, $N_\alpha^{[\beta]}(Y)$	$s = \log y$; $N(y)$, $N_{\alpha,\beta}(y)$
Lambert data	K_0, R_0 ; $K_0^{[\beta]}$, $R_0^{[\beta]}$	m_* , L_* ; m_* , L_* , u_*

Readings to avoid (batch 77).

- Source 77-51's $s = \log y$ is this report's log threshold Y , not the transition coordinate $s = -W_{-1}(-t/\alpha)$, which source 77-51 calls ℓ .
- Source 77-51's $\mu(t)$ is the mean $-F'(t)$, not the number of layers μ of Section 18; and $a_\alpha^{[\beta]}$ is not $a_{\alpha,\mu}$.

- Source 77-51's m is $K = \sqrt{2n}$; its M (with $t = \log M/M$ at $\alpha = 1$) is this report's transition site $M = e^s$, not K ; its L is r , not the L of sources 48 and 75-01.
- Source 77-51's d_j are $P_j(\pi^2/6)$ and its b_j are $\Delta_j(\pi^2/6)$: neither is source 75-06's $d_j(u)$ or $b_j(u)$, nor the cumulant polynomials $b_j(p)$; its $d = \beta + 1$ is a dimension.
- Source 77-51's $R_j(x)$ are rational functions ($= U_j$); here R is a truncation order and R_0 a Lambert datum. Its E_R includes the leading 1; this report's $\mathcal{E}_\ell$ is the single weight 2ℓ .
- Source 77-51's r is a cumulant order (and, in its Section 2, $n - M(M+1)/2$, with M there the largest integer such that $M(M+1)/2 \leq n$); here $r = \log K$. Its $C = \pi^2/6$ is c at $\alpha = 1$.

[**Merge note, batch 100.**] Sources 100-89, 100-82 and 100-93 (Sections 21 and 22) are translated as follows. New in this report: I_α , Ξ_α , ξ_r , χ_r , $\mathfrak{L}$, m_E , F_D , t_D , B_D , $\kappa_{j,D}$, t^{int} , B^{int} , Ψ_J^{int} , $n_{*,J}^{\text{int}}$, ϖ ; and, for Section 22, Λ , $R_{\geq 2}$, μ_R , B_{RR} , B_{RS} , $Q_\perp$, ρ_{RS} , $\mathbb{T}_r$, Θ , Ψ_Θ , n_Θ , the block $\mathbb{B}$ and the cells $\mathcal{I}_j$. Source 100-82 is the case $\alpha = 1$; a dash means that the source has no such object.

Notion	This report	Source 100-89	Source 100-82	Source 100-93
power in the weight	α	a	$(\alpha = 1)$	a
product, coefficients	$A_\alpha(q)$, $a_\alpha(n)$	$F_a(q)$, $A_a(n)$	$F(q)$, $a(n)$	$F_a(q)$, $A_a(n)$
$\sqrt{2n}$, $\log \sqrt{2n}$	K , r	$\sqrt{2n}$, B	K_0 , B	z , —
transition site, its log, width	M , s , H	K , b , —	K , b , —	K , b , W
log odds $\alpha \log x - tx$	$\varpi(x)$	$\log y$	$v(x)$ ($\alpha = 1$)	$f(x)$
$\log A_\alpha(e^{-t})$	$F(t)$	$L_a(t)$	$\log F(e^{-t})$	L
integral; lattice minus integral	$I_\alpha(t)$; $\Xi_\alpha(t)$	$I_a(t)$; $E_a(t)$	$I(t)$; —	—
boundary coefficients	$\xi_r(\alpha)$	$c_r(a)$	$(\xi_0(1) \text{ only})$	—
$\partial_t^r \log(1 + x^\alpha e^{-tx})$; Euler–Maclaurin order, functional	$\chi_r(x; t)$; m_E , $\mathfrak{L}$	$h_r(x; t)$; q , $\mathcal{L}_q$	—	—
compact parameter set	$\mathfrak{A}(\alpha)$; $\mathfrak{J}(\Lambda)$	$\mathcal{A}$	—	J
integral model; its saddle, variance, cumulants	F_D ; t_D , B_D , $\kappa_{j,D}$	J_D ; t_D , v_D , $\kappa_{j,D}$	—	—
Edgeworth order; block	J , $D = 2J + 2$; $1 + \sum_{\ell < J} \mathcal{E}_\ell$	$M = J - 1$, $D = 2M + 4$; $\mathcal{P}_M$	—	—

Notion	This report	Source 100-89	Source 100-82	Source 100-93
saddle of the integral, its variance	$t^{\text{int}}, B^{\text{int}}$	t_0, V	$t_n, V(t_n);$ $N(t) = -I'(t)$	—
lattice saddle; variance	$t_n; B$	$t; v$	$t; \text{Var } S_t$	$t; V$
small parameter s/M	ε ($t = \alpha\varepsilon$)	$\eta = b/K$	—	—
Bernoulli cumulant polynomials	$b_r(p)$	$\kappa_r(p)$	—	—
kernel constants; operator	$2\eta(2j); \mathcal{T}$	$\gamma_j; D_b$	$c_j; D$	—
log and inverse coefficients	$P_j(c); \Delta_j(c)$	—	$s_j; v_j$ ($P = \pi^2$)	—
log-coefficient truncation order	R	—	M	—
fixed-size phase shift; inverse correction	$s - r; \log E$	$\delta; \text{—}$	$\delta; \varepsilon$	—
inverse models and their inverses	$\Psi_J^{\text{int}}, n_{*,J}^{\text{int}}; \Psi^{\text{int}}, \nu$	H_M, ν_M	$H(n), \nu(y)$	$H_a(n), n_0$ (here Ψ_Θ, n_Θ)
log threshold; threshold index	$Y; N_\alpha(Y)$	$\log y; n_*(y)$	$Y; n_*(y)$	$Y; N_a(Y)$
Lambert data	w, R_0, K_0, n_0	W	w, C, X_0	W_0, z_L, n_L, λ_L
critical parameter	$\Lambda = \alpha(s-1)/M^{1/3}$	—	—	λ
parts ≥ 2 ; mean; variance	$R_{\geq 2}; \mu_R; B_{RR}$	—	—	$R; \mu; U$
covariance; regression variance; slope	$B_{RS}; Q_\perp; \rho_{RS}$	—	—	$C; Q; \beta$
edge moments; Bernoulli phases	$\mathbf{T}_r; \omega_k$	—	—	$T_r; \alpha_k$
theta multiplier; single Gaussian	$\Theta(\mu, Q); G_\alpha(n)$	—	—	$\Theta(\mu, Q); G_a(n)$
block, centre, half-length, length; cells	$\mathbf{B}, k_B, m_B, N_B; \mathcal{I}_j$	$\mathcal{B}, \text{—}, \text{—}, \ell; \text{—}$	block, —, —, L ; —	$B, c, M, N; \mathcal{I}_j$
window widths in proofs	$\varrho, \varsigma, e_{\max}$	—	—	δ, η, ϵ
Euler's constant	γ_E	γ	—	—

Readings to avoid (batch 100).

- In all three sources K is the transition site $M = e^s$, not $K = \sqrt{2n}$ (source 100-82's K_0 , source 100-93's z), and b is s , not the Gumbel centring b of (2.12). Sources 100-82's and 100-89's B is $r = \log \sqrt{2n}$, not the variance B ; source 100-93's B is a block of sizes (here B) and its M is that block's half-length, not the transition site.
- Source 100-89's $c_r(a)$ are the boundary coefficients $\xi_r(\alpha)$ and source 100-82's c_j are $2\eta(2j)$; neither is $c = \pi^2/(6\alpha^2)$. Source 100-89's γ_j are also $2\eta(2j)$; its γ is Euler's constant. Its $\eta = b/K$ is $\varepsilon = s/M$, not Dirichlet's $\eta(z)$.
- Source 100-82's $t_n, N(t), V(t)$ belong to the *integral* saddle (here $t^{\text{int}}, -I'_1, B^{\text{int}}$); this report's t_n is the lattice saddle. Source 100-89's t_0 is t^{int} too.
- Source 100-89's $E_a(t)$ (here Ξ_α) is neither the inverse ratio E of Section 9 nor an Edgeworth term $\mathcal{E}_\ell$; its J_D (here F_D) is not an Edgeworth order; its M is the Edgeworth order $J - 1$, not the site; its h_r is not the boundary function $h_s(u)$ of (7.1).
- The inverse models H_M (source 100-89), $H(n)$ (100-82) and $H_a(n)$ (100-93) are logarithms of coefficient models (here $\Psi_J^{\text{int}}, \Psi^{\text{int}}, \Psi_\Theta$), not the width H ; source 100-93's W is the width H , not a Lambert function.

- Source 100-93's μ is the mean count μ_R , not the number of layers μ ; its Q is $Q_\perp$, not the mean series $Q_\alpha(s)$; its λ is Λ , not a partition; its R counts parts at least two ($R_{\geq 2}$) and is neither a truncation order nor R_0 ; its U, C, V are B_{RR}, B_{RS}, B , and its C is not the divisor sum $C_\alpha(m)$; its T_r are $\mathbb{T}_r$, not the Edgeworth degree T ; its $\beta = C/V$ is ρ_{RS} , not the slot exponent β .
- Section 22's $\Theta(\mu, Q)$ is not the theta function $\vartheta(q, \delta)$ of the lattice-peak report: $\Theta(\mu, Q) = (2\pi Q)^{-1/2} \vartheta(1/Q, \mu)$. The constants c', c_3, C_3 of that section are not c .

2 Main results and the hierarchy of scales

Throughout the paper, $\alpha > 0$ is fixed. Constants in O -terms may depend on α and on the indicated truncation order. Put

$$K = \sqrt{2n}, \quad r = \log K, \quad c = \frac{\pi^2}{6\alpha^2}. \quad (2.1)$$

Theorem 2.1 (All-orders logarithmic expansion; sources 40 and 48). *There are explicit polynomials $P_j(c) \in \mathbb{Q}[c]$ such that, for every fixed integer $R \geq 1$,*

$$\log a_\alpha(n) = \alpha K \left(r - 1 + \sum_{j=1}^R \frac{P_j(c)}{r^j} + O_{\alpha,R}(r^{-R-1}) \right). \quad (2.2)$$

The first seven are

$$\begin{aligned} P_1(c) &= c, & P_2(c) &= c, \\ P_3(c) &= c - \frac{1}{2}c^2, & P_4(c) &= c + \frac{9}{10}c^2, \\ P_5(c) &= c + \frac{41}{5}c^2 + \frac{1}{2}c^3, \\ P_6(c) &= c + 27c^2 + \frac{1973}{210}c^3, \\ P_7(c) &= c + \frac{129}{2}c^2 + \frac{8109}{70}c^3 - \frac{5}{8}c^4. \end{aligned} \quad (2.3)$$

Equations (7.9)–(8.3) below generate every P_j .

In particular,

$$\begin{aligned} \log a_\alpha(n) &= \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} \\ &\quad + \left(\frac{\pi^2}{6\alpha} - \frac{\pi^4}{72\alpha^3} \right) \frac{K}{r^3} + O_\alpha(K/r^4). \end{aligned} \quad (2.4)$$

This proves (1.3) for all five OEIS sequences, and more strongly identifies the $-\alpha K$ term with an error $O_\alpha(K/r)$.

Source 48 proves Theorem 2.1 independently (its Theorem 1.1, with $P_1, \dots, P_6$ written $c_1(A), \dots, c_6(A)$), and adds three remarks. First, its constants are uniform when α ranges over a fixed compact subinterval of $(0, \infty)$; they are *not* claimed to be uniform as $\alpha \downarrow 0$. Second, the case $R = 0$ states $\log a_\alpha(n) = \alpha K(r - 1) + O_\alpha(K/r)$, which already proves more than the displayed OEIS conjecture, and the first nonzero correction is positive:

$$\lim_{n \rightarrow \infty} \frac{r}{K} (\log a_\alpha(n) - \alpha K(r - 1)) = \frac{\pi^2}{6\alpha}. \quad (2.5)$$

Third, all-order statements in this report mean: *for each fixed truncation order, the asserted remainder bound holds*. No convergence of the infinite inverse-logarithmic series is assumed (source 48).

[Merge note, batch 75.] Sources 75-01 (its Theorem 1.1, five coefficients $c_j(P)$, $P = 6c$) and 75-06 (its Theorem 6.1, eight coefficients $b_j(u)$, $u = 6c$) prove Theorem 2.1 independently; $c_j(6c) = b_j(6c) = P_j(c)$ (Table 2). Source 75-06 proves it also for the convolution powers $A_\alpha(q)^\mu$, with the same P_j (Theorem 18.1).

[Merge note, batch 77.] Source 77-51 proves Theorem 2.1 for $\alpha = 1$ (its Theorem 6.1, with $d_j = P_j(\pi^2/6)$ printed through $j = 4$ and checked through $j = 6$), and its first two orders for every α , in the resummed form

$$\log a_\alpha(n) = \alpha K(r-1) + \frac{\pi^2 K}{6\alpha(r-1)} + O_\alpha(K/r^3)$$

(Proposition 20.1, with the matching inverse).

[Merge note, batch 100.] Sources 100-82 ($\alpha = 1$; its exact record carries $P_1, \dots, P_{10}$) and 100-89 (every α , four printed terms) prove Theorem 2.1 a sixth and a seventh time (Table 8); P_9 and P_{10} are Proposition 21.1.

For a different level of precision, define

$$F(t) = \sum_{k \geq 1} \log(1 + k^\alpha e^{-tk}), \quad t > 0, \quad (2.6)$$

and let $t = t_n$ be the unique solution of $-F'(t) = n$. Set

$$s = -W_{-1}(-t/\alpha), \quad M = e^s, \quad t = \alpha s/M, \quad H = \frac{M}{\alpha(s-1)}. \quad (2.7)$$

These definitions use the large real branch $s > 1$, valid for all sufficiently large n . Let $B = F''(t)$ and $\kappa_j = (-1)^j F^{(j)}(t)$.

Theorem 2.2 (Exact-saddle coefficient expansion; sources 40 and 48). *As $n \rightarrow \infty$,*

$$M \sim K, \quad s \sim r, \quad B \sim \frac{M^3}{\alpha s}, \quad a_\alpha(n) \sim \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}}.$$

More precisely, for every fixed $J \geq 1$,

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell + O_{\alpha,J}((s/M)^J) \right), \quad (2.8)$$

where each $\mathcal{E}_\ell = O_{\alpha,\ell}((s/M)^\ell)$ is given explicitly in (6.2). The first correction is

$$\mathcal{E}_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3}. \quad (2.9)$$

Theorem 2.3 (Asymptotic inversion; sources 40 and 48). *For $Y \rightarrow \infty$, let $N_\alpha(Y)$ be the least integer n such that $a_\alpha(n) \geq e^Y$. Define*

$$w = W_0\left(\frac{Y}{\alpha e}\right), \quad R_0 = 1 + w, \quad K_0 = \frac{Y}{\alpha w}. \quad (2.10)$$

Then

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + O_\alpha(R_0^{-5}) \right). \quad (2.11)$$

A formal recurrence in Section 9 gives all inverse-logarithmic orders, with a remainder at every fixed order.

Theorem 2.4 (Logistic boundary and Gumbel extreme; sources 40 and 48). *Give $\mathcal{D}_n$ the probability law $\mathbb{P}_n(\lambda) = N(\lambda)^\alpha / a_\alpha(n)$. For each fixed $x \in \mathbb{R}$, a part near $M + xH$ has limiting occupation probability $(1+e^x)^{-1}$; any fixed finite collection of distinct such sites is asymptotically independent. If L_n is the largest part, define*

$$b = b(t_n) > \alpha/t_n, \quad \frac{b^\alpha e^{-t_n b}}{t_n} = 1. \quad (2.12)$$

Then, for every fixed real x ,

$$\mathbb{P}_n\{t_n(L_n - b) \leq x\} \longrightarrow \exp(-e^{-x}), \quad \frac{L_n}{\sqrt{2n}} \longrightarrow 1 + \frac{1}{\alpha} \quad \text{in probability.} \quad (2.13)$$

A precision warning. The error in (2.2) is $O(K/r^{R+1})$, which still grows without bound for fixed R . Consequently, exponentiating a fixed truncation need not approximate $a_\alpha(n)$ to relative error $o(1)$. Formula (2.8) does, because it retains the *exact* saddle and the *exact* infinite sums. There is also a smaller exponential scale from noncentral Fourier arcs; Section 11 identifies it. We do not conflate these three statements into an unjustified single explicit coefficient transseries.

Source 48 draws the same line: the inverse-logarithmic expansion determines the complete block on the scales $Kr, K, K/r, K/r^2, \dots$; it does *not* resolve every smaller block or every exponentially small contribution, and a finite truncation of this block generally leaves a diverging error in $\log a_\alpha(n)$. A multiplicative equivalent is instead obtained from the exact saddle and exact convergent sums. Its numerical routines are independent checks, not substitutes for the proofs; they use arbitrary-precision arithmetic but not directed-rounding interval certification (source 48).

[**Merge note, batch 75.**] Sources 75-01 and 75-06 prove Theorems 2.2, 2.3 and (75-06) 2.4 independently (Table 2). Source 75-01 sharpens the remainder of Theorem 2.2 to $O((\alpha Ms)^{-J})$ (Theorem 16.3) and localizes the threshold of Theorem 2.3 to $1 + o(1)$ (Theorem 16.4); source 75-06 proves Theorems 2.3 and 2.4 for μ layers (Section 18). Both draw the same line between the logarithmic block, the relative expansion and the unresolved exponentially small sectors.

[**Merge note, batch 100.**] Source 100-89 proves Theorem 2.2 again and gives a form with the exact integral and finitely many boundary constants in place of the lattice sum (Theorem 21.4); source 100-82's $\alpha = 1$ equivalent (21.20) uses the saddle of the integral and the constant $\frac{1}{2} \log(2\pi) - 1$. When α grows like $M^{1/3}/\log M$, the single Gaussian of Theorem 2.2 acquires a theta multiplier (Theorem 22.1).

3 A short proof of all five posted conjectures

This section already proves the formulas that are explicitly conjectural in the OEIS entries. Later sections supply the stronger refinements.

[**Merge note, batch 75.**] Sources 75-01 (Proposition 2.3 with Lemma 2.2) and 75-06 (Theorem 2.1) give a third and a fourth proof of Proposition 3.2: the same lower-bound partition, and an upper bound by Lemma 3.1 (75-01) or by the positive-part splitting of source 48's second proof (75-06, for μ layers, Theorem 18.1).

Lemma 3.1 (A useful continuum estimate). *For $s \rightarrow \infty$, $M = e^s$, and $t = \alpha s/M$,*

$$F(t) = \alpha M(s/2 - 1) + O_\alpha(M/s). \quad (3.1)$$

Proof. The function $f_t(x) = \log(1 + x^\alpha e^{-tx})$ is nonnegative and unimodal on $(0, \infty)$, tends to zero at both ends, and has maximum $O_\alpha(s)$. For a nonnegative unimodal function, comparison on unit intervals gives

$$\left| \sum_{k \geq 1} f_t(k) - \int_0^\infty f_t(x) dx \right| \leq 2 \sup_x f_t(x) = O_\alpha(s).$$

With $x = Mu$, the exponent is $\alpha[s(1-u) + \log u]$. Replacing $\log(1 + e^z)$ by $\max(z, 0)$ changes the integral by $O_\alpha(M/s)$: in a fixed neighborhood of $u = 1$, the derivative of $s(u-1) - \log u$ is comparable with s , and $\int_{\mathbb{R}} \log(1 + e^{-|v|}) dv < \infty$. Outside that neighborhood, split off $0 < u < e^{-s/2}$; its integral is exponentially small times a polynomial in s . On the remaining region away from 1, the absolute exponent grows linearly in s , except for the negligible small lower crossing. Thus the remaining error is smaller than M/s . The lower crossing changes the following signed baseline by $O(1)$:

$$\alpha M \int_0^1 (s(1-u) + \log u) du = \alpha M(s/2 - 1).$$

The lattice error $O(s)$ is also $O(M/s)$. □

Proposition 3.2 (sources 40 and 48). *For every fixed $\alpha > 0$,*

$$\log a_\alpha(n) = \alpha \sqrt{2n} (\log \sqrt{2n} - 1) + O_\alpha \left(\frac{\sqrt{n}}{\log n} \right). \quad (3.2)$$

Proof. Let m be maximal with $m(m+1)/2 \leq n$, and write $d = n - m(m+1)/2$, so $0 \leq d \leq m$. The distinct partition

$$1 + 2 + \cdots + (m-1) + (m+d)$$

has norm at least $m!$. Therefore

$$\log a_\alpha(n) \geq \alpha \log(m!) = \alpha K(r-1) + O_\alpha(r),$$

where $m = K + O(1)$ and the usual integral bounds, or Stirling's formula, give $\log(m!) = m \log m - m + O(\log m)$.

For the upper bound, nonnegative coefficients give $a_\alpha(n) \leq \exp(F(t) + nt)$ for every $t > 0$. Choose $t = \alpha r/K$, that is, $M = K$ and $s = r$ in Lemma 3.1. Since $n = K^2/2$,

$$F(t) + nt = \alpha K(r/2 - 1) + \alpha K r/2 + O_\alpha(K/r) = \alpha K(r-1) + O_\alpha(K/r).$$

The two bounds prove (3.2). □

Remark 3.3. The lower-bound partition need not maximize the norm. An exact maximizer is unnecessary. Also, a bare asymptotic equivalence to $\alpha K(r-1)$ would not by itself certify $-\alpha K$ as the second term; the error in (3.2) does certify it.

3.1 Two structural facts and a second proof (source 48)

Why an expansion at the origin cannot simply be used at the saddle.

Proposition 3.4 (source 48). *The Taylor series of the analytic logarithm of A_α at $q = 0$ has radius of convergence $3^{-\alpha/3}$, although A_α is analytic throughout $|q| < 1$.*

Proof. The zeros of $1 + k^\alpha q^k$ have modulus $k^{-\alpha/k}$. Among positive integers k , the maximum of $\log k/k$ occurs at $k = 3$: the function decreases for $k > e$, and $\log 3/3 > \log 2/2$. Hence the closest zeros of the product have modulus $3^{-\alpha/3}$. The remaining factors converge normally and cannot remove a zero. Inside this circle all factors are nonzero, so the product has a single-valued analytic logarithm with value zero at the origin. At a closest zero its logarithm cannot be analytic. This proves the assertion. $\square$

Consequently,

$$\log A_\alpha(q) = \sum_{k \geq 1} \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} k^{\alpha j} q^{kj}$$

is useful as a *formal* identity, but this expansion is not convergent at the positive saddles approaching $q = 1$. Treating it as a convergent expansion there would lose the transition that controls this problem.

The largest individual weight. Let

$$\Pi_\alpha(n) = \max_{\lambda \in \mathcal{D}_n} N(\lambda)^\alpha.$$

Proposition 3.5 (source 48). *For fixed $\alpha > 0$, with $K = \sqrt{2n}$ and $r = \log K$,*

$$\log \Pi_\alpha(n) = \alpha K(r - 1) + O_\alpha(r). \quad (3.3)$$

Proof. Write $n = m(m+1)/2 + d$ with $0 \leq d \leq m$. The distinct partition $(1, 2, \dots, m-1, m+d)$ has weight $((m-1)!(m+d))^\alpha$. Since $m = K + O(1)$, Stirling's estimate yields the required lower bound, with error $O_\alpha(r)$.

For the upper bound set $t = \alpha r/K$. Any distinct partition λ of n satisfies

$$\alpha \sum_{k \in \lambda} \log k \leq tn + \sum_{k \geq 1} (\alpha \log k - tk)_+, \quad u_+ = \max(u, 0).$$

For sufficiently large K , the positive terms are exactly those with $2 \leq k < K$, apart from a possible zero term at an integer endpoint. Summing $\log k$ by Stirling's formula and k by the arithmetic progression formula gives

$$\sum_{k \geq 1} (\alpha \log k - tk)_+ = \alpha K(r/2 - 1) + O_\alpha(r).$$

Adding $tn = \alpha Kr/2$ proves the upper bound. $\square$

Lemma 3.6 (transition width; source 48). *Let $M = e^s$, $t = \alpha s/M$, and $s \rightarrow \infty$. Then*

$$\sum_{k \geq 1} \exp(-|\alpha \log k - tk|) = O_\alpha(M/s). \quad (3.4)$$

Proof. Put $\psi(u) = \alpha \log u - tu$. On $k \geq M/2$, the derivative of ψ is at most $-t/2$ once $s \geq 4$, and $\psi(M) = 0$. Thus this part of the sum is bounded by a two-sided geometric sum of scale $1/t = O_\alpha(M/s)$.

For $1 \leq k \leq M^{1/4}$, use the bound 1, giving $O(M^{1/4})$. On $[M^{1/4}, M/2]$, concavity of ψ implies that its minimum is at an endpoint. For large s , both endpoint values exceed $c_\alpha s$ for a constant $c_\alpha > 0$. This range therefore contributes $O_\alpha(M^{1-c_\alpha})$. Both estimates are $o_\alpha(M/s)$. Adjusting integer endpoints changes nothing. $\square$

Second proof of Proposition 3.2 (source 48). The lower bound follows from $a_\alpha(n) \geq \Pi_\alpha(n)$ and Proposition 3.5 (the same partition as in the first proof). For the upper bound, positivity gives $a_\alpha(n) \leq e^{tn} A_\alpha(e^{-t})$. Use $t = \alpha r/K$ and the exact identity

$$\log(1 + e^u) = u_+ + \log(1 + e^{-|u|}).$$

The positive-part sum was evaluated in the proof of Proposition 3.5; the second sum is bounded by Lemma 3.6 with $M = K$, $s = r$. We obtain

$$tn + F(t) = \alpha K(r - 1) + O_\alpha(K/r + r),$$

and $r = o(K/r)$. This proves (3.2) without any local limit theorem, continuum integral or formal asymptotic manipulation. $\square$

This argument is a separate, elementary proof of the OEIS conjecture; the remaining sections determine what lies inside its error term (source 48).

4 The exact probabilistic saddle and its scales

For fixed $t > 0$, let X_k be independent Bernoulli random variables with

$$p_k = \mathbb{P}_t(X_k = 1) = \frac{k^\alpha e^{-tk}}{1 + k^\alpha e^{-tk}}, \quad S = \sum_{k \geq 1} k X_k. \quad (4.1)$$

The sum is finite almost surely because $\sum p_k < \infty$. Multiplying the Bernoulli probabilities gives the exact identity

$$\mathbb{P}_t(S = n) = a_\alpha(n) e^{-nt - F(t)}. \quad (4.2)$$

Differentiation under the absolutely convergent sums shows

$$\mathbb{E}_t S = -F'(t) = \sum k p_k, \quad \text{Var}_t S = F''(t) = \sum k^2 p_k (1 - p_k) > 0.$$

The mean is continuous and strictly decreasing from ∞ to 0 as t increases from 0 to ∞ . Hence the saddle equation has a unique solution for every $n > 0$.

The large crossing $k^\alpha e^{-tk} = 1$ occurs at $k = M$ in (2.7). Its logarithmic derivative there is

$$\frac{\alpha}{M} - t = -\frac{\alpha(s-1)}{M} = -H^{-1}.$$

Thus H is the width of the transition from predominantly occupied to predominantly unoccupied sites.

Lemma 4.1 (Transition estimates). *For fixed $\alpha > 0$, as $s \rightarrow \infty$,*

$$\sum_{k \geq 1} k p_k \sim M^2/2, \quad (4.3)$$

$$\sum_{k \geq 1} k^2 p_k (1 - p_k) \sim M^2 H \sim M^3/(\alpha s). \quad (4.4)$$

For each fixed integer $j \geq 2$,

$$\sum_{k \geq 1} k^j p_k (1 - p_k) = O_{\alpha,j}(M^{j+1}/s). \quad (4.5)$$

Moreover, for every fixed nonnegative integer h ,

$$\frac{1}{H^{h+1}} \sum_{k \geq 1} |k - M|^h p_k (1 - p_k) \longrightarrow \int_{\mathbb{R}} |x|^h \frac{e^x}{(1 + e^x)^2} dx. \quad (4.6)$$

Proof. For $k = M + xH$ with x in a fixed bounded interval,

$$\alpha \log k - tk = \alpha \log(1 + xH/M) - \alpha sxH/M = -x + O_{\alpha,x}(s^{-2}).$$

Here and below rounding k to an integer changes the scaled coordinate by $O(H^{-1}) = o(1)$. Thus p_k converges to $(1 + e^x)^{-1}$. Since $H \rightarrow \infty$, sums over any fixed x -interval are Riemann sums. The limiting variance density has integral 1.

We record a tail justification so that this argument is not just a local heuristic. On $k/M \in [1/2, 3/2]$, the derivative of the log odds is negative and comparable to $-1/H$ for large s ; the variance density is therefore bounded by $C_\alpha e^{-c_\alpha |k-M|/H}$. On $k \geq 3M/2$, the odds themselves are exponentially small in s and then in tk . On $k \leq M/2$, use $p_k(1 - p_k) \leq \min(1, k^{-\alpha} e^{tk})$, and split at M/s . The range $M/s < k \leq M/2$ is exponentially small in s after scaling by any of the powers of M above. On $k \leq M/s$, the crude bound is $O_\alpha(k^{-\alpha})$; its contribution to (4.6), even with $|k - M|^h \leq M^h$, is $o(H^{h+1})$. Indeed the resulting ratio is bounded by a fixed power of s times $M^{-\min(\alpha,1)}$, with an extra logarithm when $\alpha = 1$. These bounds establish uniform integrability for every fixed moment. They also prove (4.5) and (4.4).

Finally, $p_{\lfloor Mu \rfloor} \rightarrow 1$ for $0 < u < 1$ and tends to zero for $u > 1$. The same tail bounds and a Riemann sum give $M^{-2} \sum k p_k \rightarrow \int_0^1 u \, du = 1/2$. $\square$

Source 48 sets up the same Bernoulli representation (its (4.2) is identical, with the total written X_t and the variance V), and the same transition coordinate $s = -W_{-1}(-t/\alpha)$, which it calls the *upper transition coordinate*; the lower real branch is necessary because $s \rightarrow \infty$, and the branch conventions agree with the DLMF [15]. Its version of Lemma 4.1 (its Lemma 3.1) states $-F'(t) \sim M^2/2$, $B \sim M^3/(\alpha s)$ and $|\kappa_j| = O_{\alpha,j}(M^{j+1}/s)$ for $j \geq 2$, with the following proof.

Second proof of (4.3)–(4.5) (source 48). In the scaled variable $k = Mu$, the log odds equal $\alpha s(1 - u) + \alpha \log u$. For $0 < u < 1$ this tends to $+\infty$, and for $u > 1$ it tends to $-\infty$. Splitting the sum at fixed fractions of M , and using exponential decay above a fixed multiple of M , gives $M^{-2} \sum k p_k \rightarrow \int_0^1 u \, du$.

Near $k = M$, put $v = t(k - M)$. The log odds are

$$-v + \alpha \log \left(1 + \frac{v}{\alpha s} \right) \rightarrow -v.$$

The mesh in v is $t \rightarrow 0$. The derivative estimate used in Lemma 3.6 supplies an exponentially integrable bound on $k \geq M/2$. Thus a Riemann-sum limit gives

$$B \sim \frac{M^2}{t} \int_{-\infty}^{\infty} \frac{e^v}{(1 + e^v)^2} \, dv = \frac{M^2}{t} = \frac{M^3}{\alpha s}.$$

The range below $M/2$ is negligible. To see this for any fixed moment order, split it at $M^{1/4}$ as in Lemma 3.6, now multiplying by k^j . The small range is $O(M^{(j+1)/4})$; the other range is $O_\alpha(M^{j+1-c_\alpha})$. Above $M/2$, the exponential envelope gives $\sum k^j p_k(1 - p_k) = O_{\alpha,j}(M^{j+1}/s)$. This proves the cumulant bound, and normalization by the variance proves (4.8). $\square$

At the exact saddle, (4.3) gives $M \sim K$ and $s \sim r$. For cumulants, write $b_1(p) = p$ and $b_{j+1}(p) = p(1 - p)b'_j(p)$. Then

$$\kappa_j = \sum_{k \geq 1} k^j b_j(p_k), \quad |b_j(p)| \leq C_j p(1 - p) \quad (j \geq 2). \quad (4.7)$$

Consequently, with $\varepsilon = s/M$,

$$\frac{|\kappa_j|}{B^{j/2}} = O_{\alpha,j}(\varepsilon^{(j-2)/2}), \quad j \geq 3. \quad (4.8)$$

These bounds are deliberately conservative; signed cancellations can make individual corrections smaller. They suffice for all fixed Edgeworth orders.

[Merge note, batch 75.] Sources 75-01 and 75-06 determine those cancellations: $\kappa_j \sim \frac{j!}{2} M^{j+1}/(\alpha s)^{j-1}$, so the normalized cumulants are $O(\varepsilon_{\#}^{(j-2)/2})$ with $\varepsilon_{\#} = 1/(\alpha M s)$ (Lemma 16.1 and Theorem 16.3).

5 The local limit theorem, including noncentral arcs

5.1 An elementary consecutive-block inequality

Lemma 5.1 (sources 40 and 48). *There is an absolute constant $c_0 > 0$ such that for every integer $h \geq 2$, every integer a , and every $|\theta| \leq \pi$,*

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 \min(h, h^3\theta^2). \quad (5.1)$$

Proof. The geometric-sum formula gives

$$\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq \frac{1}{2} \left(h - \left| \frac{\sin(h\theta/2)}{\sin(\theta/2)} \right| \right).$$

When $h|\theta| \leq 1$, Taylor bounds for sine show that the difference in parentheses is at least $ch(h^2 - 1)\theta^2$. For $h|\theta| \geq 2\pi$, the bound $|\sin(\theta/2)| \geq |\theta|/\pi$ makes the ratio at most $h/2$. On the intermediate range $1 \leq h|\theta| \leq 2\pi$, its ratio to h is uniformly less than one. To see the uniformity, let $h \rightarrow \infty$ and use uniform convergence to $|\sin(x/2)/(x/2)| < 1$ on $1 \leq x \leq 2\pi$; the finitely many remaining h have the same strict inequality by the strict triangle inequality for a nonconstant geometric progression. Taking the minimum of the resulting positive constants proves the lemma. $\square$

Source 48 states the lemma for every sufficiently large h in the equivalent form $\sum_{k=a}^{a+h-1} \sin^2(k\theta/2) \geq c_0 h \min\{1, h^2\theta^2\}$.

Second proof (source 48). The sum is at least

$$\frac{1}{2} \left(h - \left| \sum_{k=a}^{a+h-1} e^{ik\theta} \right| \right) = \frac{1}{2} \left(h - \frac{|\sin(h\theta/2)|}{|\sin(\theta/2)|} \right).$$

For $h|\theta| \leq 1$, Taylor's formula bounds the deficit below by a constant times $h^3\theta^2$. For $1 \leq h|\theta| \leq C$, divide by h and use the strict inequality $|\sin u| < |u|$ away from zero, uniformly on a compact interval; the denominator converges uniformly to $|h\theta|/2$ in this range. For $h|\theta| \geq C$, use $|\sin(\theta/2)| \geq |\theta|/\pi$, choosing a fixed sufficiently large C . These three ranges prove the result. $\square$

5.2 Fourier decay and the central Gaussian

Let $\phi_t(\theta) = \mathbb{E}_t e^{i\theta(S - \mathbb{E}_t S)}$. The modulus identity for one Bernoulli factor implies

$$|\phi_t(\theta)| \leq \exp \left(-2 \sum_{k \geq 1} p_k(1 - p_k) \sin^2(k\theta/2) \right). \quad (5.2)$$

In a consecutive block of length $h \asymp_\alpha M/s$ centered at M , all $p_k(1 - p_k)$ are bounded below by a positive constant. On $|\theta| \leq c_1/M$, this block gives

$$|\phi_t(\theta)| \leq e^{-c_2 B \theta^2}. \quad (5.3)$$

Indeed $k \asymp M$ on the block, and $\sin(k\theta/2)$ is comparable with $k\theta$ for sufficiently small c_1 . On its complement in $[-\pi, \pi]$, Lemma 5.1 gives

$$|\phi_t(\theta)| \leq \exp(-c_3 M/s^3), \quad c_1/M \leq |\theta| \leq \pi. \quad (5.4)$$

All constants here may depend on α . Notice the power s^3 , rather than s : the random part sizes lie in a relatively narrow consecutive block, so one must control possible near-resonances. An argument checking only a neighborhood of zero would leave a genuine gap. Source 48 proves the same two bounds from the same block (its (5.3) and (5.4) are written with V for B and L for s): the worst case of Lemma 5.1 at $|\theta| = c_1/M$ is a constant times $h^3/M^2 \asymp_\alpha M/s^3$, so in particular no other root-of-unity arc contributes at an algebraic order.

At the saddle, Fourier inversion gives

$$\mathbb{P}_t(S = n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \phi_t(\theta) d\theta.$$

For bounded $x = \theta\sqrt{B}$, Taylor expansion and (4.8) give $\phi_t(x/\sqrt{B}) \rightarrow e^{-x^2/2}$. The central bound dominates this integrand by e^{-cx^2} . The noncentral integral, even after multiplication by $\sqrt{B}$, is $O(\sqrt{B}e^{-c_3 M/s^3}) = o(1)$. Thus

$$\mathbb{P}_t(S = n) \sim (2\pi B)^{-1/2}. \quad (5.5)$$

Together with (4.2), this proves the leading part of Theorem 2.2. In particular,

$$\log a_\alpha(n) = F(t_n) + nt_n + O_\alpha(s), \quad (5.6)$$

since $\log B = O(s)$.

5.3 A uniform variant used for conditioning

The same proof gives the following useful extension. Delete either a fixed number of sites $k = O(M)$, or all sites above a cutoff m for which $m/M \rightarrow \gamma > 1$ and the deleted mean and variance are $O(M)$ and $O(M^2)$. The central consecutive block remains available, the retained variance is $B(1 + o(1))$, and the cumulant bounds remain valid. If an integer target differs from the retained mean by $o(\sqrt{B})$, its point probability is

$$(2\pi B)^{-1/2}(1 + o(1)). \quad (5.7)$$

This follows in the central integral by inserting $e^{-ix(m - \mathbb{E}S)/\sqrt{B}} \rightarrow 1$; all the dominating estimates are unchanged. In the fixed-site deletion case, the deleted variance is automatically $O(M^2)$. The assertion is uniform for boundedly many deletions and shifts bounded by a fixed

multiple of M . Source 48 records the same extension from its proof of Theorem 2.2, with a factor $e^{-ih\theta}$ in the integral: uniformly for $h = o(\sqrt{B})$,

$$\mathbb{P}_t(S = -F'(t) + h) = \frac{1 + o(1)}{\sqrt{2\pi B}} \quad \text{whenever } -F'(t) + h \text{ is an integer,} \quad (5.8)$$

also after deleting finitely many summands, or after truncating well above the active interval, provided the removed variance is $o(B)$ (source 48).

6 The all-orders coefficient correction

The normalized Fourier integrand has the expansion

$$\phi_t(x/\sqrt{B}) = e^{-x^2/2} \exp\left(\sum_{j \geq 3} \frac{\kappa_j (\mathrm{i}x)^j}{j! B^{j/2}}\right). \quad (6.1)$$

Assign weight $j - 2$ to the j th normalized cumulant. Every monomial of odd weight has odd total power of x and integrates to zero. A monomial of weight 2ℓ has order $O(\varepsilon^\ell)$ by (4.8). Gaussian integration therefore gives

$$\mathcal{E}_\ell = \sum_{\substack{m_3, \dots, m_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)m_j = 2\ell}} \frac{(-1)^{T/2} (T-1)!!}{B^{T/2}} \prod_{j=3}^{2\ell+2} \frac{\kappa_j^{m_j}}{(j!)^{m_j} m_j!}, \quad T = \sum_{j=3}^{2\ell+2} j m_j. \quad (6.2)$$

Here T is automatically even. Besides (2.9), the next term is

$$\begin{aligned} \mathcal{E}_2 = & -\frac{\kappa_6}{48B^3} + \frac{7\kappa_3\kappa_5}{48B^4} + \frac{35\kappa_4^2}{384B^4} \\ & - \frac{35\kappa_3^2\kappa_4}{64B^5} + \frac{385\kappa_3^4}{1152B^6}. \end{aligned} \quad (6.3)$$

For completeness, the formal expansion above has the needed integrated remainder. On $|\theta| \leq c/M$, derivatives of order $j \geq 2$ of $\log \mathbb{E}_t e^{\mathrm{i}\theta S}$ have absolute value at most $C_{\alpha,j} M^{j+1}/s$. For $k \leq 2M$, take c sufficiently small so that the Bernoulli denominators have modulus comparable with $1 + k^\alpha e^{-tk}$. For $k > 2M$, the odds tend uniformly to zero; the denominators are bounded away from zero and the exponentially decreasing tails give the same bound. These observations justify Taylor's formula with controlled remainders.

Fix J , choose a small $\delta > 0$ depending on J , and first restrict the normalized integral to $|x| \leq \varepsilon^{-\delta}$. Expand through weight $2J - 1$, using Taylor's formula for the exponential. The terms of weight $2J$ and higher have an integrable bound of the form $C_J \varepsilon^J (1 + |x|^{C_J}) e^{-x^2/4}$, after choosing δ small enough. The omitted odd-weight terms integrate to zero on the symmetric interval. Between this interval and $c\sqrt{B}/M$, (5.3) makes the integral smaller than every power of ε . The remaining Fourier range is bounded by (5.4), with the same conclusion. This proves (2.8) and its error for every fixed J .

Remark 6.1 (What an effective finite-size bound would contain). Before absorbing flat terms into the asymptotic O -term, the proof contains an additional relative bound of the shape $C_\alpha \sqrt{B} \exp(-c_\alpha M/s^3)$ for noncentral arcs. This can be much larger than the central Edgeworth corrections at attainable n , especially for larger α . The theorem is not a claim that its first few terms have small relative error uniformly in α or for all moderate n .

6.1 The cumulative tuple form and a second proof (source 48)

Source 48 writes the corrections cumulatively. For a finitely supported tuple $\mathbf{m} = (m_3, m_4, \dots)$ of nonnegative integers put $w(\mathbf{m}) = \sum_{j \geq 3} (j-2)m_j$ and $T(\mathbf{m}) = \sum_{j \geq 3} jm_j$. Then, for every $J \geq 1$,

$$1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell = \sum_{\substack{w(\mathbf{m}) < 2J \\ T(\mathbf{m}) \text{ even}}} (-1)^{T(\mathbf{m})/2} (T(\mathbf{m}) - 1)!! \prod_{j \geq 3} \frac{1}{m_j!} \left(\frac{\kappa_j}{j! B^{j/2}} \right)^{m_j}, \quad (6.4)$$

where the empty tuple contributes 1 and $(-1)!! = 1$. This is (6.2) summed over $\ell < J$ (an even T is the same as an even weight), and (2.8) reads $a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} ((6.4) + O_{\alpha,J}((s/M)^J))$; the leading formula is a genuine multiplicative equivalent.

Second proof of Theorem 2.2 (source 48). Fourier inversion and (4.2) reduce the assertion to the probability $\mathbb{P}_t(S = n)$ at the exact saddle. Set $u = \theta\sqrt{B}$ and center the characteristic function; its expansion at zero is (6.1), interpreted to any fixed finite Taylor order with a remainder.

The Taylor expansion is uniform for the following reasons. For $|\theta| \leq c_0/M$, derivatives of order $j \geq 2$ of the logarithm of the characteristic function are $O_{\alpha,j}(M^{j+1}/s)$. For k in a fixed multiple of $[1, M]$, either the Bernoulli parameter is away from $1/2$, or the small phase keeps the factor uniformly away from zero. For k above that multiple of M , p_k is small and its factor is again uniformly away from zero. The derivatives in both ranges are bounded by $C_j k^j p_k(1 - p_k)$; summation uses Lemma 4.1. Taylor's integral remainder therefore obeys the corresponding bound from (4.8).

Explicit finite-order bookkeeping for the central integral: fix J , put $D = 2J + 2$, and write

$$Z(u) = \sum_{j=3}^D \frac{\kappa_j (iu)^j}{j! B^{j/2}}.$$

Taylor's remainder in the exponent is bounded by $C_{\alpha,J} \varepsilon^{(D-1)/2} |u|^{D+1}$. Choose $0 < \sigma < 1/6$ sufficiently small that also $\sigma < (D-1)/(2(D+1))$. On $|u| \leq \varepsilon^{-\sigma}$, both $Z(u)$ and this remainder tend uniformly to zero. Consequently replacing the exact exponential by $e^{-u^2/2} e^{Z(u)}$ has integrated error $O_{\alpha,J}(\varepsilon^{J+1/2})$, by the bound $|e^{a+b} - e^a| \leq |b|e^{|a|+|b|}$ and a fixed Gaussian moment.

Next expand e^Z through degree $2J$. Its remainder, integrated against the Gaussian, is again $O_{\alpha,J}(\varepsilon^{J+1/2})$: every coefficient in Z is $O_{\alpha,J}(\varepsilon^{1/2})$, so $|Z|^{2J+1}$ is bounded by that power of ε times a fixed polynomial in $|u|$. In the finite expansion of the retained powers, any monomial of weight at least $2J$ has integrated absolute size $O_{\alpha,J}(\varepsilon^J)$. Every monomial of smaller weight occurs in these powers, because each factor has positive weight and the largest required index is at most $2J + 1$. Odd total degree integrates to zero on the symmetric central interval. The surviving terms are precisely those in (6.4). This argument uses only finite Taylor expansions and their remainders, not uniform convergence of an infinite series.

The remainder of $|\theta| \leq c_0/M$ is smaller than every power of ε by (5.3); the rest of $[-\pi, \pi]$ is likewise negligible by (5.4). Integrating each retained polynomial over the whole real line introduces only another negligible Gaussian tail. The Gaussian moment is zero for odd T and equals $(T-1)!!$ for even T ; the phase i^T supplies the sign. At weight two the only possibilities are $m_4 = 1$ and $m_3 = 2$, giving exactly (2.9). $\square$

7 Continuum smoothing to every logarithmic order

7.1 The inverse at the occupation boundary

Define

$$h_s(u) = s(u - 1) - \log u. \quad (7.1)$$

Near $u = 1$ it is strictly increasing. Let $u_s(v)$ be its inverse with $u_s(0) = 1$, and put

$$U_m(s) = u_s^{(m)}(0). \quad (7.2)$$

The derivative operator along the inverse is

$$\frac{d}{dv} = \frac{1}{s - 1/u} \frac{d}{du}. \quad (7.3)$$

Starting with u and applying (7.3) repeatedly, then setting $u = 1$, gives an effective rational-function recursion. Equivalently, Lagrange inversion gives

$$U_m(s) = (m - 1)! [z^{m-1}] \left(\frac{z}{sz - \log(1 + z)} \right)^m. \quad (7.4)$$

The first required odd derivatives are

$$\begin{aligned} U_1(s) &= \frac{1}{s - 1}, \\ U_3(s) &= \frac{2s + 1}{(s - 1)^5}, \\ U_5(s) &= \frac{24s^3 + 58s^2 + 22s + 1}{(s - 1)^9}, \\ U_7(s) &= \frac{720s^5 + 3708s^4 + 4400s^3 + 1452s^2 + 114s + 1}{(s - 1)^{13}}. \end{aligned} \quad (7.5)$$

Write $\eta(z) = (1 - 2^{1-z})\zeta(z)$. The elementary even kernel $g(v) = \log(1 + e^{-|v|})$ has moments

$$\int_{\mathbb{R}} v^{2j} g(v) dv = 2(2j)! \eta(2j + 2), \quad j \geq 0. \quad (7.6)$$

Indeed expand the logarithm on $(0, \infty)$ and integrate each exponentially decaying term. Odd moments vanish.

Lemma 7.1 (Boundary smoothing expansion). *For every fixed $J \geq 1$,*

$$\int_0^\infty \log(1 + x^\alpha e^{-tx}) dx = M \left(\alpha(s/2 - 1) + \sum_{j=1}^J \frac{2\eta(2j)}{\alpha^{2j-1}} U_{2j-1}(s) + O_{\alpha,J}(s^{-2J-1}) \right), \quad (7.7)$$

where $t = \alpha s e^{-s}$. The expansion may be differentiated with respect to s any fixed number of times, retaining a remainder smaller than any prescribed inverse power of s by taking sufficiently many terms first.

Proof. After $x = Mu$, the integrand is $\log(1 + e^{-\alpha h_s(u)})$. The baseline is $\alpha M \int_0^1 [s(1 - u) + \log u] du = \alpha M(s/2 - 1)$. The discrepancy between this signed baseline and the positive-part baseline is confined to the small lower crossing and is exponentially small in s before multiplication by M .

Choose a fixed small $\rho > 0$ and restrict initially to $|u - 1| < \rho$. Replacing the original integrand by its positive-part baseline leaves $g(\alpha h_s(u))$. On a symmetric v -interval $|v| < \gamma s$ inside the image of this neighborhood, changing variables $v = h_s(u)$ gives

$$\int g(\alpha v) u'_s(v) dv.$$

The omitted portions have exponentially small tails. Taylor-expand $u'_s(v)$ through degree $2J - 1$. On the chosen interval, repeated use of (7.3) bounds the derivative of order $2J$ of u'_s by $O_J(s^{-2J-1})$. The Taylor remainder is thus bounded by $C_J s^{-2J-1} |v|^{2J}$, whose product with $g(\alpha v)$ is integrable. After extending the polynomial terms to the whole real line, odd terms vanish, and (7.6) supplies exactly $2\eta(2j)U_{2j-1}(s)/\alpha^{2j-1}$.

Here is a more explicit treatment of the region near zero, which must not be mistaken for part of the main boundary. On $0 < u < e^{-s/2}$, both the original integrand and the absolute baseline have integral at most $C_\alpha s e^{-s/2}$. On $e^{-s/2} \leq u \leq 1 - \rho$, the concavity of $s(1 - u) + \log u$ puts it above a positive multiple of s for large s . The smoothing error there is exponentially small. On $u \geq 1 + \rho$ the log odds are negative and decrease at rate comparable with s . All these contributions are $O_{\alpha,J}(s^{-2J-1})$ for every fixed J .

Differentiation with respect to s inserts powers of $1 - u$ and derivatives of the same logistic kernel. On the boundary interval the inverse derivatives still have polynomial bounds in s^{-1} ; away from it, exponential tail bounds remain after multiplication by any fixed polynomial. Taylor expansion to a higher order therefore gives the differentiated version with any prescribed inverse-power precision. No assumption about differentiating an arbitrary O -term is being used. $\square$

7.2 From the continuum to the lattice

As in Lemma 3.1, the difference between $F(t)$ and its continuum integral is $O_\alpha(s)$. We also need the mean. The function

$$x \mapsto \frac{x^{\alpha+1} e^{-tx}}{1 + x^\alpha e^{-tx}} = xp(x)$$

is unimodal and has maximum $O_\alpha(M)$. To verify unimodality, its derivative has the sign of $1 + (\alpha - tx)(1 - p(x))$. This is positive for $x \leq \alpha/t$. For $x > \alpha/t$, the function $(tx - \alpha)(1 - p(x))$ increases strictly from zero to infinity. Thus

$$\sum_{k \geq 1} k p_k = \int_0^\infty xp(x) dx + O_\alpha(M). \quad (7.8)$$

This exact lattice error is negligible at every fixed inverse-logarithmic order relative to the mean M^2 .

Introduce the formal Laurent series at $s = \infty$

$$d_\alpha(s) = \sum_{j \geq 1} \frac{2\eta(2j)}{\alpha^{2j}} U_{2j-1}(s). \quad (7.9)$$

The notation is asymptotic/formal: no convergence of the infinite sum in j is asserted. Each coefficient of s^{-1} uses only finitely many summands, since $U_m(s) = O_m(s^{-m})$. The expansion begins

$$\begin{aligned} d_\alpha(s) = & \frac{c}{s} + \frac{c}{s^2} + \frac{c}{s^3} + \frac{c + 7c^2/5}{s^4} + \frac{c + 77c^2/10}{s^5} \\ & + \frac{c + 49c^2/2 + 372c^3/35}{s^6} + \dots \end{aligned} \quad (7.10)$$

In the all-orders logarithmic sense,

$$F(t) \sim \alpha M(s/2 - 1 + d_\alpha(s)). \quad (7.11)$$

The lattice error $O(s)$ and the non-boundary continuum pieces are smaller than M/s^R for every fixed R . They are *not* thereby zero and must not be discarded when a multiplicative coefficient equivalent is wanted.

[Batch 100, 5 October 2026.] Source 100-89 identifies the lattice error: the difference $\Xi_\alpha(t) = F(t) - \int_0^\infty \log(1 + x^\alpha e^{-tx}) dx$ extends smoothly to $t = 0$, with explicit Taylor coefficients $\xi_r(\alpha)$, for instance $\xi_0(1) = \frac{1}{2} \log(2\pi) - 1$ (Theorem 21.2). So the bound $O_\alpha(s)$ above is in fact $O_\alpha(1)$, and the retained multiplicative factor is $e^{\xi_0(\alpha)}(1 + \xi_1(\alpha)t + \cdots)$ (Corollary 21.6).

7.3 A second route: an explicit differential operator (source 48)

Source 48 proves the two lattice estimates by the bounded-variation inequality $|\sum_{k \geq 1} f(k) - \int_0^\infty f| \leq \int_0^\infty |f'| = 2 \max f$ for a nonnegative unimodal f with $f(0) = 0$ tending to zero at infinity, applied to $\log(1 + \xi^\alpha e^{-t\xi})$ (maximum $O_\alpha(s)$) and to $\xi p(\xi)$ (maximum $O_\alpha(M)$, unimodal by the argument before (7.8)). It then replaces the inverse u_s and the Lagrange coefficients U_m by one explicit operator. Let

$$f(s) = \frac{s}{s-1}, \quad \mathcal{T} = \frac{s}{s-1} \frac{d}{ds}, \quad (7.12)$$

with η as in Section 7, and define the asymptotic series at $s = \infty$

$$\boxed{\mathcal{C}_\alpha(s) \sim 2 \sum_{j \geq 0} \frac{\eta(2j+2)}{\alpha^{2j+2}} \mathcal{T}^{2j} f(s), \quad \mathcal{B}_\alpha(s) = \mathcal{C}_\alpha(s) + \mathcal{T} \mathcal{C}_\alpha(s).} \quad (7.13)$$

The formula is locally finite after expansion in s^{-1} : for $j \geq 1$, $\mathcal{T}^{2j} f(s) = O(s^{-2j-1})$, so only finitely many summands affect any prescribed coefficient.

Lemma 7.2 (upper-transition expansion; source 48). *With $t = \alpha s e^{-s}$, in the sense of asymptotic series to every fixed finite order,*

$$F(t) \sim \frac{\alpha^2}{t} \left(\frac{s^2}{2} - s + \mathcal{C}_\alpha(s) \right), \quad (7.14)$$

$$-F'(t) \sim \frac{\alpha^2}{t^2} \left(\frac{s^2}{2} + \mathcal{B}_\alpha(s) \right). \quad (7.15)$$

More precisely, replacing $\mathcal{C}_\alpha$ or $\mathcal{B}_\alpha$ by its polynomial through s^{-R} leaves $O_{\alpha,R}(s^{-R-1})$ inside the corresponding parentheses, for every fixed $R \geq 0$.

Since $\alpha^2/t = \alpha M/s$, (7.14) is (7.11) with $d_\alpha(s) = \mathcal{C}_\alpha(s)/s$, and (7.15) is the mean relation (8.1) with $Q_\alpha(s) = 1 + 2\mathcal{B}_\alpha(s)/s^2$.

Proof. In the continuum integral of F , with integration variable ξ , put $y = t\xi/\alpha$. Since $\log(\alpha/t) = s - \log s$, the exponent becomes

$$\alpha \log \xi - t\xi = \alpha \tilde{h}_s(y), \quad \tilde{h}_s(y) = s - y + \log(y/s).$$

The zero-temperature contribution is

$$\int_0^s \alpha \tilde{h}_s(y) dy = \alpha(s^2/2 - s). \quad (7.16)$$

Subtracting this reference integral is legitimate even though $\tilde{h}_s$ becomes negative very near zero; that region is included in the error estimate.

On $y \geq s/2$, use the increasing coordinate $v = y - s - \log(y/s) = -\tilde{h}_s(y)$. If $y = y(v)$ is its inverse, then $y(0) = s$ and $y'(v) = y(v)/(y(v) - 1) = f(y(v))$. The difference from the reference integrand is $g(\alpha v) = \log(1 + e^{-\alpha|v|})$, an even exponentially decreasing function, with moments

$$\int_{-\infty}^{\infty} v^{2j} g(\alpha v) dv = \frac{2(2j)!}{\alpha^{2j+1}} \eta(2j+2), \quad (7.17)$$

by (7.6); these are the elementary Fermi–Dirac moments underlying Sommerfeld expansions [16]. Since $d/dv = f(y) d/dy$, Taylor expansion of $y'(v)$ at zero gives

$$\left. \frac{d^{2j}}{dv^{2j}} y'(v) \right|_{v=0} = \mathcal{T}^{2j} f(s).$$

On $|v| \leq s/4$, $y(v) \asymp s$, and for each fixed $a \geq 1$, $\mathcal{T}^a f(y) = O_a(s^{-a-1})$. Expanding through the odd degree $2J+1$, integrating against the even kernel, and bounding the next derivative gives an error $O_{\alpha,J}(s^{-2J-3})$. The portions with $|v| > s/4$ are exponentially small, since $y'(v)$ stays bounded on the domain $y \geq s/2$.

The range $0 < y < s/2$ also has exponentially small error *inside the scaled integral*. Split it at $e^{-s/2}$. On the small interval, integrate the bound $C_\alpha(1 + s + |\log y|)$, obtaining $O_\alpha(s e^{-s/2})$. On $[e^{-s/2}, s/2]$, concavity gives $\tilde{h}_s(y) \geq s/2 - \log s - O(1)$, so the discrepancy is $O_\alpha(s^{\alpha+1} e^{-\alpha s/2})$. These errors need not remain exponentially small in the original, unscaled variable, but they are smaller than every inverse power of s inside the present parentheses. Multiplying the integral by α/t , the moments (7.17) give (7.14); the sum–integral error is absorbed at every fixed order.

The mean formula is proved directly, rather than by differentiating an unspecified big- O term. The scaled mean integral is $\frac{\alpha^2}{t^2} \int_0^\infty y (1 + e^{\alpha y})^{-1} dy$, with reference integral $\int_0^s y dy = s^2/2$. Under the same coordinate change the difference kernel is the odd function

$$\mathcal{K}_\alpha(v) = \frac{1}{1 + e^{\alpha v}} - \mathbf{1}_{v < 0},$$

integrated against $y(v)y'(v)$. Its odd moments are

$$\int_{-\infty}^{\infty} v^{2j+1} \mathcal{K}_\alpha(v) dv = \frac{2(2j+1)!}{\alpha^{2j+2}} \eta(2j+2).$$

The elementary operator identity $\mathcal{T}(sf(s)) = f(s) + \mathcal{T}f(s)$ shows that the resulting correction is precisely $\mathcal{C}_\alpha + \mathcal{T}\mathcal{C}_\alpha$. The same cutoff and Taylor estimates apply, and (7.8) absorbs the discretization error. This proves (7.15) with controlled remainders, independently of any differentiation rule for errors. $\square$

In the parameter c , the beginning of (7.13) is

$$\mathcal{C}_\alpha(s) = c + \frac{c}{s} + \frac{c}{s^2} + \frac{c + \frac{7}{5}c^2}{s^3} + \frac{c + \frac{77}{10}c^2}{s^4} + \frac{c + \frac{49}{2}c^2 + \frac{372}{35}c^3}{s^5} + O_\alpha(s^{-6}), \quad (7.18)$$

$$\mathcal{B}_\alpha(s) = c + \frac{c}{s} + \frac{-2c + \frac{7}{5}c^2}{s^3} + \frac{-5c + \frac{7}{2}c^2}{s^4} + O_\alpha(s^{-5}). \quad (7.19)$$

In particular, the coefficient of s^{-2} in $\mathcal{B}_\alpha$ is zero, and $\mathcal{C}_\alpha(s)/s$ reproduces (7.10) term by term.

[Merge note, batch 7301.] The intake recomputed (7.18), (7.19), (8.8) and $P_1, \dots, P_6$ from (7.13) and (8.10) in exact rational arithmetic, for three values of c ; all agree.

8 Eliminating the saddle and generating all coefficients

Since $dt/ds = -\alpha(s-1)/M$, differentiating the continuum expansion and using (7.8) gives the formal mean relation

$$n \sim \frac{M^2}{2} Q_\alpha(s), \quad Q_\alpha(s) = 1 + \frac{2(d_\alpha(s) + d'_\alpha(s))}{s-1}. \quad (8.1)$$

Here d'_α denotes formal differentiation. Thus

$$r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.2)$$

Because $Q_\alpha(s) = 1 + O(s^{-2})$, this has a unique formal inverse $s = r + O(r^{-2})$. It is computed recursively, for example by iterating $s \leftarrow r - \frac{1}{2} \log Q_\alpha(s)$ in the ring of formal series in r^{-1} . Every iteration determines additional coefficients.

Substitution in $F(t) + nt$ gives

$$\mathcal{H}_\alpha(r) = \frac{s(Q_\alpha(s) + 1)/2 - 1 + d_\alpha(s)}{\sqrt{Q_\alpha(s)}}, \quad r = s + \frac{1}{2} \log Q_\alpha(s). \quad (8.3)$$

By construction,

$$\mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c) r^{-j}. \quad (8.4)$$

Equations (7.3), (7.9), (8.1), and (8.3) are an all-orders algorithm involving only rational operations and expansions of logarithms and square roots. Since even zeta values are rational multiples of powers of π^2 , the coefficients are polynomials over $\mathbb{Q}$ in c .

Proof of Theorem 2.1. Fix the desired order R . Apply Lemma 7.1 and its mean version with more than enough terms that both remainders, after all operations, are $O_{\alpha,R}(r^{-R-1})$ in the normalized entropy. The exact mean relation has an additional error $O(M)$, negligible compared with $M^2 r^{-A}$ for every fixed A . Inverting (8.2) is stable because its derivative is $1 + O(s^{-3})$. Finite Taylor substitution is consequently legitimate with remainders, not only as a formal manipulation.

We obtain

$$F(t_n) + nt_n = \alpha K \left(r - 1 + \sum_{j=1}^R P_j(c) r^{-j} + O_{\alpha,R}(r^{-R-1}) \right).$$

Equation (5.6) contributes only $O(s)$, which is $o(Kr^{-R-1})$. This proves (2.2). Direct expansion gives (2.3); the supplied symbolic script reproduces those coefficients. $\square$

One further coefficient for an independent check. The same algorithm gives

$$P_8(c) = c + \frac{259}{2} c^2 + \frac{7271}{10} c^3 + \frac{157813}{840} c^4. \quad (8.5)$$

The continuum quadrature script checks the approach of the remainder after P_7 , multiplied by r^8 , to this polynomial. It does not replace the preceding proof or evaluate integer coefficients at those enormous sizes.

[Merge note, batch 75.] Source 75-06 prints $b_8(u) = u(157813u^3 + 3664584u^2 + 3916080u + 181440)/1088640$, which equals (8.5) at $u = 6c$, and its `code/75-06-powers-formal.py`

derives it by exact arithmetic over $\mathbb{Q}[u]$. So P_8 is now also confirmed exactly, independently of source 40's algorithm, not only by the continuum diagnostic.

[Batch 100, 5 October 2026.] Source 100-82's exact record continues the list with P_9 and P_{10} (Proposition 21.1), which a rerun of source 75-06's program at order 10 confirms; it also confirms P_8 a third time.

8.1 The same elimination in operator form (source 48)

At the exact saddle, Lemma 7.2 gives, to every finite inverse-logarithmic order,

$$K \sim M \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right)^{1/2}, \quad (8.6)$$

meaning the full finite-order expansion, not merely a leading ratio. Equivalently, its formal inverse is

$$r = s + \frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(s)}{s^2} \right), \quad (8.7)$$

which is (8.2). There is a unique solution with $s = r + O(r^{-2})$ as a formal series in r^{-1} . Indeed, put $z = r^{-1}$ and $\nu = s - r$. Then $s^{-1} = z/(1 + z\nu)$. The right-hand side of the fixed-point equation

$$\nu = -\frac{1}{2} \log \left(1 + \frac{2\mathcal{B}_\alpha(r + \nu)}{(r + \nu)^2} \right)$$

has order z^2 , and changing ν by $O(z^d)$ changes the right-hand side by $O(z^{d+3})$. Iteration starting with $\nu = 0$ therefore determines every coefficient after finitely many steps. The first terms are

$$s = r - \frac{c}{r^2} - \frac{c}{r^3} + \frac{c^2}{r^4} + \frac{2c - \frac{7}{5}c^2}{r^5} + O_\alpha(r^{-6}). \quad (8.8)$$

At the saddle, addition of (7.14) and nt gives

$$F(t) + nt \sim \alpha M \left(s - 1 + \frac{2\mathcal{C}_\alpha(s) + \mathcal{TC}_\alpha(s)}{s} \right). \quad (8.9)$$

The logarithm of the Gaussian prefactor in Theorem 2.2 is $O_\alpha(s)$, and the logarithm of its relative correction is $o(1)$; both are smaller than every scale K/r^R . Thus the formal series determining Theorem 2.1 is

$$\boxed{\mathcal{H}_\alpha(r) = \frac{s - 1 + (2\mathcal{C}_\alpha(s) + \mathcal{TC}_\alpha(s))/s}{\sqrt{1 + 2\mathcal{B}_\alpha(s)/s^2}}, \quad \mathcal{H}_\alpha(r) = r - 1 + \sum_{j \geq 1} P_j(c)r^{-j}}, \quad (8.10)$$

with s substituted from (8.7). This is (8.3): with $d_\alpha = \mathcal{C}_\alpha/s$ and $Q_\alpha = 1 + 2\mathcal{B}_\alpha/s^2$ the numerators agree, since $s(Q_\alpha + 1)/2 - 1 + d_\alpha = s - 1 + (\mathcal{B}_\alpha + \mathcal{C}_\alpha)/s$.

Why the formal rule proves actual asymptotics. Fix a desired final order R . Take sufficiently many terms, for example through order s^{-R-4} , in the two estimates of Lemma 7.2. Their remainders are genuine uniform estimates on large real s . The derivative of the finite approximation in (8.7) is $1 + O(s^{-3})$, bounded away from zero. The mean value theorem transports its error to an error of the same inverse-log order. Substitution into (8.9), whose normalized derivative is bounded at the relevant order, then gives the remainder in (2.2). This is a finite calculation with controlled remainders, not an assumption that the infinite

formal series converges. Together with Theorem 2.2, this is source 48's proof of Theorem 2.1; substitution of (7.18)–(8.8) into (8.10) gives $P_1, \dots, P_6$.

No special symbolic integration routine is needed. In the variable $z = s^{-1}$, the operator is

$$\mathcal{T} = -\frac{z^2}{1-z} \frac{d}{dz}, \quad f = \frac{1}{1-z}. \quad (8.11)$$

The factors $2\eta(2j+2)/\alpha^{2j+2}$ are rational multiples of c^{j+1} , since even zeta values are rational multiples of even powers of π . Truncated polynomial arithmetic with (8.11), followed by the fixed-point substitution, generates the coefficients at any prescribed finite order. The program `code/48-a022629-coefficients.py` implements this construction and verifies the six displayed polynomials and six inverse coefficients exactly.

8.2 An entropy factor beyond the largest summand (source 48)

Combining Theorem 2.1 with Proposition 3.5 gives an additional conclusion.

Corollary 8.1 (source 48).

$$\log \frac{a_\alpha(n)}{\Pi_\alpha(n)} \sim \frac{\pi^2}{6\alpha} \frac{K}{r}. \quad (8.12)$$

Thus the total weighted enumeration exceeds its largest individual weight by a factor that tends to infinity, and whose logarithm has a precise leading constant.

For $\alpha = 1$, this explains why using only the most heavily weighted partition gets the first two large terms right but misses a multiplicatively enormous factor. The transition contains approximately M/s available sizes, and its accumulated smoothing contribution produces this factor.

9 Lambert- W inversion with an error theorem

The coefficients are strictly increasing for $n \geq 1$. Indeed, increasing the largest part of a distinct partition by one is an injection into distinct partitions of $n+1$, and strictly increases each positive norm weight. Thus the threshold index $N_\alpha(Y)$ in Theorem 2.3 is well-defined for large Y .

The dominant inversion problem is

$$Y = \alpha K(\log K - 1).$$

Writing $w = \log K - 1$ gives $we^w = Y/(\alpha e)$, proving (2.10). In particular $\log K_0 = R_0$. Set $K = K_0 E$. The all-orders formal inversion is the single equation

$$E \mathcal{H}_\alpha(R_0 + \log E) = R_0 - 1, \quad E = 1 + O(R_0^{-2}). \quad (9.1)$$

The derivative with respect to E is $R_0 + O(1)$ at $E = 1$, so successive coefficients are uniquely determined. In terms of c , they are

$$\begin{aligned} E = 1 - \frac{c}{R_0^2} - \frac{c}{R_0^3} + \frac{3c^2/2 - c}{R_0^4} - \frac{c + 2c^2/5}{R_0^5} \\ - \frac{c + 46c^2/5 + 5c^3/2}{R_0^6} + \dots, \end{aligned} \quad (9.2)$$

and hence

$$E^2 = 1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{6c^2/5 - 2c}{R_0^5} - \frac{2c + 77c^2/5 + 8c^3}{R_0^6} + \dots \quad (9.3)$$

The asymptotic index is $K_0^2 E^2/2$.

To justify inversion of the sequence rather than just of a formal expression, fix an order, use Theorem 2.1 two orders farther than needed, and bracket the target by the corresponding smooth functions with the two signs of its error bound. Their derivatives with respect to K are $\alpha r(1 + o(1)) > 0$. An entropy error $O(Kr^{-R-1})$ therefore changes K by $O(Kr^{-R-2})$, and changes $n = K^2/2$ by $O(K^2 r^{-R-2})$. These bounds transfer the formal inversion to the actual threshold by monotonicity. Rounding the brackets to integers costs $O(1)$, smaller than all of these fixed-order error scales. This proves Theorem 2.3 and its all-orders extension.

Remark 9.1 (Two different Lambert branches). The saddle phase uses W_{-1} because $s \rightarrow \infty$ solves $se^{-s} = t/\alpha \rightarrow 0^+$. The threshold inverse uses W_0 at a positive argument. Interchanging the branches would destroy the relevant scale. If inversion in elementary logarithms is desired, the usual formal solution of $w + \log w = \log(Y/(\alpha e))$ can be substituted into (9.3); retaining W_0 is shorter and preserves more precision.

9.1 The same inversion in source 48's form (source 48)

Source 48 proves Theorem 2.3 independently (its Theorem 6.1), with the threshold written in the logarithm, $N_\alpha(Y) = \min\{n \geq 1 : \log a_\alpha(n) \geq Y\}$; for a threshold on the sequence itself one uses $N_\alpha(\log \cdot)$. The minimum is well defined because the coefficients diverge (Proposition 3.2) and increase strictly for $n \geq 1$ by the injection above; the initial equality $a_\alpha(0) = a_\alpha(1) = 1$ is immaterial. Its statement displays two more orders than (2.11):

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0^2} - \frac{2c}{R_0^3} + \frac{4c^2 - 2c}{R_0^4} + \frac{\frac{6}{5}c^2 - 2c}{R_0^5} - \frac{8c^3 + \frac{77}{5}c^2 + 2c}{R_0^6} + O_\alpha(R_0^{-7}) \right), \quad (9.4)$$

the coefficients of (9.3); integer rounding changes the result by at most an additional $O(1)$, absorbed in every fixed inverse-logarithmic remainder. With $E = 1 + \delta$ and $C(r) = \sum_{j \geq 1} P_j(c)r^{-j}$, its normalized equation for the corrected inverse,

$$\delta = - \frac{(1 + \delta) \log(1 + \delta) - \delta + (1 + \delta)C(R_0 + \log(1 + \delta))}{R_0}, \quad (9.5)$$

is (9.1) solved for δ . Starting from $\delta = 0$, finite formal iteration gives the unique series with $\delta = O(R_0^{-2})$, whose coefficients are those of $E - 1$ in (9.2).

Second proof of the threshold transfer (source 48). The core equation $K(\log K - 1) = Y/\alpha$ is solved exactly by $K = K_0$, using $we^w = Y/(\alpha e)$. This is the principal, positive Lambert branch, distinct from the lower branch used for the saddle. Choose any finite truncation of Theorem 2.1 and define its continuous approximation on large real K . Its derivative is $\alpha(\log K + O_\alpha(1/\log K))$, positive and comparable to $\alpha \log K$. An error $O_{\alpha,R}(Kr^{-R-1})$ in the logarithm therefore moves its inverse by $O_{\alpha,R}(Kr^{-R-2})$. This follows directly from the mean value theorem on a fixed relative neighborhood of the core root; the preliminary leading

asymptotic puts the true root in that neighborhood. For the discrete threshold, apply these upper and lower approximations at $N_\alpha(Y)$ and $N_\alpha(Y) - 1$. Adjacent values of $K = \sqrt{2n}$ differ by $O(1/K)$, so the final bracketing introduces only an $O(1)$ uncertainty in n . Taking the forward expansion to sufficiently high fixed order proves any desired inverse order. $\square$

The first inverse correction is negative: the actual enumeration is larger than its core approximation, so it reaches a specified threshold sooner. For A022629 ($\alpha = 1$, $2c = \pi^2/3$) this begins

$$N_1(Y) = \frac{1}{2} \left(\frac{Y}{W_0(Y/e)} \right)^2 \left(1 - \frac{\pi^2}{3(1 + W_0(Y/e))^2} + O((1 + W_0(Y/e))^{-3}) \right). \quad (9.6)$$

The distinction between inversion in $Y = \log a(n)$ and inversion in $a(n)$ is essential; replacing Y by the original threshold would give a wholly wrong scale (source 48).

[Merge note, batch 75.] Source 75-01 proves Theorem 2.3 through R_0^{-5} and source 75-06 through R_0^{-8} , for μ layers; the coefficients of R_0^{-7} and R_0^{-8} in (9.3) are Δ_7 and Δ_8 of Table 5. The core equation is an instance of the dominant-block theorem of the transseries-and-inversion volume [25], and Theorem 16.4 replaces the $O(1)$ rounding bracket by a $1 + o(1)$ localization from the exact saddle.

10 Conditioned geometry: a logistic boundary and a Gumbel maximum

Conditioning the Bernoulli law (4.1) on $S = n$ gives exactly

$$\mathbb{P}_n(\lambda) = \frac{N(\lambda)^\alpha}{a_\alpha(n)}. \quad (10.1)$$

No approximation is involved in this identification.

10.1 The boundary profile

Fix distinct integer sites $k_{1,n}, \dots, k_{d,n}$ with $(k_{j,n} - M)/H \rightarrow x_j \in \mathbb{R}$. Their unconditioned probabilities tend to $p(x_j) = (1 + e^{x_j})^{-1}$ by Lemma 4.1. For any prescribed pattern $\epsilon_j \in \{0, 1\}$, independence gives

$$\begin{aligned} \mathbb{P}_n(X_{k_{j,n}} = \epsilon_j \text{ for all } j) \\ = \prod_{j=1}^d p_{k_{j,n}}^{\epsilon_j} (1 - p_{k_{j,n}})^{1-\epsilon_j} \frac{\mathbb{P}_t\{S^{\text{rem}} = n - \sum_j k_{j,n} \epsilon_j\}}{\mathbb{P}_t(S = n)}. \end{aligned} \quad (10.2)$$

The shifted target differs from the remaining mean by $O(M) = o(\sqrt{B})$. The uniform local limit theorem in Section 5.3 makes the ratio of point probabilities tend to one. This proves both the logistic limit and finite-site asymptotic independence. Similarly, a site at γM has limiting occupation 1 for $0 < \gamma < 1$, and 0 for $\gamma > 1$.

10.2 The correct centering of the maximum

The large solution of (2.12) exists uniquely for sufficiently small t . With $z = tb$, the equation is $z^\alpha e^{-z} = t^{\alpha+1}$, so

$$b(t) = -\frac{\alpha}{t} W_{-1} \left(-\frac{t^{1+1/\alpha}}{\alpha} \right). \quad (10.3)$$

Solving its logarithm, $z - \alpha \log z = -(\alpha + 1) \log t$, with $t = \alpha s e^{-s}$ yields

$$\frac{b}{M} = 1 + \frac{1}{\alpha} + \frac{-\log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha}{\alpha s} + O_\alpha\left(\frac{\log s}{s^2}\right). \quad (10.4)$$

Keeping only $b \sim (1 + 1/\alpha)M$ is not adequate for the Gumbel centering: the omitted shift has size $M \log s/s$, larger than $1/t$. Keeping all the displayed terms in (10.4) does give an $o(1/t)$ centering error. The exact definition (2.12) or (10.3) avoids this truncation bookkeeping altogether.

Proof of the maximum assertion in Theorem 2.4. Fix $x \in \mathbb{R}$ and put $m = \lfloor b + x/t \rfloor$. Then $tm \rightarrow \infty$ and $m/M \rightarrow 1 + 1/\alpha > 1$. In the tail $k > m$, all odds tend uniformly to zero. A sum–integral comparison followed by integration by parts gives

$$\sum_{k>m} p_k \sim \sum_{k>m} k^\alpha e^{-tk} \sim \frac{m^\alpha e^{-tm}}{t} \longrightarrow e^{-x}. \quad (10.5)$$

For the second equivalence, substitute $v = tx$ in the integral and use $\int_z^\infty v^\alpha e^{-v} dv = z^\alpha e^{-z}(1 + O_\alpha(z^{-1}))$. The endpoint discretization has relative size $O(t)$. The last limit follows from $b^\alpha e^{-tb}/t = 1$ and $m/b \rightarrow 1$. Also, $\sup_{k>m} p_k = O(t)$, so $\sum_{k>m} p_k^2 = o(1)$. Therefore the unconditioned probability of no part above m is

$$\prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}. \quad (10.6)$$

It remains to justify conditioning, rather than silently replacing one ensemble by another. The removed mean and variance satisfy

$$\sum_{k>m} k p_k = O(M), \quad \sum_{k>m} k^2 p_k (1 - p_k) = O(M^2).$$

These bounds follow from the same tail-integral calculation with one or two extra powers of k . The remaining sum $S_{\leq m}$ thus has mean $n - O(M)$ and variance $B(1 + o(1))$. Its central block around M is intact. The uniform local limit estimate gives

$$\frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow 1.$$

Independence of the retained and removed sites now gives

$$\mathbb{P}_n(L_n \leq m) = \prod_{k>m} (1 - p_k) \frac{\mathbb{P}_t(S_{\leq m} = n)}{\mathbb{P}_t(S = n)} \longrightarrow e^{-e^{-x}}.$$

This proves the Gumbel limit. Its tightness, (10.4), $M \sim K$, and $(tM)^{-1} = 1/(\alpha s) \rightarrow 0$ imply $L_n/K \rightarrow 1 + 1/\alpha$ in probability. $\square$

The two locations are genuinely different. The occupation boundary lies near $M \sim \sqrt{2n}$, while the maximum lies near $(1 + 1/\alpha)M$. For $\alpha = 1$, a typical largest part is asymptotic to $2\sqrt{2n}$, not to $\sqrt{2n}$. Individual occupancies far above the boundary are small, but there are enough possible sites to move the extreme by a macroscopic multiple of M . This is why a boundary profile alone does not determine the largest part.

10.3 Explicit centring in $\log \sqrt{2n}$ (source 48)

Source 48 proves the Gumbel law independently, with the centring written in $K = \sqrt{2n}$ and $r = \log K$ instead of the exact root $b(t_n)$. For positive integer α the measure $\mathbb{P}_n$ is the size distribution obtained by choosing uniformly among the coloured objects; for every $t > 0$ it is exactly the conditional distribution of $\{k : X_k = 1\}$ given $S = n$ (source 48).

Corollary 10.1 (Gumbel law with explicit centring; sources 48 and 40). *Define*

$$\beta_\alpha(r) = (\alpha + 1)r - \log r + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha. \quad (10.7)$$

For every fixed real x ,

$$\mathbb{P}_n \left(\frac{\alpha r}{K} L_n - \beta_\alpha(r) \leq x \right) \longrightarrow \exp(-e^{-x}). \quad (10.8)$$

For A022629 ($\alpha = 1$), an equivalent form is

$$\frac{L_n - \left(2K - \frac{K}{r} \log(r/2)\right)}{K/r} \implies G, \quad \mathbb{P}(G \leq x) = e^{-e^{-x}}. \quad (10.9)$$

Given Theorem 2.4, the corollary amounts to $t_n b(t_n) - \beta_\alpha(r) \rightarrow 0$: by (10.4), $t_n b = \alpha s b/M$ agrees with $(\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha$ up to $O_\alpha(\log s/s)$, and $s = r + O(r^{-2})$, $t_n = (\alpha r/K)(1 + O(r^{-2}))$ by (8.6) and (8.8). (The intake compared the two centrings numerically for $\alpha = 1, 2, 5$; the difference decays like $\log s/s$.) Source 48 proves it directly.

Proof (source 48). Work first with the unconditioned Bernoulli variables at the exact saddle, with s, M, t as in (2.7). Set

$$d = (\alpha + 1)s - \log s + \alpha \log(\alpha + 1) - (\alpha + 1) \log \alpha, \quad m = \left\lfloor \frac{d + x}{t} \right\rfloor.$$

Then $tm \rightarrow \infty$ and $m/M \rightarrow (\alpha + 1)/\alpha > 1$. Summation versus integration for a decreasing tail, followed by one integration by parts (or its standard real-power analogue), gives

$$\sum_{k>m} k^\alpha e^{-tk} \sim t^{-\alpha-1} (tm)^\alpha e^{-tm}. \quad (10.10)$$

For real $\alpha > 0$, the relative error from the integral asymptotic is $O_\alpha((tm)^{-1})$; the relative discretization error is $O_\alpha(t)$. Now $\log(1/t) = s - \log s - \log \alpha$ and $\log(d + x) = \log s + \log(\alpha + 1) + o(1)$. Hence

$$(\alpha + 1) \log(1/t) + \alpha \log(tm) - tm = -x + o(1),$$

and the tail in (10.10) tends to e^{-x} . The maximum individual odds on this tail are $O_\alpha(t)$ and tend to zero, so replacing odds by p_k changes the sum by $o(1)$, and $\sum_{k>m} p_k^2 = o(1)$. Independence gives

$$\mathbb{P}_t(X_k = 0 \text{ for every } k > m) = \prod_{k>m} (1 - p_k) \longrightarrow e^{-e^{-x}}.$$

It remains to justify conditioning on total size, which cannot be omitted. Let $T = \sum_{k>m} kX_k$ and $S^{\leq m} = S - T$. The same tail calculation with extra factors of k gives $\mathbb{E}_t T = O_\alpha(M)$ and $\text{Var}_t(T) = O_\alpha(M^2)$. By Lemma 4.1, $\mathbb{E}_t T/\sqrt{B} \rightarrow 0$ and $\text{Var}_t(T)/B \rightarrow 0$. The entire active interval near M lies below m for large n . Thus the shifted local limit estimate (5.8) applies

to the truncated sum, and $\mathbb{P}_t(S^{\leq m} = n)/\mathbb{P}_t(S = n) \rightarrow 1$. Independence of the tail and the truncated sum now gives

$$\mathbb{P}_n(L_n \leq m) = \mathbb{P}_t(T = 0) \frac{\mathbb{P}_t(S^{\leq m} = n)}{\mathbb{P}_t(S = n)} \rightarrow e^{-e^{-x}}.$$

This proves the result in the exact saddle coordinates. Finally, (8.6) and (8.8) imply $M/K = 1 + O_\alpha(r^{-2})$, $s = r + O_\alpha(r^{-2})$, and $t = (\alpha r/K)(1 + O_\alpha(r^{-2}))$. The change in the normalized extreme-value coordinate is $o(1)$, since L_n/M is tight by the limit just proved. Replacing s, M, t by $r, K, \alpha r/K$ therefore proves (10.8); tightness of the Gumbel coordinate also gives $L_n/K \rightarrow 1 + 1/\alpha$. For $\alpha = 1$, $\beta_1(r) = 2r - \log(r/2)$, which is (10.9). $\square$

The upper transition is near $M \sim K$, but the largest part is near $(1 + 1/\alpha)K$. There is no contradiction: a long tail of individually rare occupancies produces an extreme well beyond the transition (source 48).

Source 48 also proves the logistic part of Theorem 2.4 (its Proposition 7.2): for fixed distinct reals $u_1, \dots, u_d$ and $k_j = \lfloor M + u_j/t \rfloor$ at the exact saddle, the inclusion indicators converge jointly to independent Bernoulli variables with probabilities $(1 + e^{u_j})^{-1}$. Since $tH = s/(s-1) \rightarrow 1$, its coordinate $u = t(k - M)$ and source 40's $x = (k - M)/H$ have the same limit, and its proof is the argument around (10.2): the log odds at k_j are $-u_j + o(1)$, and fixing finitely many indicators moves the remaining target by $O_\alpha(M)$ and removes variance $O_\alpha(M^2)$, both negligible.

[Merge note, batch 75.] Source 75-06 proves both parts of Theorem 2.4 independently, for any number μ of layers (Theorem 18.4); at $\mu = 1$ its centring equation is (2.12).

11 A proved exponential scale from Fourier resonances

The bound (5.4) suggests an exponentially small scale not visible in either the logarithmic block or the central Edgeworth series. We can identify its constant at each fixed integer resonance.

Theorem 11.1 (Fixed-resonance modulus). *For every fixed nonzero integer ℓ , with ϕ_t as in Section 5,*

$$-\log \left| \phi_t \left(\frac{2\pi\ell}{M} \right) \right| \sim \frac{2\pi^4\ell^2}{3} \frac{H^3}{M^2} \sim \frac{2\pi^4\ell^2}{3\alpha^3} \frac{M}{s^3}. \quad (11.1)$$

Proof. The exact modulus formula is

$$\log |\phi_t(\theta)| = \frac{1}{2} \sum_{k \geq 1} \log \left(1 - 4p_k(1 - p_k) \sin^2(k\theta/2) \right).$$

At $\theta = 2\pi\ell/M$, write $k = M + xH$. For bounded x ,

$$\sin^2(\pi\ell k/M) = \pi^2\ell^2 x^2 H^2/M^2 + O_\ell(x^4 H^4/M^4).$$

The logarithm is therefore $-4p_k(1 - p_k)$ times this sine square to leading order. Lemma 4.1, applied with moments 2 and 4, gives

$$-\log |\phi_t(2\pi\ell/M)| \sim \frac{2\pi^2\ell^2 H^3}{M^2} \int_{\mathbb{R}} x^2 \frac{e^x}{(1 + e^x)^2} dx.$$

One may first restrict to $|x| \leq s^\delta$ with small fixed $\delta > 0$; Taylor remainders are then uniformly controlled. Outside this interval, the exponential boundary tails and the lower-endpoint bounds in Lemma 4.1 are negligible at the displayed scale. The quadratic logarithm error is $O(H^5/M^4) = o(H^3/M^2)$. Finally,

$$\int_{\mathbb{R}} x^2 \frac{e^x}{(1+e^x)^2} dx = 4 \sum_{j \geq 1} \frac{(-1)^{j-1}}{j^2} = \frac{\pi^2}{3},$$

by expansion on the positive half-line. This proves the theorem. $\square$

For fixed α , the scale $\exp[-C_\alpha M/s^3]$ is smaller than every power of s/M , because M/s^3 dominates $\log M$. However, its constant deteriorates as α^{-3} . This makes the fixed- α asymptotic theorem compatible with large visible resonances at moderate n .

The theorem concerns the *modulus at a specified point*. It does not yet determine the shifted secondary saddles, their phases, or the contribution of their full neighborhoods to the coefficient integral. Calling it a complete multi-saddle transseries would be incorrect. It does identify a rigorous exponential scale and a concrete next target for exponential improvement.

[Batch 100, 5 October 2026.] In the critical window of growing powers, $\alpha \asymp M^{1/3}/\log M$, the action H^3/M^2 of Theorem 11.1 stays bounded, and source 100-93 determines all three: the shifted centres $2\pi j B_{RS}/B$, the phases $-2\pi j \mu_R$ and the full-cell contributions, with relative error $o(1)$ (Theorem 22.1 and Section 22.7). For fixed α they remain undetermined.

12 Reproducible computation and finite-size diagnostics

12.1 Exact calculations

The supplied `code/40-norm-moments-verify.py` computes the five integer sequences through $n = 5000$ by descending polynomial multiplication. It compares their initial sixteen terms against the consulted OEIS entries and independently checks $n \leq 150$ by a logarithmic-derivative recurrence. Explicitly, define

$$C_\alpha(m) = \sum_{d|m} (-1)^{m/d+1} d^{1+\alpha m/d}. \quad (12.1)$$

Then $a_\alpha(0) = 1$ and

$$na_\alpha(n) = \sum_{m=1}^n C_\alpha(m) a_\alpha(n-m). \quad (12.2)$$

This follows by expanding $qA'_\alpha(q)/A_\alpha(q)$ formally. The two algorithms have different arithmetic paths, so their agreement is a useful implementation check. It is not a proof of the asymptotic theorems.

All five exact prefix checks, all recurrence checks through 150, and strict monotonicity checks through 5000 passed. The complete computed lists are included as plain text. They were independently generated here, not copied from OEIS b-files.

12.2 The central approximation is excellent for one column, not all

Let $G = e^{F(t_n)+nt_n}/\sqrt{2\pi B}$, with the infinite sums evaluated by high-precision truncation. The following are numerical diagnostics, not interval-certified bounds. Values shown are rounded from the included CSV.

α	n	$a_\alpha(n)/G$	$a_\alpha(n)/(G(1 + \mathcal{E}_1))$	$a_\alpha(n)/(G(1 + \mathcal{E}_1 + \mathcal{E}_2))$
1	100	0.989159702	0.999826713	0.999908591
1	1000	0.997526322	0.999995495	0.999999984
1	5000	0.999084078	0.999999367	0.999999999
2	5000	0.999545155	1.000036354	1.000036527
3	5000	0.961625508	0.961944240	0.961944315
4	5000	1.230518103	1.230825332	1.230825386
5	5000	1.581097698	1.581414178	1.581414223

At $\alpha = 1, n = 5000$, two central corrections reduce the relative error to about 1.1×10^{-9} . At $\alpha = 5, n = 5000$, the central approximation is plainly not accurate, and adding central corrections does not repair it. This is evidence about finite-size behavior, not a contradiction of a fixed-parameter limit. In particular, these tests must not be advertised as numerical confirmation of rapid convergence for all five columns.

[Merge note, batch 75.] Source 75-01's tests at $\alpha = 2, 3, n = 2500$ show the same effect, with residuals that do not change from $J = 1$ to $J = 3$: a noncentral contribution, not slow Edgeworth convergence (Section 16.3).

The resonance diagnostic computes the exact product modulus at $\theta = 2\pi/M$, using the same numerical saddle:

α	n	$ \phi_t(2\pi/M) $
1	5000	2.33482×10^{-15}
2	5000	2.72621×10^{-5}
3	5000	0.0173352
4	5000	0.150645
5	5000	0.361805

The increase in these noncentral moduli is consistent with the observed failure of a central-only approximation. Their values do not by themselves predict the sign of the coefficient error; the omitted phases matter.

[Batch 100, 5 October 2026.] A numerical decomposition of these residuals into Fourier cells is recorded in Section 16.3; it is evidence, not a formula.

12.3 Logarithmic expansions are not monotone numerical improvements

For $\alpha = 1, n = 5000$, the exact logarithm is approximately 401.567349464797. Subtracting the leading expression $K(r - 1)$ leaves 41.05033086599. Subtracting the expansion through P_3 instead leaves -2.72431695876 , while truncating through P_7 leaves -7.67561845406 . The additional terms are not a promise of monotone improvement at a fixed small r . The missing prefactor and endpoint contributions are also on scales suppressed by the logarithmic theorem but relevant numerically.

An independent continuum check evaluates the integral and mean in Lemma 7.1, without using the rational-function expansion, and compares the normalized remainder after P_7 with P_8 :

α	s	Scaled remainder $r^8(\text{entropy}/(\alpha K) - \mathcal{H}_{\alpha,7}(r))$
1	24	7881.08213
1	48	5997.09948
1	96	5424.47363
Predicted limit $P_8(\pi^2/6)$:		4963.76905
3	24	15.7012553
3	48	9.85270737
3	96	9.49113831
Predicted limit $P_8(\pi^2/54)$:		9.15764483

The approach is slow, but has the expected direction in these samples. These are continuum diagnostics at nonintegral implied n , not exact integer-sequence computations at exponentially large indices.

12.4 What the numerical implementation guarantees

The integer arithmetic and recurrence comparisons are exact. For the saddle calculations, the program uses arbitrary-precision arithmetic and chooses a cutoff from a geometric majorant for $\sum k^{\alpha+6}e^{-tk}$. This bounds the tails relevant to all cumulants through order six, apart from harmless fixed constants. The test used 50 requested decimal digits with guard digits; the saved mean residuals are far smaller than the displayed precision. Nevertheless, the majorant and Newton iteration are evaluated in floating point, not outward-rounded interval arithmetic. Accordingly, the CSV values are diagnostics rather than certificates of rigorous numerical error bounds.

12.5 Source 48's computations for A022629 (source 48)

Two exact coefficient constructions. For an integer exponent $\alpha \geq 1$, multiply the product factors successively. Starting with $a_0 = 1$ and $a_n = 0$ for $n > 0$, process each k by updating in *decreasing* order of n : $a_n \leftarrow a_n + k^\alpha a_{n-k}$ ($n \geq k$). Descending order enforces at most one use of size k ; increasing order would compute a different partition problem. The independent check is the logarithmic-derivative recurrence (12.2). That formal identity is valid regardless of the analytic restriction in Proposition 3.4: only finitely many terms contribute to a fixed degree. The program `code/48-a022629-verify.py` compares the two constructions through degree 100, checks the initial OEIS values and strict monotonicity, and computes the whole exact table through $n = 6400$ for $\alpha = 1$ (`data/48-a022629-exact-coefficients.csv`).

[**Merge note, batch 7301.**] On the common range $n \leq 5000$ this table equals source 40's `data/40-norm-moments-A022629_computed.txt` entry for entry; the intake also recomputed both from the product through $n = 6400$.

Saddle diagnostics. The exact saddle is found by Newton iteration, $t_{\text{new}} = t + (-F'(t) - n)/B(t)$, with a positivity safeguard. The first four cumulants use $b_2 = p(1-p)$, $b_3 = p(1-p)(1-2p)$ and $b_4 = p(1-p)\{1-6p(1-p)\}$. For $\beta \geq 0$ the explicit tail bound

$$\sum_{k>k_0} k^\beta e^{-tk} \leq e^t t^{-\beta-1} \Gamma(\beta+1, t(k_0+1)) \quad (12.3)$$

follows by integrating $(u+1)^\beta e^{-tu}$ over $u \geq k_0$. It controls truncation of the log product, mean, variance, and fixed-order cumulants after multiplication by a suitable polynomial constant.

This addresses the infinite sums, but the implementation does not turn floating-point rounding into certified interval arithmetic.

n	$\log a(n)$	Gaussian relative error	Corrected relative error
100	31.57034412	$1.096 \cdot 10^{-2}$	$1.733 \cdot 10^{-4}$
400	80.97262500	$4.427 \cdot 10^{-3}$	$1.377 \cdot 10^{-5}$
1600	197.60929471	$1.849 \cdot 10^{-3}$	$2.527 \cdot 10^{-6}$
6400	467.20404426	$7.888 \cdot 10^{-4}$	$4.704 \cdot 10^{-7}$

Table 1: Computed diagnostics for A022629 ($\alpha = 1$), source 48. Relative error means approximation divided by the exact integer coefficient, minus one. The corrected approximation includes the two cumulant terms of (2.9). Calculations used 45 decimal digits.

At $n = 6400$, the bare logarithmic core underestimates $\log a(n)$ by about 45.36. Adding the first four inverse-logarithmic terms instead overestimates it by about 3.55. Thus that four-term logarithmic approximation is still poor as a *relative coefficient* approximation at this size, despite being a valid asymptotic expansion; the corrected exact-saddle formula has relative error approximately $4.7 \cdot 10^{-7}$. This is why Theorems 2.1 and 2.2 are stated separately. The recorded file `data/48-a022629-numerical_checks.json` gives the saddle, variance, cumulant correction, mean residual, cutoff, and tail estimate as well as the errors in Table 1. Its decimal values are computational evidence only; the exact integers and symbolic polynomial checks are exact computations. No Lean executable was run for either source.

[Merge note, batch 73O1.] The $n = 100$ row agrees with source 40’s $\alpha = 1$ row in Section 12 ($1/0.989159702 - 1 \approx 1.096 \cdot 10^{-2}$, $1/0.999826713 - 1 \approx 1.733 \cdot 10^{-4}$), and source 48’s saddle at $n = 100$ agrees with `data/40-norm-moments-run_alpha1.txt` to all printed digits.

13 Further research questions and proposed projects

The proved results leave several substantial and concrete directions.

1. An effective multi-saddle expansion. Locate the secondary saddles near $2\pi\ell/M$, compute their phases and Gaussian factors, and sum the relevant arcs with a uniform remainder. Theorem 11.1 supplies the first exponential action but not the full contribution. The moderate- n failures for $\alpha = 3, 4, 5$ make this a practical as well as theoretical priority.

[Batch 100, 5 October 2026.] Answered at leading order in the critical window of growing powers only (Section 22, source 100-93): there the satellites near $2\pi j/M$ are located, their phases and Gaussian factors computed, and all cells summed, with relative error $o(1)$ but no effective remainder. At fixed α the project stays open; Section 16.3 records an uncertified cell-by-cell decomposition of the residuals at $\alpha = 2, \dots, 5$.

2. A complete two-scale coefficient transseries. Refine the continuum-to-lattice passage sufficiently to retain endpoint terms, Euler–Maclaurin corrections, the central determinant, and the resonance sectors in one consistent normalization. Determine which of these expansions are Gevrey and whether optimal truncation produces a useful exponential improvement. The present all-logarithmic series alone does not settle this.

[Batch 100, 5 October 2026.] Source 100-89 supplies the endpoint and Euler–Maclaurin part in one normalization: the lattice-minus-integral difference is smooth with explicit coefficients

(Theorem 21.2), and the relative expansion keeps the exact integral and finitely many of them (Theorem 21.4). The resonance sectors, the Gevrey question and optimal truncation remain open (question 100.1).

3. The crossover as the tilt vanishes. The fixed- α expansion is singular as $\alpha \downarrow 0$, while the unweighted distinct-partition count has a different growth law. A natural crossover parameter is $\alpha \log K$. Derive a uniform family of saddle profiles when this parameter stays bounded, and match it to both the unweighted regime and the fixed-positive-tilt regime. The shrinking-tilt analysis in Gikunda's thesis provides an important comparison point [10], but matching the full regimes still requires work.

4. Growing moments and concentration near maximal norm. Allow $\alpha = \alpha(n) \rightarrow \infty$. The fluctuation width H and resonance action show that several thresholds can occur before the model collapses onto maximal-norm partitions. Determine the sharp parameter ranges for a local limit theorem, for single-saddle dominance, and for localization near an extremal partition. The fixed- α proof does not justify substituting an arbitrary growing parameter.

[Batch 100, 5 October 2026.] Source 100-93 finds one sharp range: for $\alpha \asymp M^{1/3}/\log M$ ($\Lambda = \alpha(s-1)/M^{1/3}$ in a compact subset of $(0, \infty)$) a local theorem holds with a theta multiplier, and single-saddle dominance fails (Theorem 22.1, Corollary 22.7). Whether single-saddle dominance holds for every $\alpha = o(M^{1/3}/\log M)$, and the localization near maximal-norm partitions as $\Lambda \rightarrow \infty$, stay open (questions 100-93.4 and 100-93.5).

5. The joint law of the largest several parts. The independent tail calculation suggests a Poisson process with intensity $e^{-x} dx$ for the rescaled upper parts. Prove its conditioned point-process limit and obtain joint distributions of the largest j parts, including second-order corrections to their centering. The maximum theorem is the one-dimensional starting point, not a substitute for tightness of the process.

[Batch 100, 5 October 2026.] Question 3 of Part II of a271619-strict-twice-partitions (Section 27 there) asks the same for the occupied lengths of A358836, $\prod_k (1 + q^k/(q; q)_k)$, where the one-threshold Gumbel law of the largest occupied length and its independence from the Gaussian block count are proved (its Theorem 24.1); the point process is open there too.

6. Fluctuations of the number of parts and of the log norm. A multivariate conditioned local limit theorem should describe the number of occupied sites and the logarithm of the norm under fixed positive tilt. Conditioning removes a covariance direction and can cancel a leading variance term; computing the surviving scale requires more than plugging the unconditioned variance into a Gaussian formula.

[Batch 100, 5 October 2026.] In the critical window of Section 22 the variance of the number of parts at least two, conditioned (by regression) on the size, is $Q_{\perp} \sim \pi^2 H^3/(3M^2)$, while its unconditioned variance is $\sim H$: the cancellation anticipated here, exact in (22.11) and proved for growing α only. The fixed- α question is open.

7. More general weights. Replace k^{α} by $k^{\alpha}(\log k)^{\beta}$ away from the harmless small indices, or by a smooth regularly varying weight. Identify hypotheses that preserve a single large crossing, the logistic boundary, and a computable inverse expansion. Arithmetic thinning of the allowed part sizes would also require a replacement for the consecutive-block lemma and may introduce persistent lattice obstructions.

[Batch 77, 2 October 2026.] A different generalization, integer slot multiplicities k^β with the weight k^α kept, is proved by source 77-51 (Theorem 20.2); the weight perturbations asked for here remain open.

[Batch 100, 5 October 2026.] The weight $p(k)$ of A271619, $\prod_k(1 + p(k)q^k)$, treated in Part I of **a271619-strict-twice-partitions** (Report 182 of the session bundle; labels **stp:sum:**), is not regularly varying ($\log p(k) \sim \pi\sqrt{2k/3}$). There the single large crossing is replaced by two competing effective charges that exchange dominance near the triangular phase 9/16 (its Theorems 4.1 and 9.1), and a single central minor-arc bound provably fails (its Section 6). This illustrates the failure of the single-crossing picture for stretched-exponential weights; it does not answer this question, whose regularly varying weights remain open.

8. Certified numerics and formal verification. Give explicit constants in the Fourier bounds, add interval-enclosed saddle solving, and certify a chosen relative-error target at a stated n and α . For Lean, a sensible modular route is: the formal Euler product and recurrence; the Bernoulli coefficient identity; the consecutive-block inequality; an abstract triangular-array local-limit theorem; and finally the boundary smoothing lemma. The algebraic generation of P_j and the inverse coefficients can be separately certified by finite polynomial identities. This would connect directly to the repository’s transseries infrastructure without pretending that formal algebra alone proves analytic asymptotics.

13.1 Source 48’s proposed projects (source 48)

Source 48 lists eight next steps, stated for A022629 and the real family. They are proposed research tasks, not claims that the corresponding problems are all globally open. Its numbers are kept; overlaps with the projects above are marked.

48.1 Resolve all subdominant asymptotic blocks (overlaps project 2). The upper-transition calculation determines every K/r^j term. Determine the smaller contributions from the lower endpoint and the sum–integral discrepancy, including any powers of K , logarithms, and constants that appear for different α . Prove a fully explicit multiplicative equivalent with no implicit exact saddle sums. Distinguish this task from merely computing more terms in (2.2).

[Batch 100, 5 October 2026.] Source 100-89 determines the lower-endpoint and sum–integral part: $\Xi_\alpha(t) = \sum_{r \leq D} \xi_r(\alpha)t^r + O(t^{D+1})$ with explicit ξ_r (Section 21.2), and Corollary 21.6 is a multiplicative equivalent with no lattice sum, though it keeps the exact integral and its saddle. A fully explicit equivalent without saddle quantities remains open.

48.2 The transition as $\alpha \downarrow 0$ (overlaps project 3). At $\alpha = 0$ the model is ordinary distinct-part enumeration; for every fixed $\alpha > 0$ the leading logarithm has the different scale $\alpha K \log K$. Develop uniform asymptotics in a double limit, for example when $\alpha \log n$ stays bounded. The present proof explicitly does not cover this regime, and the change in the active interval makes it a useful test case for uniform asymptotic methods.

48.3 The number of parts after conditioning (overlaps project 6). The finite-dimensional occupation profile is simple, but conditioning on total size creates long-range correlations. Determine the mean, variance, and limiting law of the total number of parts to several orders. A naive unconditioned variance calculation is not sufficient because number and size are strongly correlated in the narrow transition region.

48.4 Joint extremes and point processes (overlaps project 5). Upgrade Corollary 10.1 to joint laws of the largest several parts and convergence of the rescaled tail point process. Prove quantitative error bounds, uniform tail estimates, and asymptotics of the expectation and variance of the largest part. Distributional convergence alone does not justify taking expectations.

48.5 Stable weight perturbations (overlaps project 7). Replace k^α by $k^\alpha(\log k)^\beta$ or by $k^\alpha \exp(c_1/k + c_2/k^2 + \dots)$, with a positive convention for small k . Determine which coefficients and extreme-value centring terms are universal and which remember the perturbation. Establish explicit uniform hypotheses rather than assuming regular variation is automatically enough.

48.6 Optimal saddle remainder scales. The absolute cumulant bound gives the safe parameter s/M . Signed cumulant cancellations can make actual errors much smaller, as Table 1 suggests. Determine the sharp first relative correction, its complete expansion, and the optimal error after each correction. This would improve both numerical evaluation and rigorous inverse brackets.

[Batch 75, 2 October 2026.] Partly answered by Section 16 (sources 75-01 and 75-06): the sharp signed scales $\kappa_j \sim \frac{j!}{2} M^{j+1} / (\alpha s)^{j-1}$, the sharp first correction $\mathcal{E}_1 \sim -3/(8\alpha M s)$, and the error $O((\alpha M s)^{-j-1})$ after each correction, with the threshold localization of Theorem 16.4. A complete expansion of $\mathcal{E}_1$ in $1/s$, and optimality beyond these orders, remain open.

48.7 Effective inversion and formalization (overlaps project 8). Turn the asymptotic O constants into explicit bounds, combine them with finite exact verification, and produce a certified algorithm for $N_\alpha(Y)$. A suitable ProveIt formalization can separate finite coefficient algebra, the positive product measure, the block minor-arc estimate, remainder transport, and integer-threshold bracketing. None of those formalization tasks should be replaced by unchecked numerical fits.

[Batch 75, 2 October 2026.] Theorem 16.4 (source 75-01) localizes $N_\alpha(Y)$ to within $1 + o(1)$ of an exact-saddle inverse, but with implicit constants; the explicit bounds asked for here are still open.

48.8 OEIS and literature follow-through. Submit a concise, independently reviewed proof of (1.3) and the first sharper correction to the OEIS editors. Continue the literature audit for products with coefficients *inside* their factors, rather than mistaking $\prod(1 + k^\alpha q^k)$ for $\prod(1 + q^k)^{k^\alpha}$. Compare any earlier theorem at the level of hypotheses and error terms before making a priority claim.

[Batch 77, 2 October 2026.] The intermediate products $\prod(1 + k^\alpha q^k)^{k^\beta}$ are treated in Section 20.3 (source 77-51), which also cites Naranjo and Ramírez [27]; the broader literature audit asked for here is still open.

14 Conclusion

The five checked OEIS conjectures share a single analytic mechanism. Filling parts up to a moving boundary produces the dominant factorial-scale entropy; smoothing that boundary produces the π^2 correction and its all-orders inverse-logarithmic successors. A conditioned Bernoulli model supplies the local probability that converts this entropy into an actual

coefficient, and also gives logistic occupation and a Gumbel extreme. Lambert- W inversion turns the same expansion into a quantitative threshold law.

The resulting package contains explicit proofs rather than numerical extrapolation. It also records an important limitation: central asymptotics can converge extremely slowly for higher moments because noncentral resonances are only suppressed on the scale $\exp[-C_\alpha K/(\log K)^3]$. That proved scale and the visible finite-size discrepancies identify a well-defined next research problem: a phase-sensitive, exponentially improved multi-saddle expansion.

Source 48 reaches the same conclusions for A022629 by partly different routes: an elementary proof with a stronger additive remainder, an explicit differential operator that generates the entire dominant logarithmic block by finite exact algebra, a Lambert-based inverse for the integer threshold, a precise comparison with the largest individual partition weight, and a Gumbel law centred explicitly in $\log \sqrt{2n}$. The remaining distinctions—formal versus convergent series, logarithmic versus relative accuracy, and proved mathematics versus formal-machine verification—are essential parts of the result rather than editorial qualifications (source 48).

[Merge note, batch 75.] Sources 75-01 and 75-06 reach the same conclusions a third and a fourth time and add what follows: the sharp saddle scale and threshold (Section 16), the eventual shape of the sequence (Section 17), and the convolution powers (Section 18).

[Merge note, batch 77.] Source 77-51 reaches them a fifth time and adds the slot multiplicities k^β and A266891 (Section 20).

[Merge note, batch 100.] Sources 100-82 and 100-89 reach them a sixth and a seventh time and identify the lattice-minus-integral sector (Section 21); source 100-93 shows that once α grows like $M^{1/3}/\log M$ the resonances of Section 11 become part of the leading order (Section 22).

15 Addendum (batch 75): two further independent sources

Batch 75 of ProveIt's incoming-reports intake delivered two more manuscripts on the family $\prod_{k \geq 1} (1 + k^\alpha q^k)$, both dated October 1, 2026 and both pinned to the ProveIt commit 29aca108e. Neither cites sources 40 or 48, and neither cites the other.

- **Source 75-01** (batch-75 manuscript 01), *Norm-weighted distinct partitions: A proof of the A022629 growth conjecture, all-order asymptotics, Lambert inversion, and eventual Jensen hyperbolicity*: the family for real power $s > 0$ (here α), five printed logarithmic coefficients, an exact-saddle expansion with a sharp remainder, two inversions, eventual strict log-concavity, fixed-degree Jensen hyperbolicity, and exact data to $n = 10000$.
- **Source 75-06** (batch-75 manuscript 06), *Norm-weighted distinct partitions: A022629, all-order expansions, and inversion*: the two-parameter family $\prod_{k \geq 1} (1 + k^r q^k)^s$, real power $r > 0$ (here α) and integer number of layers $s \geq 1$ (here μ), eight printed logarithmic and seven printed inverse coefficients, the relative saddle expansion, and logistic and Gumbel laws, all for every number of layers.

These are the third and fourth independent proofs in this report of Proposition 3.2 and Theorems 2.1, 2.2, 2.3 (and, for source 75-06, 2.4), with identical coefficients. Following Section 1.4, none of these results is printed again. Table 2 records where each source proves them; Section 15.2 keeps the parts of their proofs and remarks that differ from the routes already printed. The genuinely new material follows in Sections 16 (sharp saddle scales and the near-integer threshold), 17 (log-concavity and Jensen polynomials) and 18 (convolution

powers), and the two sources' research questions in Section 19. Their notation is translated by the second table of Section 1.5. Source 75-01 is not a version of source 48, although their archive names differ only in letter case (A022629_research_package.zip and A022629_Research_Package.zip): their texts share 0.87 % of their word 8-grams in each direction, and their theorem sets are different.

15.1 Same results, further independent derivations

Result in this report	Source 75-01	Source 75-06	Relation
Proposition 3.2 (two-term growth; the five OEIS conjectures)	Prop. 2.3 with Lemma 2.2	Thm. 2.1 (μ layers)	same lower-bound partition $1, \dots, k-1, k+d$ as sources 40 and 48; third and fourth routes
Lemma 3.1 Strict monotonicity (Section 9) Recurrence (12.2)	Lemma 2.2 Prop. 2.1 (8), $b_s(m) = C_\alpha(m)$ used in Section 4.2	proof of Thm. 2.1 Section 7.1 (10.1), factor μ	same estimate same injection; 75-06 adds $\mu \geq 2$ (Section 18) identical
Lemma 4.1, (4.5)	Section 4.1	Lemma 3.1	same; signed version new (Lemma 16.1)
Lemma 5.1, (5.3), (5.4) Theorem 2.2, (6.2), (2.9), (6.3) (5.6)	Section 4.1 Thm. 4.1, E_r, E_1, E_2 identical proof of Thm. 1.1	Lemma 5.1, (5.3)–(5.4) Thm. 5.2, Q_j, Q_1 identical Cor. 5.3	same geometric-sum proof and bounds same expansion; 75-01's remainder is sharper (Theorem 16.3) same
Lemma 7.1, (7.4)–(7.6)	Lemma 5.1, $u_L^{(m)}(0)$ with $a = s-1$	Prop. 4.1, $Y_v^{(m)}(0) = U_m(s)$	same; 75-01's Taylor remainder by Cauchy bounds (Section 15.2)
Mean relation (8.1), (7.15) (8.2), (8.3), (8.8)	$B_s = Q_\alpha/2$ Section 5.2, $G_s = \mathcal{H}_\alpha$	Prop. 4.2, $1 + d(v) = Q_\alpha$ (6.4)–(6.7)	same; 75-06 proves the mean directly, as source 48 does identical construction
Theorem 2.1, $P_1, \dots, P_8$ (2.5)	Thm. 1.1, $c_j(6c) = P_j(c)$, $j \leq 5$ (6)	Thm. 6.1, $b_j(6c) = P_j(c)$, $j \leq 8$ (6.3)	identical; 75-06's b_8 is an exact check of (8.5) same
Theorem 2.3, (9.3), (9.4)	Thm. 6.1, through R_0^{-5}	Thm. 7.1, $\Delta_2, \dots, \Delta_8$	identical through R_0^{-6} ; Δ_7, Δ_8 new (Theorem 18.3)
Theorem 2.4, logistic part	—	Thm. 8.2	same at $\mu = 1$
Theorem 2.4, Gumbel part, $L_n/K \rightarrow 1 + 1/\alpha$	—	Thm. 9.1	same centring equation at $\mu = 1$
Remark on the two Lambert branches	Section 1.3	Section 7.2	same
Signed cumulant scales and $\mathcal{E}_1 \sim -3/(8\alpha Ms)$	Lemma 3.1, Thm. 4.1	Prop. 5.4	each other's duplicate; new here (Section 16)

Table 2: Results of sources 75-01 and 75-06 that this report already proves. Statement numbers are those of the delivered manuscripts; equation numbers in parentheses are the sources' own.

[Merge note, batch 75.] The intake checked the agreements of Table 2 in exact rational arithmetic: $b_j(6c) = P_j(c)$ for $j \leq 8$ and $c_j(6c) = P_j(c)$ for $j \leq 5$; and, solving (9.1) afresh with $P_1, \dots, P_8$, the coefficients of E^2 through R_0^{-8} , which equal source 75-06's $d_j(6c)$ for $j \leq 8$ and source 75-01's inverse coefficients for $j \leq 5$. Source 75-01's exact A022629 table (`data/75-01-jensen-exact_s1.json`) equals both earlier tables on their ranges; its tables for $\alpha = 2, 3$ to $n = 2500$ equal source 40's `data/40-norm-moments-A092484_computed.txt` and `data/40-norm-moments-A265840_computed.txt` there and are not shipped. Source 75-06's relative errors at $n = 100, 500, 1000, 5000, 10000$ (Table 3) coincide with source 75-01's to all printed digits, and their $n = 100$ values with those of sources 40 and 48.

Stale statements and priority caveats. Both pins are the ProveIt commit 29aca108e (1 October 2026, 19:03; full identifier in Appendix D). It already contained the archives of sources 40 and 48, unopened, and predates their placement (9df4ba51a). Source 75-01's sentence "A repository code search for A022629 returned no match" was accurate for that snapshot and is stale now: the repository's treatment is this report. Source 75-06 makes no such statement, but its repository reading was blind to the same two archives. Likewise, the proof of the five Kotešovec conjectures that both sources present as their principal contribution is, in this report, the third and fourth proof; their new contributions are those of Sections 16–18. Both keep explicit priority caveats, which this report keeps. Source 75-01: its search "located no prior proof of these precise fixed- s statements", which is "not a claim that no such proof exists anywhere, nor a claim that a longstanding major conjecture in partition theory has been settled"; Kumar and Rana's 2026 article on statistical inequalities for norm moments [18] "should also be considered in any eventual publication-level priority review"; the Hermite-limit mechanism is Griffin, Ono, Rolén and Zagier's [17], and no new general Hermite-limit theorem is claimed. Source 75-06: its targeted search "does not establish bibliographic priority"; Lagarias and Sun's reciprocal-supernorm ensemble [20] is a different ensemble and not a proof of these results; the conditioning method is Fristedt's [19], applied with every estimate proved for this model. Source 75-06 read the repository's Analysis/Transseries README, and source 75-01 the repository map and the Lambert- W drafts of the Fabius tree [24]; neither uses a repository theorem as a premise, and neither claims Lean or Rocq verification.

15.2 Second routes and remarks that differ

Analytic inverse and Cauchy bounds (source 75-01). Source 75-01's proof of Lemma 7.1 controls the Taylor remainder of the boundary inverse by analyticity rather than by repeated use of (7.3). The local inverse $u_s(v)$ of h_s is analytic on $|v| \leq cs$ for a fixed small $c > 0$: either by the inverse function theorem with uniform bounds on $u \in [1/2, 3/2]$, or by solving $u - 1 = (v + \log u)/s$ by contraction on a fixed complex disk around 1. Cauchy's estimates then bound the remainder after m Taylor terms of u'_s by a constant times $s^{-m-1}|v|^m$, and truncating the v -integral at cs loses an exponentially small quantity. Derivatives in s are handled by differentiating the localized integral and its explicit remainder before estimating, so no differentiability of an unqualified remainder is assumed.

The mean without differentiating a remainder (source 75-06). Source 75-06 proves the mean relation (8.1) directly, as source 48 does (Lemma 7.2): the odd kernel $(1+e^U)^{-1} - \mathbf{1}_{U<0}$ has moments $2(2j+1)!\eta(2j+2)$, and the identity $\partial_t X = -XX'$ for the upper inverse $X(U)$ of $tx - \alpha \log x$ shows that the directly justified moment series equals the formal negative t -derivative of the boundary expansion. Its first terms reproduce (7.19), including the vanishing coefficient of s^{-4} in $Q_\alpha(s) - 1$ (equivalently of s^{-2} in $\mathcal{B}_\alpha$), which source 75-06 calls "a useful

algebraic check”.

Coefficient structure (source 75-06). The formal construction gives $P_j(0) = 0$ and $\deg_c P_j \leq \lfloor (j+1)/2 \rfloor$. Source 75-06 tracks this by assigning degree 2 to c : a term of c -degree a in the normalized entropy cannot begin before $r^{-(2a-1)}$; the initial inverse-derivative series has that filtration, and multiplication, composition and the square-root reversion preserve it. These identities in a formal parameter do not make the analytic theorem uniform as $\alpha \downarrow 0$. $P_1, \dots, P_8$ attain the bound (checked by the intake).

Prefactor bookkeeping (source 75-06). At the saddle, $\log a_\alpha(n) = F(t_n) + nt_n - \frac{1}{2} \log(2\pi B) + O(s/M)$, so the probability prefactor contributes only $O(s)$ to the logarithm, below every scale K/r^R (source 75-06’s Corollary 5.3; compare (5.6)).

Two finite-size readings of the logarithmic block. For $\alpha = 1$, $n = 10000$, source 75-01 records $\log a_1(10000) = 613.21132139736055145\dots$. The leading expression $K(r-1)$ is low by about 54.3504; adding the first inverse-logarithmic correction leaves about 7.37119; the five-term truncation is high by about 4.65395. Source 75-06 records the normalized first correction $\mathcal{R}(n) = (r/K)(\log a_1(n) - K(r-1))$, whose limit is $\pi^2/6 \approx 1.6449$ by (2.5): exact values give $\mathcal{R}(100) \approx 1.5450$, $\mathcal{R}(1000) \approx 1.8190$ and $\mathcal{R}(10000) \approx 1.9030$, and saddle evaluations at $n = 10^6, 10^8, 10^{10}$ give about 1.8800, 1.8274 and 1.7900. The last three are saddle estimates, not comparisons with exact coefficients. Both sources draw the conclusion of Section 12: the logarithmic block is not a high-accuracy numerical exponent at these sizes, and plots alone can obscure the correct next coefficient.

16 Sharp saddle scales and near-integer thresholds

This section is source 75-01’s Sections 3, 4 and 6.2, translated into the notation of this report, with source 75-06’s Proposition 5.4 as a second proof of the cumulant scales. Throughout, t , s , M and H are as in (2.7), and, beside the safe parameter $\varepsilon = s/M$ of (4.8), we use the *sharp* parameter

$$\varepsilon_\# = \frac{1}{\alpha M s} = \frac{\varepsilon}{\alpha s^2}. \quad (16.1)$$

16.1 Signed cumulant scales

Lemma 16.1 (Signed cumulant scales; sources 75-01 and 75-06). *For fixed $\alpha > 0$, as $s \rightarrow \infty$ with $t = \alpha s/M$,*

$$\kappa_1 \sim \frac{M^2}{2}, \quad \kappa_j \sim \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}} \quad (j \geq 2 \text{ fixed}). \quad (16.2)$$

Consequently, at the exact saddle,

$$\frac{\kappa_j}{B^{j/2}} = O_{\alpha,j}(\varepsilon_\#^{(j-2)/2}), \quad j \geq 3. \quad (16.3)$$

Proof (source 75-01). Sum–integral comparison for $x^j b_j(p(x))$, where $p(x) = (1 + e^{tx - \alpha \log x})^{-1}$, has error $O_{\alpha,j}(M^j)$: differentiate the summand in x , split at $M/2$ and $2M$, and use exponential decay beyond $2M$. For $j \geq 2$, b_j has a factor $p(1-p)$ and its other factors are bounded on $[0, 1]$. For $j = 1$ the bulk integral is $\int_0^M x \, dx = M^2/2$ up to lower-order errors, which is (4.3).

For $j \geq 2$ the upper moving edge supplies the leading term. On $x \geq M/2$ the substitution $y = tx - \alpha \log x$ is increasing for large s ; let $x(y)$ be its inverse there and $w(y) = e^y/(1 + e^y)^2$. Since $p = (1 + e^y)^{-1}$ and $dp/dy = -p(1 - p)$, the recursion $b_{j+1}(p) = p(1 - p)b'_j(p)$ gives

$$b_j((1 + e^y)^{-1}) = (-1)^{j-2} w^{(j-2)}(y), \quad j \geq 2.$$

Integration by parts $j - 2$ times transfers all these derivatives to $x(y)^j x'(y)$, and the signs cancel. The boundary terms at $M/2$ are exponentially smaller in s than the edge term, and those at infinity vanish. For bounded y ,

$$x(y) = M\{1 + O_{\alpha,y}(s^{-1})\}, \quad x'(y) = \frac{M}{\alpha s}\{1 + O_{\alpha,y}(s^{-1})\}.$$

Repeated differentiation, using $x' = 1/(t - \alpha/x)$, gives

$$\frac{d^{j-2}}{dy^{j-2}} \{x(y)^j x'(y)\} = \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}} (1 + o_{\alpha,j}(1)) :$$

the leading operator is $t^{-1}d/dx$, and differentiating its denominator costs a relative $O_{\alpha,j}(s^{-1})$. The kernel w has integral one and exponential tails; localization first to $|y| \leq C \log s$, then $C \rightarrow \infty$, justifies integrating the expansion.

The region below $M/2$ contributes $O_{\alpha,j}(M^{j+1-c_\alpha} s^{C_j})$ for some $c_\alpha > 0$ (split again at $x = M^{1/2}$ and use $p(1 - p)$ above it), which is smaller than M^{j+1}/s^{j-1} , and so is the sum-integral error. This proves (16.2), and dividing by $B^{j/2}$, with $B = \kappa_2 \sim M^3/(\alpha s)$, gives (16.3). $\square$

Second proof of (16.2) (source 75-06). After the same reduction to the upper branch and the same $j - 2$ integrations by parts, the integrand is $\frac{d^{j-2}}{dy^{j-2}}(x(y)^j x'(y)) w(y)$. On bounded y -intervals $x \sim M$ and $x' \sim A := M/(\alpha s)$, and every occurrence of a higher derivative of x costs a further factor $O(1/s)$. So the leading term comes from the affine function $x = M + Ay$:

$$\frac{d^{j-2}}{dy^{j-2}} ((M + Ay)^j A) \Big|_{y=0} = \frac{j!}{2} M^2 A^{j-1} = \frac{j!}{2} \frac{M^{j+1}}{(\alpha s)^{j-1}}.$$

Exponential decay of w , whose integral is 1, permits restriction to a growing bounded neighbourhood followed by dominated estimates. $\square$

Remark 16.2 (Why the absolute cumulant bound is not sharp; source 75-01). Taking absolute values before the integrations by parts loses the cancellations in b_j and gives M^{j+1}/s , which is (4.5), instead of M^{j+1}/s^{j-1} . The signed estimate is what identifies $\varepsilon_\# = (\alpha Ms)^{-1}$, rather than $\varepsilon = s/M$, as the algebraic saddle parameter. The weaker absolute estimate is still the right tool for a high-order Taylor remainder, but not for the retained coefficients.

16.2 The sharp exact-saddle remainder

Theorem 16.3 (Sharp exact-saddle expansion; source 75-01). *For fixed $\alpha > 0$ and every fixed integer $J \geq 0$, at the exact saddle $t = t_n$,*

$$a_\alpha(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 + \sum_{\ell=1}^J \mathcal{E}_\ell + O_{\alpha,J}(\varepsilon_\#^{J+1}) \right), \quad (16.4)$$

with $\mathcal{E}_\ell$ as in (6.2). Here $\mathcal{E}_\ell = O_{\alpha,\ell}(\varepsilon_\#^\ell)$ and

$$\mathcal{E}_1 \sim -\frac{3}{8\alpha Ms} \sim -\frac{3}{8\alpha Kr}. \quad (16.5)$$

Proof (source 75-01). The Fourier reduction and the bounds (5.3) and (5.4) are those of Section 5; outside $|\theta| \leq c_0/M$ the integral is smaller than $B^{-1/2}\varepsilon_{\#}^A$ for every fixed A . This is an eventual estimate: its unspecified α -dependent constant is not a useful finite- n accuracy certificate.

In the central interval put $x = \theta\sqrt{B}$. The exponent is $-x^2/2 + \sum_{j=3}^{R-1} \kappa_j(ix)^j/(j!B^{j/2})$ plus a Taylor remainder. For each fixed $R \geq 2$ the absolute R th derivative on $|\theta| \leq c_0/M$ is $O_{\alpha,R}(M^{R+1}/s)$: for $k \leq 2M$ the complex Bernoulli denominators stay comparable to their positive values, and for $k > 2M$ the odds are small and exponentially decreasing. Hence the normalized remainder is at most $C_{\alpha,R}(s/M)^{R/2-1}|x|^R$. For the expansion through order J choose $R = 2J + 6$; then $(s/M)^{J+2} = o_{\alpha}(\varepsilon_{\#}^{J+1})$, since $s^{2J+3}/M \rightarrow 0$.

Restrict first to $|x| \leq \sqrt{C \log M}$. By (16.3) and the remainder bound, the exponential may be expanded with an integrable Gaussian majorant. Give the j th normalized cumulant weight $j - 2$: a monomial of weight w is $O(\varepsilon_{\#}^{w/2})$ times a fixed polynomial in x . Retain weights through $2J + 1$. Odd weights give odd polynomials, which integrate to zero on the symmetric interval; the first discarded even weight is $2J + 2$, which contributes $O_{\alpha,J}(\varepsilon_{\#}^{J+1})$. Expanding sufficiently many ordinary Taylor terms makes this a finite remainder estimate. The rest of $|\theta| \leq c_0/M$ is algebraically negligible by (5.3), for C large. The Gaussian moments $(2\pi)^{-1/2} \int e^{-x^2/2}(ix)^T dx = (-1)^{T/2}(T-1)!!$ for even T give exactly (6.2), and the bound on $\mathcal{E}_{\ell}$ follows from its weight. Finally, Lemma 16.1 gives

$$\mathcal{E}_1 = \frac{\kappa_4}{8B^2} - \frac{5\kappa_3^2}{24B^3} \sim \frac{12}{8\alpha Ms} - \frac{45}{24\alpha Ms} = -\frac{3}{8\alpha Ms}.$$

□

Theorem 16.3 sharpens Theorem 2.2: after the corrections $\mathcal{E}_1, \dots, \mathcal{E}_{J-1}$ the relative error is $O(\varepsilon_{\#}^J) = O((\alpha Ms)^{-J})$ rather than $O((s/M)^J)$, a gain of $(\alpha s^2)^J$, and the first correction is negative, of exact order $3/(8\alpha Kr)$, so the bare Gaussian factor overestimates $a_{\alpha}(n)$. At one layer, source 75-06's Proposition 5.4 states the first refinement

$$a_{\alpha}(n) = \frac{\exp(F(t_n) + nt_n)}{\sqrt{2\pi B}} \left(1 - \frac{3 + o(1)}{8\alpha Ms}\right),$$

deduced from its remainder $O(\varepsilon^2) = o(\varepsilon_{\#})$; Section 18 gives its version for μ layers. The second correction $\mathcal{E}_2$ is (6.3) (source 75-01 prints the same display), and source 75-01's program computes $\mathcal{E}_3$ from the general formula (6.2) rather than from a hard-coded expression.

[Batch 75, 2 October 2026.] This theorem and Lemma 16.1 partly answer source 48's project 48.6, "Optimal saddle remainder scales" (Section 13): they give the sharp signed scales, the sharp first relative correction, and the error order $\varepsilon_{\#}^{J+1}$ after each correction. Neither source gives the complete expansion of $\mathcal{E}_1$ in powers of $1/s$, nor any optimality statement beyond these orders; that part of the question stays open.

16.3 Saddle accuracy at finite size

Let $G_J(n) = G \sum_{\ell=0}^J \mathcal{E}_{\ell}(t_n)$ with $\mathcal{E}_0 = 1$ and G as in Section 12. Table 3 gives source 75-01's signed relative errors $G_J(n)/a_1(n) - 1$, computed with 40 decimal digits in the finite sums, together with the two extra rows of source 75-06. A 60-digit rerun at $n = 10000$ changes the recorded errors by at most 7.0×10^{-38} ; this stability test is not an interval-arithmetic certificate, and neither source's floating-point computation is directed-rounding certified.

For larger powers the picture changes. At $\alpha = 2$, $n = 2500$, source 75-01 finds $G_J/a_2 - 1 \approx -2.2561 \cdot 10^{-4}$, $-2.2601 \cdot 10^{-4}$ and $-2.2601 \cdot 10^{-4}$ for $J = 1, 2, 3$; at $\alpha = 3$, $n = 2500$, the

n	$J = 0$	$J = 1$	$J = 2$	$J = 3$
100	$1.09591 \cdot 10^{-2}$	$1.73317 \cdot 10^{-4}$	$9.14176 \cdot 10^{-5}$	$9.01486 \cdot 10^{-5}$
200*	$6.90519 \cdot 10^{-3}$	$2.59790 \cdot 10^{-5}$		
500	$3.84109 \cdot 10^{-3}$	$1.07637 \cdot 10^{-5}$	$1.45599 \cdot 10^{-7}$	$8.06153 \cdot 10^{-8}$
1000	$2.47981 \cdot 10^{-3}$	$4.50477 \cdot 10^{-6}$	$1.57836 \cdot 10^{-8}$	$-2.77601 \cdot 10^{-9}$
2000*	$1.60998 \cdot 10^{-3}$	$1.92292 \cdot 10^{-6}$		
2500	$1.40254 \cdot 10^{-3}$	$1.46485 \cdot 10^{-6}$	$3.61741 \cdot 10^{-9}$	$1.17092 \cdot 10^{-11}$
5000	$9.16761 \cdot 10^{-4}$	$6.32944 \cdot 10^{-7}$	$1.05783 \cdot 10^{-9}$	$2.22571 \cdot 10^{-12}$
10000	$6.02038 \cdot 10^{-4}$	$2.75808 \cdot 10^{-7}$	$3.11999 \cdot 10^{-10}$	$4.66600 \cdot 10^{-13}$

Table 3: Signed relative errors $G_J(n)/a_1(n) - 1$ for A022629 ($\alpha = 1$), source 75-01; rows marked * are source 75-06's ($J \leq 1$ only). At $n = 100, 500, 1000, 5000$ and 10000 source 75-06's $J = 0, 1$ values agree with these to every printed digit.

errors for $J = 0, 1, 2, 3$ are about 0.05070, 0.05016, 0.05016 and 0.05016. Source 75-01 reads this as slow convergence (“Higher powers can converge slowly”) and states that the residuals “have not been decomposed here into separate oscillatory sectors”.

[Merge note, batch 75.] The residual is independent of J once $\mathcal{E}_1$ is included, so it is not a truncation effect of the central Edgeworth series, which the corrections already resolve to 10^{-9} . It is the noncentral contribution that Theorem 11.1 and Remark 6.1 identify, invisible to every central correction. It agrees in sign and order of magnitude with source 40's rows $\alpha = 2, 3$ at $n = 5000$ in Section 12 (there $G(1 + \mathcal{E}_1)/a_\alpha - 1 \approx -3.6 \cdot 10^{-5}$ and $+0.040$), whose growth with α the resonance moduli of that section track. Neither source computes the phases that would turn this identification into a formula.

[Batch 100, 5 October 2026.] A decomposition into Fourier cells supports this identification numerically. The intake of batch 100 computed $\sqrt{2\pi B} \mathbb{P}_t(S = n)$ at the exact saddles of Section 12, $n = 5000$, cell by cell over $|M\theta - 2\pi j| \leq \pi$, $|j| \leq 3$ (composite Gauss–Legendre quadrature of the exact characteristic function in double precision; uncertified):

α	$j = 0$	$j = \pm 1$	$j = \pm 2$	$j = \pm 3$	total	ratio of Section 12
2	0.999509	$+3.66 \cdot 10^{-5}$	$-4.7 \cdot 10^{-8}$	$-3.7 \cdot 10^{-9}$	0.999545	0.999545
3	0.999669	-0.0383	$+2.8 \cdot 10^{-4}$	$-2.2 \cdot 10^{-5}$	0.961625	0.961626
4	0.999750	$+0.2244$	$+0.0068$	$-2.7 \cdot 10^{-4}$	1.230648	1.230518
5	0.999800	$+0.652$	-0.080	$+0.013$	1.584610	1.581098

At $\alpha = 5$ the cells $|j| > 3$ are not negligible. The cell $j = 0$ is the central Edgeworth contribution ($\approx 1 + \mathcal{E}_1$); the residuals sit in the cells around $\pm 2\pi/M$, with signs that the modulus of Theorem 11.1 does not give. This is numerical evidence only. Source 100-93's theta formula does not explain it: at fixed α the lower bulk is not frozen and $\Lambda \rightarrow 0$, and its formula with the exact moments gives 1.0000055, 0.98057 and 1.19429 at $\alpha = 3, 4, 5$ (Section 22.7). A formula for these cells is question 100.6.

16.4 Near-integer localization of the threshold

For real $x > 0$ let $t(x)$ be the unique solution of $-F'(t(x)) = x$, so that $t_n = t(n)$. The inverse function theorem gives the exact rules

$$\frac{dt}{dx} = -\frac{1}{B}, \quad \frac{d\kappa_j(t(x))}{dx} = \frac{\kappa_{j+1}}{B}. \quad (16.6)$$

Define the Gaussian inverse

$$\begin{aligned}\Psi_0(x) &= xt(x) + F(t(x)) - \tfrac{1}{2} \log(2\pi B(t(x))), \\ \Psi'_0(x) &= t(x) - \frac{\kappa_3}{2B^2} = t(x) + O_\alpha(M^{-2}) \sim \frac{\alpha s}{M} > 0,\end{aligned}\tag{16.7}$$

and let $n_*(Y)$ be the unique sufficiently large solution of $\Psi_0(n_*(Y)) = Y$. As in Theorem 2.3, $N_\alpha(Y)$ is the least n with $\log a_\alpha(n) \geq Y$ (source 75-01 writes $\mathcal{N}_s(A)$ with $A = e^Y$).

Theorem 16.4 (Near-integer localization; source 75-01). *For fixed $\alpha > 0$, as $Y \rightarrow \infty$,*

$$|N_\alpha(Y) - n_*(Y)| \leq 1 + o(1).\tag{16.8}$$

More precisely, at an exact sequence value, with s taken at t_n ,

$$n_*(\log a_\alpha(n)) - n = -\frac{3}{8\alpha^2 s^2}(1 + o_\alpha(1)).\tag{16.9}$$

If Ψ_0 is replaced by $\Psi_1 = \Psi_0 + \mathcal{E}_1(t(x))$, the displacement at $Y = \log a_\alpha(n)$ is $O_\alpha(M^{-1}s^{-3})$.

Proof (source 75-01). Theorem 16.3 with $J = 1$ gives $\log a_\alpha(n) = \Psi_0(n) + \mathcal{E}_1(t_n) + O_\alpha(\varepsilon_\sharp^2)$. Divide this logarithmic error by the slope (16.7) and apply the mean value theorem in an $o(1)$ neighbourhood of n : the leading displacement is

$$\frac{-3/(8\alpha Ms)}{\alpha s/M} = -\frac{3}{8\alpha^2 s^2},$$

which proves (16.9). With Ψ_1 the logarithmic error is $O_\alpha(\varepsilon_\sharp^2)$, giving $O_\alpha(M^{-1}s^{-3})$ after division by the slope. For general Y put $m = N_\alpha(Y)$, so that $\log a_\alpha(m-1) < Y \leq \log a_\alpha(m)$. Monotonicity of the eventual inverse n_* places $n_*(Y)$ between $m-1+o(1)$ and $m+o(1)$, proving (16.8). $\square$

No universal rounding instruction follows from (16.8): a height arbitrarily close to a coefficient threshold can require deciding on which side it lies. An effective finite-input certificate would also need explicit error constants and validated numerical evaluation; source 75-01's inverse routine solves the approximating equation, not that certification problem.

[Merge note, batch 75.] The two steps are instances of the transseries-and-inversion volume [25]: the passage from the logarithmic residual to the displacement is its residual-to-root conversion (`p0:thm:backward-error`) with slope bounds $\Psi'_0 \asymp \alpha s/M$, and the passage to the integer threshold is its staircase theorem (`p0:thm:staircase`, parts (1)–(2)), whose separation condition is exactly the obstruction to a universal rounding rule just described. That separation step is formalized in Lean as `Fabius.staircase_separation` (with `Fabius.staircase_ceil`) [26]; nothing else in this section is formalized. Compared with Section 9, whose brackets carry the fixed-order inverse-logarithmic errors plus an $O(1)$ rounding, the exact-saddle inverse localizes the threshold to $1 + o(1)$, at the price of evaluating the exact saddle. It does not answer project 48.7, which asks for explicit constants.

[Batch 100, 5 October 2026.] Source 100-89's Theorem 21.8 gives, for every truncation order, exact recovery at sequence values and the threshold off an $O(t^{J-1})$ -neighbourhood of the integers, at the conservative scale; it agrees that no rounding rule holds for arbitrary thresholds. In the critical window of growing powers the theta factor moves the threshold by an oscillating amount of order at most $M^{2/3}$, and of exactly that order away from two phases per period (Remark 22.10; dated note, 5 October 2026: first written “of order $M^{2/3}$ ”).

For computation one can solve directly in t : with $Z(t) = F(t) + t\kappa_1(t) - \frac{1}{2}\log(2\pi B(t))$ and $Z'(t) = -tB + \kappa_3/(2B)$, Newton iteration for $Z(t) = Y$, initialized by (2.10), returns $n_* = \kappa_1(t)$. This uses only positive real sums, not the inversion of a long truncated power series. At $Y = \log a_1(10000)$, source 75-01 finds

$$n_*(Y) = 9999.9838291094228371832520930128038\dots,$$

a displacement of about -0.0161708905772 , which is $1.02387586805\dots$ times the predicted leading value $-3/(8s^2)$ at the saddle. This is consistent with (16.9), but a single ratio is not a proof of its limit. (The recorded saddle tables `data/75-01-jensen-saddle_s2_dps40.csv` and `data/75-01-jensen-saddle_s3_dps40.csv` also list this scaled displacement for $\alpha = 2, 3$: about 1.69 and 0.83 at $\alpha = 2$, $n = 1000, 2500$, and about 52 and 118 at $\alpha = 3$, the finite-size residual of Section 16.3 seen through the inverse.)

17 Eventual log-concavity and Jensen hyperbolicity

This section is source 75-01's Section 7. It needs errors much smaller than those of the logarithmic block, and differentiating a pointwise O -term for $\log a_\alpha(n)$ would not be legitimate. Instead a smooth approximant is built from exact cumulants, and the discrete error is controlled separately.

Let

$$D_r(t) = [z^r] \log\left(1 + \sum_{j \geq 1} \mathcal{E}_j(t) z^j\right), \quad \mathcal{L}_J(x) = \Psi_0(x) + \sum_{r=1}^J D_r(t(x)). \quad (17.1)$$

Thus $D_1 = \mathcal{E}_1$, $D_2 = \mathcal{E}_2 - \mathcal{E}_1^2/2$, and every D_r is a finite expression. (Here r is a summation index, not $\log K$.) For each fixed J , Theorem 16.3 gives

$$\log a_\alpha(n) = \mathcal{L}_J(n) + O_{\alpha,J}(\varepsilon_\#^{J+1}). \quad (17.2)$$

Equations (16.6) and (16.2) give, for fixed r and k ,

$$\frac{d^k}{dx^k} \mathcal{E}_r(t(x)), \quad \frac{d^k}{dx^k} D_r(t(x)) = O_{\alpha,r,k}(\varepsilon_\#^r M^{-2k}) : \quad (17.3)$$

differentiate each finite cumulant monomial with $d\kappa_j/dx = \kappa_{j+1}/B$; every differentiation costs $O_{\alpha,j}(M^{-2})$. If $S(x) = xt(x) + F(t(x))$, then $S' = t$ and $S'' = -1/B$, so

$$\mathcal{L}'_J(x) = t + O_\alpha(M^{-2}), \quad \mathcal{L}''_J(x) = -B^{-1} + O_\alpha(M^{-4}), \quad \mathcal{L}_J^{(k)}(x) = O_{\alpha,J,k}(sM^{1-2k}) \quad (k \geq 2), \quad (17.4)$$

locally uniformly when x changes by a bounded amount.

17.1 Strict log-concavity

Theorem 17.1 (Eventual strict log-concavity; source 75-01). *For every fixed $\alpha > 0$,*

$$\log \frac{a_\alpha(n)^2}{a_\alpha(n-1)a_\alpha(n+1)} \sim \frac{1}{B(t_n)} \sim \frac{\alpha \log \sqrt{2n}}{(2n)^{3/2}}. \quad (17.5)$$

In particular $a_\alpha(n)$ is strictly log-concave for all sufficiently large n .

Proof (source 75-01). Take $J = 3$ in (17.2); the error is $O_\alpha(\varepsilon_\#^4) = o_\alpha(s/M^3)$. Taylor's formula and (17.4) give

$$2\mathcal{L}_3(n) - \mathcal{L}_3(n-1) - \mathcal{L}_3(n+1) = B^{-1} + O_\alpha(M^{-4}) + O_\alpha(sM^{-7}).$$

Adding the three discrete errors does not change the leading term, and $B \sim M^3/(\alpha s)$, $M \sim K$, $s \sim r$ give (17.5). $\square$

17.2 Hermite limits of Jensen polynomials

For integers $d \geq 1$ and $n \geq 0$ define the Jensen polynomials

$$\mathcal{J}_\alpha^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} a_\alpha(n+j) X^j, \quad (17.6)$$

and the Hermite polynomials H_d by $e^{Xz-z^2} = \sum_{d \geq 0} H_d(X) z^d / d!$, so that $H_0 = 1$, $H_1 = X$, $H_2 = X^2 - 2$, $H_3 = X^3 - 6X$. (H_d is not the transition width H .)

Theorem 17.2 (Fixed-degree Jensen hyperbolicity; source 75-01). *For every fixed $\alpha > 0$ and fixed integer $d \geq 1$, let $\delta_n = (2B(t_n))^{-1/2}$. Then, coefficientwise in X ,*

$$\frac{\mathcal{J}_\alpha^{d,n}(e^{-t_n}(\delta_n X - 1))}{a_\alpha(n) \delta_n^d} \rightarrow H_d(X). \quad (17.7)$$

Consequently $\mathcal{J}_\alpha^{d,n}$ has d distinct negative real roots for all sufficiently large n .

Proof (source 75-01). Choose J with $J+1 > 3d/2$; then the discrete error in (17.2) is $o_\alpha(\delta_n^d)$. Put $\tau_n = \mathcal{L}'_J(n)$ and $\tilde{\delta}_n^2 = -\frac{1}{2}\mathcal{L}''_J(n) > 0$ for large n . Taylor expansion for each $0 \leq j \leq d$ gives

$$\log \frac{a_\alpha(n+j)}{a_\alpha(n)} = \tau_n j - \tilde{\delta}_n^2 j^2 + \sum_{k=3}^d \gamma_k(n) j^k + o_\alpha(\tilde{\delta}_n^d), \quad \gamma_k = \frac{\mathcal{L}_J^{(k)}(n)}{k!},$$

where the next Taylor derivative contributes $o(\delta_n^d)$, and $\gamma_k(n)/\tilde{\delta}_n^k = O_{\alpha,k}((Ms)^{1-k/2}) \rightarrow 0$ for $k \geq 3$. These are the hypotheses of the Hermite-limit criterion of Griffin, Ono, Rolén and Zagier [17, Theorem 3].

The mechanism can also be seen directly. Expand the exponential of the last display as a polynomial in j through degree d . The coefficient of j^k , divided by $\tilde{\delta}_n^k$, tends to the coefficient of z^k in e^{-z^2} . In the normalized Jensen sum the d th finite difference annihilates every polynomial of degree below d , and the surviving coefficients combine to $H_d(X)$; the local $o(\tilde{\delta}_n^d)$ error is exactly what this cancellation needs. Finally (17.4) gives $\tilde{\delta}_n/\delta_n \rightarrow 1$ and $(\tau_n - t_n)/\delta_n \rightarrow 0$, so the affine change of normalized variable tends to the identity and $\tau_n, \tilde{\delta}_n$ may be replaced by t_n, δ_n . Hermite polynomials have distinct real roots; on disjoint small intervals around them the endpoint signs alternate, coefficientwise convergence preserves those signs, and degree d allows no further roots. The Jensen polynomial has positive coefficients and a nonzero constant term, so its real roots are negative. $\square$

Neither the theorem nor its proof gives a bound uniform in d . It does not assert that every Jensen polynomial of every degree is hyperbolic at one fixed index, and it should not be conflated with infinite iterated log-concavity of the sequence. Source 75-01's contribution here is the verification of the Griffin–Ono–Rolén–Zagier hypotheses, including an error scale small enough to survive finite differences; the Hermite-limit mechanism itself is theirs.

17.3 Exact finite data

Source 75-01 evaluates the positive product recursion exactly through $n = 10000$ for $\alpha = 1$ and through $n = 2500$ for $\alpha = 2, 3$, and checks the signed recurrence (12.2) against it through $n = 200$ for each of $\alpha = 1, 2, 3$. For $\alpha = 1$, among the tested indices $1 \leq n \leq 9999$, strict log-concavity fails exactly at

$$n = 1, 2, 4, 6, 8, 11, 13, 16, 17, 19, 23, 26, 29, 32, 34, 38, 40, 41, 43, 46, 53, 59, 64, 76,$$

so every tested index from 77 through 9999 is strictly log-concave. This does *not* prove that 77 is the eventual threshold. For $\alpha = 2$ the tested range $1 \leq n \leq 2499$ has 162 failures, the last at $n = 1033$; for $\alpha = 3$ it has 904 failures, persisting to $n = 2499$, the end of the range (counts from `data/75-01-jensen-exact_audit_s2.json` and `data/75-01-jensen-exact_audit_s3.json`). These observations caution against uniformity in the power.

Exact rational real-root counting gives the numbers of real roots of $\mathcal{J}_1^{d,n}$ in Table 4; every polynomial there was also checked to be squarefree.

n	$d = 2$	$d = 3$	$d = 4$	$d = 5$
100	2	1	0	1
1000	2	3	4	1
5000	2	3	4	5

Table 4: Real roots of the Jensen polynomials $\mathcal{J}_1^{d,n}$ (source 75-01): the onset of hyperbolicity depends on the degree.

[Batch 75, 2 October 2026.] Source 75-06’s research question on shape inequalities (Section 19, item 75-06.9) asks for eventual log-concavity; Theorem 17.1 answers that part for one layer. Ratio monotonicity, higher finite-difference inequalities, and the case of $\mu \geq 2$ layers stay open.

18 Convolution powers

This section is source 75-06’s two-parameter extension (its Sections 1.2, 2 and 4–9 for more than one layer), translated: source 75-06’s power r is α here and its number of layers s is μ . For fixed $\alpha > 0$ and a fixed integer $\mu \geq 1$ put

$$A_\alpha(q)^\mu = \prod_{k \geq 1} (1 + k^\alpha q^k)^\mu = \sum_{n \geq 0} a_{\alpha,\mu}(n) q^n, \quad K_\mu = \sqrt{2n/\mu}, \quad r_\mu = \log K_\mu. \quad (18.1)$$

For $\mu = 1$ this is $a_\alpha(n)$, with $K_1 = K$, $r_1 = r$. An object is an ordered μ -tuple of distinct partitions of total size n , weighted by the product of their α th norm powers; for integral α the weights count colourings. The analysis also applies to real $\alpha > 0$, when the coefficients are positive reals.

OEIS identifications (source 75-06). For $\alpha = 1$ and $\mu = 2, 3$ the coefficients are [A022630](#) and [A022631](#) [21]; for $\alpha = 1$ and fixed $\mu \geq 1$ they are the positive columns of the array [A297321](#) [22]; for integer $\alpha \geq 1$ and $\mu = 1$ they are the positive columns of [A292189](#) [6]. Every statement below is column by column, with the column index fixed; no uniform assertion is made for a growing index.

[Merge note, batch 75.] The intake confirmed on 2 October 2026 that the OEIS names of [A022630](#) and [A022631](#) are $\prod_{m \geq 1} (1 + mq^m)^2$ and $(\cdots)^3$, that their first fifteen terms equal the coefficients computed from the product, and that [A297321](#)’s column k is $\prod_{j \geq 1} (1 + jx^j)^k$. Source 75-06’s proposed-notes file further says [A297321](#) cross-references its first 32 columns to [A022629](#)–[A022660](#); the intake did not check that.

The μ -layer model. Let $X_{k,i}$ ($k \geq 1$, $1 \leq i \leq \mu$) be independent Bernoulli variables with parameter p_k of (4.1), and $S = \sum_{k,i} kX_{k,i}$. Then $\log \mathbb{E}e^{\cdot}$ is $\mu F(t)$, the cumulants are $\mu \kappa_j$,

and $a_{\alpha,\mu}(n) = e^{\mu F(t) + nt} \mathbb{P}_t(S = n)$; the conditional law given $S = n$ is the weighted law on μ -layer objects of size n . The saddle equation is $-\mu F'(t) = n$.

[**Merge note, batch 75.**] So the μ -layer saddle at n is the one-layer saddle $t(n/\mu)$ of Section 16.4 at the real argument n/μ , and s, M, H are its one-layer values there. This is why the logarithmic coefficients below do not depend on μ . Source 75-06 does not phrase it this way; its proofs treat μ layers directly.

Theorem 18.1 (Growth and all orders for μ layers; source 75-06). *Fix $\alpha > 0$ and an integer $\mu \geq 1$.*

1. $\log a_{\alpha,\mu}(n) = \mu\alpha K_\mu(r_\mu - 1) + O_{\alpha,\mu}(K_\mu/r_\mu)$.
2. For every fixed $R \geq 1$, with the polynomials $P_j(c)$ of Theorem 2.1,

$$\log a_{\alpha,\mu}(n) = \mu\alpha K_\mu \left(r_\mu - 1 + \sum_{j=1}^R \frac{P_j(c)}{r_\mu^j} + O_{\alpha,\mu,R}(r_\mu^{-R-1}) \right). \quad (18.2)$$

Proof (source 75-06). (1) Lower bound: split n into μ integers $N_1, \dots, N_\mu$ differing by at most one, and use the partition $1, 2, \dots, k-1, k+d$ of Proposition 3.2 in each layer. Each layer contributes $\alpha \log k! = \alpha\sqrt{2N_i}(\log \sqrt{2N_i} - 1) + O_\alpha(\log n)$, and the sum is $\mu\alpha K_\mu(r_\mu - 1) - O_{\alpha,\mu}(r_\mu)$. Upper bound: with $t = \alpha r_\mu / K_\mu$, positivity gives $\log a_{\alpha,\mu}(n) \leq nt + \mu F(t)$. Write $\log(1 + e^u) = u_+ + \log(1 + e^{-|u|})$ with $u = \alpha \log x - tx$: the positive part integrates to $\alpha K_\mu(r_\mu/2 - 1) + O_\alpha(1)$; on $x \geq K_\mu/2$ the slope bound $|u'| \geq \alpha(r_\mu - 2)/K_\mu$ makes the smoothing integral $O_\alpha(K_\mu/r_\mu)$; on $0 < x \leq \sqrt{K_\mu}$ use $\log 2$; on $\sqrt{K_\mu} \leq x \leq K_\mu/2$ convexity gives $u \leq -\alpha r_\mu/2 + O_\alpha(1)$ and a contribution $O(K_\mu^{1-\alpha/2})$. With the sum–integral error $O_\alpha(r_\mu)$ this gives $\mu F(t) = \mu\alpha K_\mu(r_\mu/2 - 1) + O(K_\mu/r_\mu)$, and $nt = \mu\alpha K_\mu r_\mu/2$.

(2) The boundary and mean expansions of Sections 7 and 8 hold for μF and $-\mu F'$ with the factor μ ; at the saddle $K_\mu^2 = M^2 Q_\alpha(s)$, which is (8.2) with r_μ for r , and $\mu F(t) + nt = \mu\alpha K_\mu \mathcal{H}_\alpha(r_\mu)$ to every fixed order. The local probability contributes $O(s)$ to the logarithm (source 75-06's Corollary 5.3), below every scale K_μ/r_μ^R . Taking enough guard terms in the two independently proved expansions gives the remainder. $\square$

Theorem 18.2 (Relative expansion for μ layers; source 75-06). *Fix $\alpha > 0$, an integer $\mu \geq 1$ and $J \geq 0$. At the exact saddle $-\mu F'(t_n) = n$, with $\varepsilon = s/M$,*

$$a_{\alpha,\mu}(n) = \frac{e^{\mu F(t_n) + nt_n}}{\sqrt{2\pi\mu B}} \left(1 + \sum_{\ell=1}^J \mathcal{E}_\ell^{(\mu)} + O_{\alpha,\mu,J}(\varepsilon^{J+1}) \right), \quad (18.3)$$

where $\mathcal{E}_\ell^{(\mu)}$ is (6.2) with every κ_j (including $B = \kappa_2$) replaced by $\mu\kappa_j$, and $\mathcal{E}_\ell^{(\mu)} = O(\varepsilon^\ell)$. Moreover $\mu\kappa_j \sim \mu \frac{j!}{2} M^{j+1} / (\alpha s)^{j-1}$ and

$$\mathcal{E}_1^{(\mu)} \sim -\frac{3}{8\mu\alpha Ms}, \quad a_{\alpha,\mu}(n) = \frac{e^{\mu F(t_n) + nt_n}}{\sqrt{2\pi\mu B}} \left(1 - \frac{3 + o(1)}{8\mu\alpha Ms} \right). \quad (18.4)$$

The proof is that of Theorem 2.2 and Lemma 16.1: a consecutive block of length $\asymp M/s$ around M in any one layer gives (5.3) and (5.4), and the Taylor estimates are summed over the μ copies.

[**Merge note, batch 75.**] Each monomial of (6.2) has $\sum_j m_j - T/2 = -\ell$, so $\mathcal{E}_\ell^{(\mu)}(t) = \mu^{-\ell} \mathcal{E}_\ell(t)$: the μ -layer corrections are the one-layer ones at $t(n/\mu)$, divided by μ^ℓ . Source 75-06 states the remainder with the safe parameter ε ; the sharp remainder of Theorem 16.3 is proved only for $\mu = 1$.

Monotonicity and exact computation (source 75-06). For $\mu = 1$ and $n \geq 1$, $a_{\alpha,1}(n+1) > a_{\alpha,1}(n)$ by the injection of Section 9; convolution with a positive nondecreasing sequence preserves nondecrease, and the extra summand and the strict inequalities give eventual strict increase for every fixed μ . For integers $\alpha, \mu \geq 1$ the coefficients through degree N follow from the finite product: for each k , repeat μ times the descending update $A[n] \leftarrow A[n] + k^\alpha A[n-k]$ ($n = N, \dots, k$), which costs $O(\mu N^2)$ integer operations and $O(N)$ stored integers (an arithmetic count, not a bit-complexity bound). Logarithmic differentiation gives

$$n a_{\alpha,\mu}(n) = \mu \sum_{m=1}^n C_\alpha(m) a_{\alpha,\mu}(n-m), \quad (18.5)$$

with C_α as in (12.1), also as an identity of real formal power series; the alternating divisor sum can be badly conditioned in floating point. Source 75-06's `code/75-06-powers-checks.py` checks the product against (18.5) through $n = 120$ for $\alpha, \mu \in \{1, 2, 3\}$ and the two-layer product against the self-convolution through $n = 200$.

Theorem 18.3 (Threshold inversion for μ layers; source 75-06). *Let $N_{\alpha,\mu}(Y)$ be the least n with $\log a_{\alpha,\mu}(n) \geq Y$, and put*

$$w = W_0\left(\frac{Y}{\mu\alpha e}\right), \quad R_0 = 1 + w, \quad K_0 = \frac{Y}{\mu\alpha w} = e^{R_0}, \quad n_0 = \frac{\mu K_0^2}{2}.$$

For every fixed $R \geq 2$, as $Y \rightarrow \infty$,

$$N_{\alpha,\mu}(Y) = n_0 \left(1 + \sum_{j=2}^R \frac{\Delta_j(c)}{R_0^j} + O_{\alpha,\mu,R}(R_0^{-R-1}) \right), \quad (18.6)$$

with the polynomials of Table 5.

j	$\Delta_j(c)$	source 75-06's $d_j(u)$, $u = 6c$
2	$-2c$	$-u/3$
3	$-2c$	$-u/3$
4	$4c^2 - 2c$	$u(u-3)/9$
5	$\frac{6}{5}c^2 - 2c$	$u(u-10)/30$
6	$-2c - \frac{77}{5}c^2 - 8c^3$	$-u(20u^2 + 231u + 180)/540$
7	$-2c - 57c^2 - \frac{1336}{105}c^3$	$-u(668u^2 + 17955u + 3780)/11340$
8	$-2c - 138c^2 - \frac{6226}{35}c^3 + 16c^4$	$u(140u^3 - 9339u^2 - 43470u - 3780)/11340$

Table 5: Inverse coefficients. $\Delta_2, \dots, \Delta_6$ are those of (9.3) and (9.4); Δ_7 and Δ_8 are new. Source 75-06's $d_j(u)$ are unrelated to $d_\alpha(s)$.

Proof (source 75-06). Write $B(r) = \sum_{j \geq 1} P_j(c)r^{-j}$ and $K_\mu = K_0 e^h$. The formal inverse equation is $e^h(R_0 + h - 1 + B(R_0 + h)) = R_0 - 1$, that is (9.1) with $E = e^h$, or

$$h = -\log\left(1 + \frac{h + B(R_0 + h)}{R_0 - 1}\right).$$

It has a unique formal solution $h = O(R_0^{-2})$, since the right side has formal derivative $O(1/R_0)$ in h , and $N_{\alpha,\mu}/n_0 = e^{2h}$ gives Table 5. For the integer threshold, use a high finite truncation of Theorem 18.1: its smooth main term has n -derivative asymptotic to $\alpha r_\mu/K_\mu > 0$; moving

n by a large constant times n_0/R_0^{R+1} changes it by order K_0/R_0^R , enough to bracket the retained residual and the forward remainder; monotonicity then brackets the threshold between neighbouring integers, and the $O(1)$ rounding is absorbed. Only explicit smooth approximants are differentiated. $\square$

[Merge note, batch 75.] The intake recomputed $\Delta_2, \dots, \Delta_8$ by solving (9.1) with $P_1, \dots, P_8$ in exact rational arithmetic; they equal source 75-06's $d_j(6c)$. As in Section 9, the core $K(\log K - 1) = Y/(\mu\alpha)$ is, after taking logarithms with $w = \log K - 1$, the equation $w + \log w = \log(Y/(\mu\alpha e))$, an instance of the dominant-block theorem `p0:thm:lambert-core` of [25] on the principal branch; the corrections are its perturbed inversion (`p0:thm:perturbed-inversion`) and the integer step its staircase theorem. The principal branch W_0 inverts the growth; the lower branch W_{-1} parametrizes the moving boundary.

[Batch 100, 5 October 2026.] The next two inverse coefficients, Δ_9 and Δ_{10} , are in Proposition 21.1 (source 100-82's exact record, confirmed by source 75-06's program at order 10); that record also reproduces Δ_7 and Δ_8 .

Theorem 18.4 (Boundary profile and largest part for μ layers; source 75-06). *Let $\mathbb{P}_n$ be the weighted law on μ -layer objects of total size n , at the saddle of Theorem 18.2.*

1. *Fix distinct reals $x_1, \dots, x_d$ and layers $i_1, \dots, i_d$, and let $k_{n,j} = \lfloor M + Hx_j \rfloor$. The occupation indicators of these site-layer pairs converge jointly to independent Bernoulli variables with parameters $(1 + e^{x_j})^{-1}$; the same holds for any fixed collection of distinct site-layer pairs with convergent rescaled positions, including different layers at one site. A site $\lfloor yM \rfloor$ is occupied with probability tending to 1 for fixed $0 < y < 1$ and to 0 for fixed $y > 1$.*
2. *Let L_n be the largest occupied size in any layer, and b the large solution of*

$$\frac{\mu}{t} b^\alpha e^{-tb} = 1, \quad b = -\frac{\alpha}{t} W_{-1}\left(-\frac{t}{\alpha} \left(\frac{t}{\mu}\right)^{1/\alpha}\right). \quad (18.7)$$

Then $\mathbb{P}_n\{t_n(L_n - b) \leq x\} \rightarrow \exp(-e^{-x})$ for every real x , and $L_n/K_\mu \rightarrow 1 + 1/\alpha$ in probability; for the A022629 ensemble the limit is 2.

At $\mu = 1$ this is Theorem 2.4, and its proof is the proof given there, with the following changes (source 75-06). The local limit estimate of Section 5.3 survives deleting finitely many site-layer variables at sizes $O(M)$ (choose a longest surviving subinterval of the central block, still a fixed fraction of M/s long), or every variable above a cutoff m with $m/M \rightarrow \gamma > 1$ and deleted variance $O(M^2)$; the target may move by $o(\sqrt{\mu B})$. For the maximum, $tb \rightarrow \infty$ and $b/M \rightarrow (\alpha + 1)/\alpha$, the tail intensity is $\mu \sum_{k>m} p_k = (\mu/t)m^\alpha e^{-tm}(1 + O(1/(tm)) + t) + o(1) \rightarrow e^{-x}$ for $m = b + x/t$, and the no-tail probability is $\prod_{k>m} (1 - p_k)^\mu \rightarrow e^{-e^{-x}}$; the deleted tail has mean $O(M)$ and variance $O(M^2)$. The two scales differ for the reason given after Theorem 2.4: a site at yM with $y > 1$ is occupied with probability about $M^{-\alpha(y-1)}$, there are order M/s sites in an extreme window, and the balance occurs at $y = 1 + 1/\alpha$.

Finite-size checks (source 75-06). At $n = 500$ the first-corrected relative errors for $(\alpha, \mu) = (1, 2), (1, 3), (2, 1)$ are about $6.36 \cdot 10^{-6}$, $4.72 \cdot 10^{-6}$ and $1.76 \cdot 10^{-3}$. The less impressive last case is kept deliberately: finite- n behaviour depends on the parameters, and the theorems are not uniform small- n guarantees.

19 Research questions of sources 75-01 and 75-06

Both sources close with research directions, kept here with their own numbers; overlaps with Section 13 are marked, and questions answered in this report are re-scoped. They are proposed tasks, not claims that the problems are globally open.

Source 75-01.

- 75-01.1 Is 77 the exact log-concavity threshold for A022629?** Theorem 17.1 reduces this to effective analytic bounds plus a finite check, but its constants are implicit; an explicit minor-arc bound is the main bottleneck. For the other columns, find the minimal thresholds and their dependence on α .
- 75-01.2 How fast may the Jensen degree grow?** Find a nontrivial $d(n) \rightarrow \infty$ for which hyperbolicity is still guaranteed; this needs simultaneous control of high cumulants, remainders and Hermite root spacing.
- 75-01.3 The crossover as $\alpha \downarrow 0$** (overlaps project 3 and 48.2): connect the central limit regime near zero tilt to the moving edge, for example when $\alpha \log n = O(1)$.
- 75-01.4 Growing moment order** (overlaps project 4): the diagonal A292190 of A292189 is covered by no theorem here; determine where a continuum edge stops being accurate and maximizing partitions take over. [Batch 100, 5 October 2026.] The window $\alpha \asymp M^{1/3}/\log M$ is now treated (Section 22); the diagonal has $\Lambda \rightarrow \infty$ and is still not covered (question 100-93.5).
- 75-01.5 The subdominant Fourier sectors** (overlaps project 1): locate the noncentral saddle contributions, their phases and amplitudes, and explain the finite-size residuals at $\alpha = 2, 3$ (Section 16.3). Theorem 11.1, which source 75-01 did not have, supplies the action at each fixed resonance. [Batch 100, 5 October 2026.] The noncentral contributions are computed in the critical window of growing powers (Section 22); for the fixed- α residuals only an uncertified cell decomposition exists (Section 16.3).
- 75-01.6 Extend the weight class** (overlaps project 7 and 48.5), e.g. $k^\alpha(\log(k+1))^\beta$ or regularly varying weights, preserving the cumulant scale, the local expansion and eventual Jensen hyperbolicity.
- 75-01.7 Joint limit laws under norm weighting** (overlaps project 6, 48.3 and 48.4): number of parts and largest part, with the correct conditional fluctuation scales.
- 75-01.8 A validated inverse** (overlaps 48.7 and project 8): combine the tail bound below with interval evaluation and explicit constants in Theorem 16.3, returning the exact threshold or a certified pair of candidates without all earlier coefficients. Source 75-01's tail bound: for $b \geq 0$ and $k_0 + 1 \geq 2b/t$, $\sum_{k > k_0} k^b e^{-tk} \leq (k_0 + 1)^b e^{-t(k_0 + 1)} / (1 - e^{-t/2})$.
- 75-01.9 Formalize the finite algebra, then the analysis** (overlaps project 8): the coefficient recurrence, the Bernoulli-cumulant recursion, the formula for $\mathcal{E}_r$, the Lagrange coefficients and the formal inverse recurrence first; then the block bound, the sum-integral estimates and the local Fourier remainder. No such formalization is claimed.
- 75-01.10 The signed companion.** A022661, $\prod_{k \geq 1} (1 - kq^k)$, asks whether its coefficient at $n = 9$ is its only zero (Robert Israel, June 2, 2015) [23]. The positivity method does not settle it; competing phases make it a substantially different target.

Source 75-06.

- 75-06.1 Complete the lower-scale transseries** (overlaps project 2 and 48.1): determine the endpoint, discretization and determinant terms jointly; appending only $-\frac{1}{2}\log(2\pi B)$ would be incorrect.
- 75-06.2 Large-order growth and optimal truncation** (overlaps project 2): growth of $P_j(c)$ for fixed c , a Gevrey bound, control by the complex singularities of the boundary inverse, Borel summability and resurgence.
- 75-06.3 Weak norm weighting** (overlaps project 3 and 48.2): $\alpha = \alpha(n) \downarrow 0$ with $\alpha \log n$ bounded or slowly growing.
- 75-06.4 Growing moments and growing layer counts** (overlaps project 4): α or μ depending on n ; the width $M/(\alpha(s-1))$ may stop diverging, and diagonal entries of both arrays are not covered. *[Batch 100, 5 October 2026.]* For one layer, the window in which that width is of order $M^{2/3}$ is Section 22.
- 75-06.5 Slowly varying activities** (overlaps project 7 and 48.5): $k^\alpha \ell(k)$, persistence of the logistic and Gumbel laws, multiscale forward and inverse expansions.
- 75-06.6 Fluctuations in the number of parts** (overlaps project 6 and 48.3): a two-dimensional local theorem for $\prod(1 + zk^\alpha q^k)^\mu$; no conditional variance formula is asserted. *[Batch 100, 5 October 2026.]* In the critical window, for one layer, the regression variance $Q_\perp$ of the number of parts at least two is computed (Section 22.2); the two-dimensional local theorem is still open (question 100-93.7).
- 75-06.7 The full extreme point process** (overlaps project 5 and 48.4), including multiplicities across layers and joint convergence with the boundary occupations.
- 75-06.8 Quantitative error bounds and certified computation** (overlaps project 8 and 48.7), including when saddle evaluation beats the quadratic exact product.
- 75-06.9 Shape inequalities and discrete regularity**: eventual log-concavity, ratio monotonicity, higher finite-difference inequalities. *[Batch 75, 2 October 2026.]* Eventual strict log-concavity and fixed-degree Jensen hyperbolicity are proved for $\mu = 1$ by source 75-01 (Theorems 17.1 and 17.2); the rest stays open.
- 75-06.10 Large deviations for the norm**: a joint theory for size and log norm, with rate functions under uniform and norm-tilted measures, first uniformly in α on compact intervals.
- 75-06.11 Formal verification in ProveIt** (overlaps project 8): exact identities and rational reversion first; formalizing the elementary proof would give a checked resolution of the leading conjecture before the probability theory.
- 75-06.12 OEIS and bibliographic follow-through** (overlaps 48.8): formula updates with exact quantifiers, and a broader search (norm-weighted strict partitions, multiplicative statistics, variable-activity fermionic products) before any novelty claim; independent review is the next validation step.

20 Addendum (batch 77): a fifth independent source and slot multiplicities

Batch 77 of ProveIt’s incoming-reports intake delivered one more manuscript on this family: **source 77-51** (batch-77 manuscript 51), *Asymptotic expansions and inversion for fixed power weighted partitions*, dated October 1, 2026, author line “Research report”, archive `oeis-fixed-power-asymptotics-result.zip` (arrival commit 096ee7b87; 11-page PDF, not shipped). It names no ProveIt commit, records no repository search, and cites none of sources 40, 48, 75-01 and 75-06. It treats A022629 in detail and the two-parameter family

$$A_{\alpha}^{[\beta]}(q) = \prod_{k \geq 1} (1 + k^{\alpha} q^k)^{k^{\beta}}, \quad \alpha > 0 \text{ real, } \beta \geq 0 \text{ an integer,}$$

whose case $\beta = 0$ is this report’s $A_{\alpha}(q)$. Its symbols are translated by the third table of Section 1.5.

For this report, source 77-51 is a fifth independent proof of Proposition 3.2 for every fixed α , of Theorems 2.1 and 2.3 for $\alpha = 1$ (all orders) and for every fixed α (first two orders), and of Theorem 2.2 for every fixed α , with identical coefficients. Table 6 records them, following Section 1.4; they are not printed again. Its new material is Section 20.3 (slot multiplicities k^{β} , $\beta \geq 1$) and Section 20.4 (OEIS A266891), the resummed “thermal” forms of Proposition 20.1, a further citation (Section 20.5), and four research questions (Section 20.6). Material from this source is marked (source 77-51).

Stale statements and priority caveats. Source 77-51’s abstract and README present the proofs of the A022629 and A092484 conjectures as its contribution (“proves the logarithmic OEIS conjectures for A022629, A092484 and A266891”); in this report they are the fifth proofs, and only the A266891 statement is new. It makes no priority claim: “neither an exponentially complete transseries nor a literature-wide priority claim is asserted”; “Our proof does not depend on an absence-of-literature assertion”; its comparison with Gikunda’s thesis “neither discounts the thesis nor claims that no other source treats fixed tilts”. Its asymptotic constants “depend on fixed parameters and truncation order” and “have not been made effective”; uniformity at zero tilt and proof-assistant verification are not asserted, and “No external submission or publication” was made (its README).

Neighbouring reports of batch 77. `oeis-sequence-asymptotics/a301746-divisor-weighted-asymptotics`, written on the same day in the same style, applies the same exact-saddle sum and consecutive-block lemma to $\prod (1 + q^k)^{d(k)^2}$; its independent audit cites “the independently reviewed A022629 proof”, which is source 77-51’s audit (77-51-fp-ind-AUD IT.md here). `oeis-sequence-asymptotics/a291698-moving-fugacity-partitions` treats distinct partitions with one global fugacity $u = n^{\alpha}$ attached to every part, $[q^n] \prod_k (1 + uq^k)$, rather than the per-part weights k^{α} of this report: a different weighting, with no result in common.

20.1 Same results, a fifth independent derivation

[Merge note, batch 77.] The intake checked the coefficient agreements in exact rational arithmetic: with $P = \pi^2 = 6c$, source 77-51’s $d_3 = P/6 - P^2/72 = c - c^2/2$, $d_4 = P/6 + P^2/40 = c + \frac{9}{10}c^2$, $b_4 = P(P-3)/9 = 4c^2 - 2c$ and $b_5 = P(P-10)/30 = \frac{6}{5}c^2 - 2c$ are P_3 , P_4 , Δ_4 , Δ_5 at $\alpha = 1$, and its family theorem at $\beta = 0$ reproduces $P_1 = P_2 = c$ and $\Delta_2 = \Delta_3 = -2c$ for every

Result in this report	Source 77-51	Relation
Proposition 3.2 (two-term growth; the A022629 and A092484 conjectures)	Thm. 1.1, first display, and Section 2; Thm. 8.1 at $\beta = 0$	same lower-bound partition $1, \dots, m-1, m+d$; the upper bound by a cutoff sum is a fifth route (Section 20.2)
Strict monotonicity (Section 9)	Section 2, end	same injection; extended to $\beta \geq 1$ in Section 20.3
Lemma 4.1, (4.5)	Lemma 3.1	same conservative scales: $-F' \sim M^2/2$, $B \asymp M^3/s$, $ \kappa_j = O(M^{j+1}/s)$
Lemma 5.1 and the noncentral bounds	Lemma 3.2 and the arc bounds of Section 3	same inequality, same geometric-sum proof
Theorem 2.2, (6.2), (2.9)	Thm. 1.2, E_R , E_1 ; Thm. 8.1 at $\beta = 0$	same expansion with $J = R + 1$ ($E_R = 1 + \sum_{\ell \leq R} \mathcal{E}_\ell$); remainder $O(w^{-R-1})$, $w = M/s = 1/\varepsilon$: the conservative scale, <i>not</i> the sharp $O((\alpha Ms)^{-J})$ of Theorem 16.3
(2.5) and the first two orders of Theorem 2.1	Thm. 1.1, second display; Thm. 8.1 at $\beta = 0$	same, in resummed form (Proposition 20.1)
Theorem 2.1 at $\alpha = 1$, $P_1, \dots, P_4$	Thm. 6.1, $d_1, \dots, d_4$ printed, checked through d_6	identical: $d_j = P_j(\pi^2/6)$; same generation (Section 20.2)
Theorem 2.3 at $\alpha = 1$, $\Delta_2, \dots, \Delta_5$	Thm. 6.1, $b_2, \dots, b_5$ printed, checked through b_7 ; $m_* = K_0$, $L_* = R_0$	identical: $b_j = \Delta_j(\pi^2/6)$
Threshold localization (Theorem 16.4, source 75-01)	Section 7	weaker: N lies between the roundings of $x_R \mp C_R w^{-R}$, with x_R the inverse of the exact-saddle model; $N = x_0 + O(1)$
Recurrence (12.2)	Section 9	same (an “independent recurrence uses logarithmic differentiation”)
Gikunda and Granovsky–Stark comparisons (Section 1.2)	Section 9	the same distinctions, in other words; Naranjo–Ramírez is new (Section 20.5)

Table 6: Results of source 77-51 that this report already proves. Statement and section numbers are those of the delivered manuscript.

α (Proposition 20.1). Its relative errors (Table 7) are $a_1(n)/G_J - 1$, the reciprocal convention of Table 3; converted, they agree with that table at $n = 200, 1000$ and 2000 to within one unit in the last printed digit.

20.2 Second routes and forms that differ

Proposition 20.1 (Thermal forms; source 77-51). *For fixed $\alpha > 0$, with K, r, c as in (2.1) and K_0, R_0 as in (2.10),*

$$\log a_\alpha(n) = \alpha K(r-1) + \frac{\alpha c K}{r-1} + O_\alpha(K/r^3), \quad (20.1)$$

$$N_\alpha(Y) = \frac{K_0^2}{2} \left(1 - \frac{2c}{R_0(R_0-1)} + O_\alpha(R_0^{-4}) \right). \quad (20.2)$$

Here $\alpha c = \pi^2/(6\alpha)$; for $\alpha = 1$ source 77-51 writes $m(L-1) + Cm/(L-1)$ and $\frac{m_*^2}{2}(1 - \pi^2/(3L_*(L_*-1)))$, with $m = K, L = r, C = \pi^2/6, m_* = K_0 = Y/W_0(Y/e), L_* = R_0$.

Proof. Expanding $1/(r-1)$ and $1/(R_0(R_0-1))$ in powers of $1/r$ and $1/R_0$ shows that the two displays agree with Theorem 2.1 ($P_1 = P_2 = c$) and with (2.11) $(-2c, -2c)$ through the orders kept, so they follow from those theorems. Source 77-51 proves them directly (source 77-51): at $t = \alpha s/M$ the first-order case of the boundary smoothing (Lemma 7.1) gives $F(t) = \alpha M(s/2 - 1) + \pi^2 M/(6\alpha(s-1)) + O(M/s^3)$, through the inverse branch $x(v)$ of $tx - \alpha \log x$ with $x'(0) = M/(\alpha(s-1))$ and the kernel integral $\int_{\mathbb{R}} \log(1 + e^{-|v|}) dv = \pi^2/6$; the mean relation gives $M = K(1 + O(r^{-2}))$; replacing M by K in the hard exponent and in the thermal term costs $O(K/r^3)$; and the exact-saddle theorem adds only $O(r)$. For the inverse it substitutes $K = K_0\{1 - c/(R_0(R_0-1))\}$: the thermal term cancels the first variation of $\alpha K(\log K - 1)$ to error $O(K_0/R_0^3)$, and division by the slope $\sim \alpha R_0$ gives the relative error $O(R_0^{-4})$. $\square$

[Merge note, batch 77.] The resummed forms have the same order of accuracy as the two-term truncations of Theorem 2.1 and (2.11): the term $\alpha c K/r^3$ hidden in (20.1) is not $\alpha P_3(c)K/r^3$. They are compact displays, not sharper estimates. Source 77-51 calls (20.1) “a sharper thermal correction” relative to its first display, $\log a_n = m(L-1) + O(m/L)$. As in Section 9, the core $K(\log K - 1) = Y/\alpha$ is the case $a = b = 1$ of the Lambert-core theorem `p0:thm:lambert-core` of [25], after taking logarithms with $w = \log K - 1$.

An elementary upper bound by a cutoff sum (source 77-51). For $\alpha = 1$ choose $t = r/K$; positivity gives $\log a_1(n) \leq nt + F(t)$. With $g(x) = \log x - tx$ and $K' = \lfloor K \rfloor$, source 77-51 shows $F(t) = \sum_{k=1}^{K'} g(k) + O(K/r)$. For large K , concavity gives $g(k) \geq 0$ on $2 \leq k \leq K'$, and the $k = 1$ correction is bounded. On $2 \leq k \leq K/2$, $\log(1 + e^{-g(k)}) \leq e^{tk}/k \leq \sqrt{K}/k$, a total $O(\sqrt{K} \log K)$. On $K/2 < k \leq K$, $\log x \geq -2(1-x)$ for $1/2 \leq x \leq 1$ gives $g(k) \geq (r-2)(1 - k/K)$, and a geometric sum gives $O(K/r)$. For $k > K$, $\log(1 + e^{g(k)}) \leq (k/K)e^{-r(k-K)/K}$, with the same bound. Finally $nt + \sum_{k \leq K'} g(k) = \log K'! + t\{n - K'(K'+1)/2\} = K(r-1) + O(r)$. This is a discrete form of the positive-part argument of Section 3.1; source 77-51 adds that solving $K(\log K - 1) = Y$ exactly, with error $O(K/\log K)$ and slope $\log K$, proves the first inverse form “without saddle analysis”.

Coefficient generation in source 77-51’s variables (source 77-51). Source 77-51 generates d_j and b_j by the same two steps as Sections 7 and 8, at $\alpha = 1$. Its rational

functions $R_1(x) = 1/(x-1)$, $R_{j+1}(x) = (R_j(x) + xR'_j(x))/(x-1)$ are the derivatives U_j of (7.4) ($x^{(j)}(0) = MR_j(s)$); its series $T(L) = \sum_{j \geq 0} 2\eta(2j+2)R_{2j+1}(L)$ is the smoothing series of Lemma 7.1; and its stationarity equation for $\rho = M/K$ in powers of $h = 1/r$ is the elimination of the saddle. With $P = \pi^2$ it records

$$T(L) = \frac{P}{6(L-1)} + \frac{7P^2(2L+1)}{360(L-1)^5} + O(L^{-6}), \quad \rho = 1 - \frac{Ph^2}{6} - \frac{Ph^3}{6} + \frac{P^2h^4}{24} + O(h^5),$$

and its inverse equation $\tau[h^{-1} + \log \tau - 1 + D((h^{-1} + \log \tau)^{-1})] = h^{-1} - 1$, $D(h) = \sum_j d_j h^j$, $\tau^2 = 1 + \sum_{j \geq 2} b_j h^j$, is (9.1) with $E = \tau$, $h = 1/R_0$. A separately written rational-series checker agrees through d_6 and b_7 ; the formal series are justified, as here, by truncation and the implicit function theorem, “without interchanging infinite asymptotic series”. No convergence of either formal hierarchy is asserted.

Inverses of the exact-saddle model (source 77-51). Source 77-51 inverts $A_R(x) = e^{F(t_x) + xt_x} (2\pi B(t_x))^{-1/2} (1 + \sum_{\ell \leq R} \mathcal{E}_\ell(t_x))$, whose logarithmic slope is asymptotic to t_x , and places the threshold between the roundings of $x_R(Y) \mp C_R w^{-R}$, $w = M/s$. It “does not prescribe an exact rounding choice when $x_R(y)$ is very close to an integer”, and “the asymptotic constants are not supplied as effective numerical interval bounds”. Theorem 16.4 (source 75-01) is the sharper statement; the two-ceiling step is the staircase theorem `p0:thm:staircase` of [25].

20.3 Slot multiplicities k^β

This subsection is source 77-51’s Section 8, translated (its $a_n^{(\alpha, \beta)}$, $F_{\alpha, \beta}$, d , m_0 , L_0 , ℓ , $\mathcal{W}$, u_* , m_* , L_* are written as below). Fix $\alpha > 0$ and an integer $\beta \geq 0$, and let

$$A_\alpha^{[\beta]}(q) = \prod_{k \geq 1} (1 + k^\alpha q^k)^{k^\beta} = \sum_{n \geq 0} a_\alpha^{[\beta]}(n) q^n. \quad (20.3)$$

Here k^β is the number of labelled slots of size k , each used at most once, and a used slot of size k has weight k^α . The coefficients are nonnegative real weights; for integer α they are coloured counts. Fractional slot multiplicities are outside the theorem.

[Merge note, batch 77.] For $\beta = 0$ this is A_α . For $\beta \geq 1$ it lies between the two products that Section 1.1 distinguishes: several parts of one size may occur, as in $\prod (1 + q^k)^{k^\alpha}$, but each carries the internal weight k^α , as in A_α . It is also not the convolution power A_α^μ of Section 18, which has a constant number μ of slots at every size. The notation $a_\alpha^{[\beta]}$ is chosen so as not to collide with $a_{\alpha, \mu}$.

Theorem 20.2 (Slot multiplicities; source 77-51). Put $K^{[\beta]} = ((\beta + 2)n)^{1/(\beta+2)}$ and $r^{[\beta]} = \log K^{[\beta]}$. Then

$$\log a_\alpha^{[\beta]}(n) = \alpha(K^{[\beta]})^{\beta+1} \left(\frac{r^{[\beta]}}{\beta+1} - \frac{1}{(\beta+1)^2} \right) + \frac{\pi^2 (K^{[\beta]})^{\beta+1}}{6\alpha(r^{[\beta]} - 1)} + O_{\alpha, \beta}((K^{[\beta]})^{\beta+1}/(r^{[\beta]})^3). \quad (20.4)$$

Define the exact saddle t_n and the cumulants $\kappa_j^{[\beta]}$, $B^{[\beta]} = \kappa_2^{[\beta]}$ from $F^{[\beta]}(t) = \log A_\alpha^{[\beta]}(e^{-t})$ as in Section 4, and s , $M = e^s$ by $t = \alpha s/M$ as in (2.7). With $\mathcal{W} = M^{\beta+1}/s$, for every fixed $R \geq 0$,

$$a_\alpha^{[\beta]}(n) = \frac{e^{F^{[\beta]}(t_n) + nt_n}}{\sqrt{2\pi B^{[\beta]}}} \left(1 + \sum_{\ell=1}^R \mathcal{E}_\ell^{[\beta]} + O_{\alpha, \beta, R}(\mathcal{W}^{-R-1}) \right), \quad (20.5)$$

where $\mathcal{E}_\ell^{[\beta]}$ is (6.2) with κ_j replaced by $\kappa_j^{[\beta]}$. For $Y = \log y$ set

$$u_* = \frac{(\beta+1)^2 Y / \alpha}{W_0((\beta+1)^2 Y / (\alpha e))}, \quad K_0^{[\beta]} = u_*^{1/(\beta+1)}, \quad R_0^{[\beta]} = \log K_0^{[\beta]}. \quad (20.6)$$

The least index $N_\alpha^{[\beta]}(Y)$ with $a_\alpha^{[\beta]}(n) \geq e^Y$ obeys

$$N_\alpha^{[\beta]}(Y) = \frac{(K_0^{[\beta]})^{\beta+2}}{\beta+2} \left[1 - \frac{(\beta+2)\pi^2}{6\alpha^2 R_0^{[\beta]}(R_0^{[\beta]} - 1)} + O_{\alpha,\beta}((R_0^{[\beta]})^{-4}) \right]. \quad (20.7)$$

The exact-saddle inverse has pre-rounding uncertainty $O_{\alpha,\beta,R}(\mathcal{W}^{-R-1}/t)$.

At $\beta = 0$, $K^{[0]} = K$, $r^{[0]} = r$, $\mathcal{W} = M/s = 1/\varepsilon$, and the theorem is Proposition 20.1 with Theorem 2.2.

Monotonicity (source 77-51). Choose the least-labelled selected slot at the largest occupied size k , and move it to size $k+1$ with the same label. The label exists because $(k+1)^\beta \geq k^\beta$, and no slot at the new largest size was occupied. The unique new largest slot records the inverse operation. The map is injective and multiplies its weight by $((k+1)/k)^\alpha > 1$. Thus the coefficient sequence is strictly increasing for $n \geq 1$, including noninteger α .

Uniform estimates for a fixed pair (source 77-51). To prove the estimates uniformly for a fixed pair (α, β) , take $0 < \delta < \min(1/2, (\beta+1)/(2\alpha))$, and let p_k be the Bernoulli parameter of a slot of size k , as in (4.1). For $k \leq \delta M$ and fixed $j \geq 0$,

$$k^{\beta+j} p_k (1 - p_k) \leq M^{\alpha\delta} k^{\beta+j-\alpha}.$$

Its sum is $O(M^{\max(\alpha\delta, \beta+j+1-\alpha+\alpha\delta)} \log M)$, and that exponent is less than $\beta+j+1$. This handles large α , for which a naive split at $M/2$ is insufficient. On $\delta M \leq k \leq M$,

$$\alpha \log k - tk \geq \alpha(s - 1/\delta)(1 - k/M),$$

so a geometric sum applies. Above M , use $p_k \leq (k/M)^\alpha e^{-\alpha s(k-M)/M}$. It follows that

$$\sum_{k \geq 1} k^{\beta+j} p_k (1 - p_k) = O_{\alpha,\beta,j}(M^{\beta+j+1}/s). \quad (20.8)$$

The active interval still has consecutive length comparable to M/s , but each size now has $k^\beta \asymp M^\beta$ independent slots. Thus $B^{[\beta]} \asymp M^2 \mathcal{W}$ and $\kappa_j^{[\beta]} = O_j(M^j \mathcal{W})$. The analytic-disk argument remains valid; in its upper tail the small quantity is $O(M^{-\alpha})$. The trigonometric exponent of Lemma 5.1 is multiplied by M^β , giving noncentral suppression $e^{-cM^{\beta+1}/s^3}$ (with an absolute constant c , not this report's $c = \pi^2/(6\alpha^2)$). The proof of the exact-saddle expansion then applies with $1/\varepsilon$ replaced by $\mathcal{W}$.

Thermal and inverse constants (source 77-51). Sum-integral errors are $O(M^\beta s)$ for $x^\beta \log(1 + x^\alpha e^{-tx})$ and $O(M^{\beta+1})$ for the mean integrand. These follow by integrating absolute derivatives, using (20.8). The hard integrals of the log generating function and of the mean are

$$F_{\text{hard}} = \alpha M^{\beta+1} \left(\frac{s}{(\beta+1)(\beta+2)} - \frac{1}{(\beta+1)^2} \right), \quad \frac{M^{\beta+2}}{\beta+2}.$$

Small-size and distant-tail contributions are smaller than every inverse-logarithmic order, by the δ split just proved. On the central branch $v = tx - \alpha \log x$,

$$x'(0) = \frac{M}{\alpha(s-1)}, \quad x''(0) = -\frac{M}{\alpha^2(s-1)^3}, \quad x'''(0) = \frac{M(2s+1)}{\alpha^3(s-1)^5}.$$

The even soft-cutoff kernel multiplies $x(v)^{\beta}x'(v)$, whose second derivative is $O(M^{\beta+1}/s^3)$. Its constant term gives

$$F^{[\beta]}(t) = F_{\text{hard}} + \frac{\pi^2 M^{\beta+1}}{6\alpha(s-1)} + O(M^{\beta+1}/s^3).$$

For the mean, the odd kernel multiplies $x(v)^{\beta+1}x'(v)$, whose first derivative is $O(M^{\beta+2}/s^2)$. Hence $M = K^{[\beta]}(1 + O((r^{[\beta]})^{-2}))$. The hard exponent has zero derivative at $K^{[\beta]}$ and second derivative $\alpha(r^{[\beta]} - 1)(K^{[\beta]})^{\beta-1}$, so the displacement costs $O((K^{[\beta]})^{\beta+1}/(r^{[\beta]})^3)$. This proves (20.4).

The main entropy $H(m) = \alpha m^{\beta+1}(\log m/(\beta+1) - 1/(\beta+1)^2)$ has derivative $\alpha m^{\beta} \log m$. With $u = m^{\beta+1}$, the equation $H(m) = Y$ is $u(\log u - 1) = (\beta+1)^2 Y/\alpha$, giving (20.6). The thermal term changes m by the relative amount $-\pi^2/[6\alpha^2 R_0^{[\beta]}(R_0^{[\beta]} - 1)] + O((R_0^{[\beta]})^{-4})$, proving (20.7). The exact-saddle inverse follows as before from a logarithmic slope asymptotic to t .

[Merge note, batch 77.] The core $u(\log u - 1) = (\beta+1)^2 Y/\alpha$ is, with $w = \log u - 1$, the equation $w + \log w = \log((\beta+1)^2 Y/(\alpha e))$: the case $a = b = 1$ of `p0:thm:lambert-core` of [25], on the principal branch, as for $\beta = 0$. No novelty is claimed for the inversion. Source 77-51 gives every inverse-logarithmic order only for A022629 ($\alpha = 1, \beta = 0$); for $\beta \geq 1$ only the two orders of (20.4) and (20.7) are proved.

20.4 The OEIS instances, and A266891

For $(\alpha, \beta) = (1, 0)$ the theorem gives A022629, and for $(2, 0)$ A092484, whose entry records the conjecture $\log a_n \sim \sqrt{2n}\{\log(2n) - 2\}$ [2]; both are proved above by Theorem 2.1.

Corollary 20.3 (OEIS A266891; source 77-51). *For $(\alpha, \beta) = (1, 1)$, that is for $\prod_{k \geq 1} (1 + kq^k)^k$ (A266891), with $K^{[1]} = (3n)^{1/3}$ and $r^{[1]} = \log K^{[1]}$,*

$$\log a_1^{[1]}(n) = (K^{[1]})^2 \left(\frac{r^{[1]}}{2} - \frac{1}{4} \right) + \frac{\pi^2 (K^{[1]})^2}{6(r^{[1]} - 1)} + O((K^{[1]})^2/(r^{[1]})^3). \quad (20.9)$$

Its leading term is exactly the A266891 conjecture $n^{2/3}(2\log(3n) - 3)/(4 \cdot 3^{1/3})$, credited to Kotešovec on 8 May 2018 [28]. These are specializations of one family theorem. Its constants are not asserted to be uniform as $\alpha \downarrow 0$ or $\beta \rightarrow \infty$ (source 77-51).

[Merge note, batch 77.] The intake computed the coefficients of $\prod_{k \geq 1} (1 + kq^k)^k$ by the finite product: 1, 1, 4, 13, 29, 81, 188, 456, 1030, 2405, 5295, 11611, 25246. The first ten equal the A266891 prefix recorded in source 77-51's `77-51-fp-ind-SOURCE_CHECK.md`; the intake did not re-fetch the OEIS entry. No other ProveIt file mentions A266891.

20.5 Related work and computational checks (source 77-51)

Naranjo and Ramírez [27] discuss A022629 explicitly in their combinatorial treatment of coloured partitions; source 77-51's proof “does not depend on an absence-of-literature assertion”. Its comparisons with Gikunda's thesis [10] (Proposition 4.2.5 assumes a shrinking tilt $a = O(r^\delta)$, which a fixed $\alpha > 0$ does not satisfy) and with Granovsky and Stark [14] (their framework

assumes internal weights at most one, so it cannot be applied directly with $k^\alpha > 1$) are those of Section 1.2.

[**Merge note, batch 77.**] Source 77-51 cites the thesis at a ResearchGate address (publication 402798449) and its supplementary source check at another (publication 402798310); this report keeps the DOI of [10].

Source 77-51's package contains proof-source snapshots, independent mathematical audits, and separate exact and numerical programs. One exact recurrence multiplies the finite product through the requested degree; an independent one uses logarithmic differentiation. The A022629 producer check verifies 2,001 exact coefficients, the first 43 listed terms, 1,999 strict inequalities and 2,000 factorial lower bounds. The independent core checker compares 251 coefficients by two methods and checks Gaussian corrections through $R = 4$ and 48 inverse-derivative identities; independent rational-series calculations reproduce $d_1, \dots, d_6$ and $b_2, \dots, b_7$. Table 7 gives its high-precision comparisons for A022629. They are not interval certificates or proof dependencies. The signed relative error is the exact coefficient divided by the approximant, minus one. The exact-saddle approximation is already useful in this range;

n	Leading saddle	With $\mathcal{E}_1$
50	$-1.75854 \cdot 10^{-2}$	$-8.75761 \cdot 10^{-4}$
200	$-6.85783 \cdot 10^{-3}$	$-2.59784 \cdot 10^{-5}$
1000	$-2.47368 \cdot 10^{-3}$	$-4.50475 \cdot 10^{-6}$
2000	$-1.60739 \cdot 10^{-3}$	$-1.92291 \cdot 10^{-6}$

Table 7: Relative errors $a_1(n)/G_J - 1$ for A022629, $J = 0, 1$, source 77-51.

the inverse-logarithmic hierarchy serves a different, much coarser scale. No finite truncation is guaranteed to improve every small- n value (source 77-51).

20.6 Research questions of source 77-51

Source 77-51 closes with four problems, kept with its own numbers; overlaps with Sections 13 and 19 are marked.

77-51.1 Effective constants for the Fourier remainder and the inverse brackets, so that a given large threshold can be located with certified numerical error (overlaps project 8, 48.7, 75-01.8 and 75-06.8).

77-51.2 The beyond-all-inverse-logarithmic contributions from discrete sum–integral corrections and noncentral Fourier arcs: the argument suppresses them in bounds and does not identify a complete exponential transseries (overlaps projects 1 and 2, 48.1, 75-01.5 and 75-06.1).

[Batch 77, 2 October 2026.] Theorem 11.1 (source 40) gives the action of each fixed resonance; the full contribution stays open.

[Batch 100, 5 October 2026.] The sum–integral corrections are identified by Theorem 21.2 (source 100-89); the noncentral arcs only in the critical window (Section 22).

77-51.3 The transition as the positive tilt tends to zero with n , connecting the fixed-tilt cutoff to the shrinking-tilt regime of [10]; the constants of the present proof are not uniform there (overlaps project 3, 48.2, 75-01.3 and 75-06.3).

77-51.4 Slot multiplicities beyond exact integer powers, under explicit regularity and lattice conditions. Arbitrary fractional exponents in (20.3) need not retain coefficient positivity and are not covered by the Bernoulli interpretation (related to project 7, 48.5, 75-01.6 and 75-06.5, which vary the weight rather than the number of slots).

Source 77-51 establishes its displayed statements by ordinary analysis. Finite computations corroborate the formulas, but “no numerical fit, exhaustive test or proof-assistant verification is used to infer the asymptotic theorems”.

21 Addendum (batch 100): a smooth boundary sector and exact-range inversion

Batch 100 of ProveIt’s incoming-reports intake delivered three more manuscripts on this family. They are Research Reports 82, 89 and 93 of the session bundle that arrived in commit 60f54ea06 and was placed in 36571ae0e, and they form one author series: all three carry the author line “Research note prepared for Vladimir Reshetnikov with OpenAI” (also in their PDF metadata), and each builds on the one before.

- **Source 100-82** (Report 82), *Asymptotics and Inverse Growth for Weighted Distinct Partitions* (subtitle: the sequence OEIS A022629), 1 October 2026, 7 pages: $\alpha = 1$ only; a relative equivalent at the saddle of the *integral* with the explicit constant $\frac{1}{2} \log(2\pi) - 1$, all logarithmic orders, and the Lambert- W inverse. Its JSON record carries ten forward and nine inverse coefficients (inverse orders 2–10).
- **Source 100-89** (Report 89), *Relative Expansions and Exact Range Inversion for Power Weighted Distinct Partitions*, 1 October 2026, 9 pages: every real $\alpha > 0$, uniformly on compact sets; the lattice-minus-integral difference is smooth at $t = 0$, with explicit Taylor coefficients; an all-orders relative expansion with the exact integral in the exponent; an all-orders exact-range inverse. Its README calls source 100-82 “the earlier A022629 report”.
- **Source 100-93** (Report 93), *Critical Theta Asymptotics and Inversion for Power Weighted Distinct Partitions*, 2 October 2026, 13 pages: α growing with n in a critical window. It calls source 100-89 its “fixed-power companion” and pins it by the SHA-256 of its archive (225b2070...), which matches the delivered archive).

Sources 100-82 and 100-89 name no ProveIt commit and record no repository search. Source 100-93 names the ProveIt commit 6591a56b3 (1 October 2026, 18:55 PDT), which predates the placement 9df4ba51a (19:13 PDT) of this report’s first two sources; none of the three saw this report. They are called by their bundle report numbers, because the cluster-local numbers 01–03 of batch 100 collide with those of other batch-100 clusters.

This section merges sources 100-89 and 100-82. Source 100-89 is the base: it has the weaker hypotheses (every real $\alpha > 0$), and source 100-82 is its case $\alpha = 1$. Section 22 is source 100-93. For this report, sources 100-82 and 100-89 are the sixth and seventh proofs of the A022629 conjecture, and source 100-89 is the sixth for A092484 and A265840–A265842 (sources 40, 48, 75-01, 75-06 and 77-51 treat every real $\alpha > 0$). Their shared results are recorded once, in Table 8, and not reprinted. Their new material is: the two coefficients P_9, P_{10} and the two inverse coefficients Δ_9, Δ_{10} (Proposition 21.1); the smooth boundary sector and its constants (Section 21.2); the all-orders relative expansion with the exact integral and its first-order form at the integral saddle (Section 21.3); and an all-orders exact-range inverse (Section 21.4).

Material from these sources is marked (source 100-82), (source 100-89) or (source 100-93); text added in this merge is marked **[Merge note, batch 100.]**; dated notes on earlier text read *[Batch 100, 5 October 2026.]*; proofs and statements supplied by this write rather than by a source are marked [write]. Their notation is translated by the fourth table of Section 1.5.

21.1 Same results, a sixth and a seventh derivation

Result in this report	Source 100-89	Source 100-82	Relation
Proposition 3.2 and the conjectures (1.3)	Section 7, (30), by minimizing the phase	Theorem 1.2 (the case $R = 0$)	same leading statement; 100-89 states only the equivalence $\sim$
Strict monotonicity (Section 9)	proof of Theorem 6.1	Section 6	same injection (source 100-93's Lemma 6.2 is a third copy)
Lemma 4.1, (4.5)	(19)	Lemma 3.1 (integral moments, $\sim M^{r+1}/s$)	same scales; 100-82's integral form by the same logistic rescaling
Lemma 5.1, (5.3), (5.4)	(21)–(23)	(10)–(11)	same bounds; a third proof of the block inequality (Section 21.5)
Theorem 2.2, (6.4), (2.9), (6.3)	(24), its $\mathcal{P}_M$ of (4) and the next block	—	identical polynomial ($\mathcal{P}_M = 1 + \sum_{\ell \leq M} \mathcal{E}_\ell$); remainder $O(t^{M+1})$, the conservative scale, <i>not</i> the sharp one of Theorem 16.3
(5.6)	Section 7	Section 5	same
Lemma 7.1, (7.14)	(31)	Lemma 4.1 ($\alpha = 1$)	same expansion, with $2\eta(2j)$ and the operator $\mathcal{T}$ (Section 21.5)
(8.2), (8.3)	Section 7	(13)–(14)	same elimination, as a minimized phase (Section 21.5)
Theorem 2.1, $P_1, \dots, P_8$	Section 7, $P_1, \dots, P_4$	Theorem 1.2, $s_1, \dots, s_4$; JSON	identical: $s_j = P_j(\pi^2/6)$; P_9, P_{10} new
(2.5)	Section 7	$s_1, \dots, s_{10}$	same
Theorem 2.3, (9.3), Table 5	Section 7 (leading term)	Theorem 1.2 Theorem 1.3, $v_2, \dots, v_4$; JSON	identical: $v_j = \Delta_j(\pi^2/6)$; Δ_9, Δ_{10} new
		$v_2, \dots, v_{10}$	
Theorem 16.4	Theorem 6.1	Section 6, the resummed inverse	100-89: all orders at sequence values (Theorem 21.8); 100-82: weaker
Two Lambert branches (Section 9)	Section 7	Theorem 1.3	same principal branch
Recurrence (12.2)	its numerical check	—	same
Gikunda,	Section 7.2	Section 7	same distinctions; three citations new
Granovsky–Stark,			(Section 21.7)
Naranjo–Ramírez			
(Section 1.2,			
Section 20.5)			

Table 8: Results of sources 100-89 and 100-82 that this report already proves. Statement, section and equation numbers are those of the delivered manuscripts.

[**Merge note, batch 100.**] The intake compared the coefficients in exact rational arithmetic. Source 100-82’s JSON record `data/100-82-integral-check_formal_series.json` gives s_j and v_j as polynomials in $P = \pi^2$ at $\alpha = 1$. With $P = 6c$ they equal $P_1, \dots, P_8$ of (2.3) and (8.5) and $\Delta_2, \dots, \Delta_8$ of Table 5; source 100-89’s four printed terms are $P_1, \dots, P_4$. The substitution is legitimate for every α : the generation is algebraic in the factors $2\eta(2j)/\alpha^{2j}$, which are rational multiples of $(6c)^j$ (source 100-89’s README makes the same point, “ $\pi^2 \rightarrow \pi^2/a^2$ ”).

Proposition 21.1 (Two further forward and inverse coefficients; source 100-82, checked here). *The next coefficients of (2.2) and of (9.3) (equivalently of (18.6)) are*

$$P_9(c) = c + \frac{1162}{5}c^2 + \frac{9305}{3}c^3 + \frac{3668017}{1050}c^4 + \frac{7}{8}c^5, \quad (21.1)$$

$$P_{10}(c) = c + \frac{1926}{5}c^2 + \frac{51747}{5}c^3 + \frac{11264369}{350}c^4 + \frac{2861817}{440}c^5, \quad (21.2)$$

$$\Delta_9(c) = -2c - 276c^2 - 1344c^3 - \frac{37148}{105}c^4, \quad (21.3)$$

$$\Delta_{10}(c) = -2c - \frac{2459}{5}c^2 - \frac{18455}{3}c^3 - \frac{227819}{35}c^4 - 32c^5. \quad (21.4)$$

Proof. These are exact computations from the all-orders algorithm of Sections 8 and 9. Source 100-82 prints only $s_1, \dots, s_4$ and $v_2, \dots, v_4$ in its article; the values above are its JSON record’s s_9, s_{10}, v_9, v_{10} (“ten completed forward and inverse orders”, with saddle and reversion residuals checked through order twelve) after $P = 6c$. [write] Independently, this write reran source 75-06’s program `code/75-06-powers-formal.py` on a copy with `--order 10`: it generates the coefficients by a different construction, in exact arithmetic over $\mathbb{Q}[u]$ with $u = 6c$ for every α , and its forward coefficients through order 10 and inverse coefficients through order 10 agree exactly with source 100-82’s after $P = 6c$ (that program asserts only its first eight against its own table; the orders 9 and 10 are its output, not an internal check). $\square$

The coefficient structure of Section 15.2 continues: $P_9(0) = P_{10}(0) = 0$ and $\deg_c P_9 = \deg_c P_{10} = 5 = \lfloor (j+1)/2 \rfloor$, the bound attained. At $\alpha = 1$ these are source 100-82’s $s_9 = \frac{P}{6} + \frac{581P^2}{90} + \frac{9305P^3}{648} + \frac{3668017P^4}{1360800} + \frac{7P^5}{62208}$ and its s_{10}, v_9, v_{10} at $P = \pi^2$.

Stale statements and priority caveats. Source 100-82’s abstract says its logarithmic expansion “proves and strengthens the conjecture currently listed in OEIS A022629”, and source 100-89’s that its corollary “settles the current conjectures in OEIS A092484 and A265840–A265842”. Both were true of the entries as consulted (the intake did not re-fetch them) and blind to this report; here they are further proofs, and only the material listed above is new. Source 100-89’s “The closest located source is Gikunda’s 2026 thesis” likewise predates this report. Both keep explicit caveats, kept here: source 100-82, “The bounded literature check is not an exhaustive priority determination”, “No submission or publication has been made”; source 100-89, “The literature comparison below is bounded and makes no exhaustive priority claim”, and “These specific comparisons are not an exhaustive priority search”. Both call themselves “unrefereed ordinary mathematics, not a Lean formalization” and credit Euler–Maclaurin summation, cumulant expansions, saddle inversion, and the probabilistic local-limit architecture as classical.

21.2 The smooth boundary sector

This subsection is source 100-89’s Sections 1–3, translated. Throughout, $\mathfrak{A} \subset (0, \infty)$ is an arbitrary fixed compact set, and uniform assertions in α refer to $\alpha \in \mathfrak{A}$. For $t > 0$ let

$$I_\alpha(t) = \int_0^\infty \log(1 + x^\alpha e^{-tx}) dx, \quad \Xi_\alpha(t) = F(t) - I_\alpha(t), \quad (21.5)$$

so Ξ_α is the lattice-minus-integral difference, and write $\varpi(x) = \alpha \log x - tx$ for the log odds, so that $p(x) = 1/(1 + e^{-\varpi(x)})$. Section 7 bounds it by $O_\alpha(s)$ and stresses that it “must not be discarded” when a multiplicative equivalent is wanted; source 100-89 identifies it.

Theorem 21.2 (Smooth matched boundary; source 100-89). Ξ_α has a C^∞ extension from $t > 0$ to $t = 0$, uniformly with every fixed derivative for $\alpha \in \mathfrak{A}$. Its Taylor coefficients $\xi_r(\alpha) = \Xi_\alpha^{(r)}(0)/r!$ are real analytic on $(0, \infty)$ and extend holomorphically to $\Re \alpha > 0$. For every fixed $D \geq 0$,

$$\Xi_\alpha(t) = \sum_{r=0}^D \xi_r(\alpha) t^r + O_{\mathfrak{A},D}(t^{D+1}). \quad (21.6)$$

Smoothness does not imply that the Taylor series $\sum_r \xi_r(\alpha) t^r$ converges, and no convergence is asserted (source 100-89). The holomorphy is asserted by source 100-89 with a sketch; Lemma 21.3 proves it.

Proof of the smoothness (source 100-89). Put $f(x; t) = \log(1 + x^\alpha e^{-tx})$ and $\chi_r(x; t) = \partial_t^r f(x; t)$. Choose an even integer $m_E \geq 2$ with $m_E > r + 1$. For $t > 0$, Euler–Maclaurin summation from $x = 1$ gives

$$\begin{aligned} \Xi_\alpha^{(r)}(t) &= \frac{\chi_r(1; t)}{2} - \int_0^1 \chi_r(x; t) dx - \sum_{j=1}^{m_E/2} \frac{B_{2j}}{(2j)!} \partial_x^{2j-1} \chi_r(1; t) \\ &\quad - \frac{1}{m_E!} \int_1^\infty \tilde{B}_{m_E}(x) \partial_x^{m_E} \chi_r(x; t) dx, \end{aligned} \quad (21.7)$$

with the Bernoulli numbers B_{2j} and the periodic Bernoulli polynomial $\tilde{B}_{m_E}$. All boundary terms at infinity vanish for $t > 0$, and differentiation in t is legitimate by exponential decay. With the Bernoulli cumulant polynomials b_r of (4.7) and $p = p(x)$,

$$\chi_r(x; t) = (-x)^r b_r(p), \quad r \geq 1.$$

For $r \geq 2$, b_r is divisible by $p(1-p)$; for $r = 1$, $b_1 = p$, but every positive spatial derivative of p has that factor, and since $m_E > r$ every Leibniz term differentiates the cumulant at least once. Using $\varpi'(x) = \alpha/x - t$ and its derivatives,

$$|\partial_x^{m_E} \chi_r(x; t)| \leq C x^{r-m_E} (1+tx)^{m_E} e^{-|\varpi(x)|}, \quad x \geq 1, r \geq 1, \quad (21.8)$$

with C uniform on $\mathfrak{A}$: the dominant $-x$ of χ_1 is killed by $m_E \geq 2$. For $r = 0$ the term with no derivative on p is $p \partial_x^{m_E} \varpi = O(x^{-m_E})$ and the others carry $p(1-p)$, so

$$|\partial_x^{m_E} f(x; t)| \leq C x^{-m_E} + C t^{m_E} e^{-|\varpi(x)|}, \quad (21.9)$$

the limiting logarithmic growth has not been discarded. On $1 \leq x \leq 1/t$ the right side of (21.8) is at most $C x^{r-m_E}$, integrable since $m_E > r + 1$, and the derivative converges pointwise to its value at $t = 0$. On the rest, uniformly on $\mathfrak{A}$,

$$\int_{1/t}^\infty x^r e^{-|\varpi(x)|} dx = O((t^{-1} \log(1/t))^{r+1}) : \quad (21.10)$$

use $e^{-|\varpi|} \leq 1$ up to $T = C \mathfrak{A} t^{-1} \log(1/t)$, with the constant chosen so that $x^\alpha \leq e^{tx/2}$ for $x \geq T$, and $e^{-|\varpi(x)|} \leq e^{-tx/2}$ beyond. Multiplied by t^{m_E} this is $o(1)$. This proves uniform L^1 convergence of the derivative in (21.7) for $r \geq 1$, and (21.9) does so for $r = 0$. The local terms and the integral over $[0, 1]$ converge uniformly because α stays away from zero. Hence every $\Xi_\alpha^{(r)}(t)$ has a finite uniform limit as $t \downarrow 0$; the fundamental theorem of calculus, applied successively, gives the smooth extension, and boundedness of the next derivative gives (21.6). $\square$

Let $\mathfrak{L}_{m_E}[h]$ denote the right side of (21.7) with $\chi_r(\cdot; t)$ replaced by a function h . Then

$$\xi_r(\alpha) = \frac{\mathfrak{L}_{m_E}[\chi_r(\cdot; 0)]}{r!}, \quad m_E > r + 1, \quad (21.11)$$

a convergent formula, independent of m_E (source 100-89). For complex α with $\Re \alpha > 0$, source 100-89 uses the determinations $\alpha \log x + \log(1 + x^{-\alpha})$ on $x > 1$ and $\log(1 + x^\alpha)$ on $x < 1$, which match at $x = 1$, and states that the spatial derivatives in (21.11) have locally uniform integrable bounds, so that the coefficients are holomorphic on $\Re \alpha > 0$. “This statement concerns the limiting coefficients; it does not assume that the positive- t product is zero-free in a fixed complex neighborhood of a .” The bounds are asserted, not proved; the following lemma supplies them.

Lemma 21.3 (Holomorphy of the boundary coefficients; [write]). *For every $r \geq 0$, the right side of (21.11) is holomorphic in α on $\Re \alpha > 0$.*

Proof. Fix a compact set $\mathfrak{K} \subset \{\Re \alpha > 0\}$ and put $\sigma_{\mathfrak{K}} = \min_{\mathfrak{K}} \Re \alpha > 0$. For $x > 0$, $|x^\alpha| = x^{\Re \alpha}$ and $\arg x^\alpha = \Im \alpha \log x$, so $x^\alpha = -1$ would need $x = 1$, where $x^\alpha = 1$; hence $1 + x^{\pm \alpha} \neq 0$. On $x \geq 1$ write $y = x^{-\alpha}$, so $|y| \leq 1$ with equality only at $x = 1$, where $y = 1$. The continuous function $|1 + y|$ on $[1, X] \times \mathfrak{K}$ is positive and tends to 1 as $x \rightarrow \infty$ uniformly on $\mathfrak{K}$; hence $|1 + y| \geq \rho_{\mathfrak{K}} > 0$ on $[1, \infty) \times \mathfrak{K}$. Likewise $|1 + x^\alpha| \geq \rho_{\mathfrak{K}}$ on $(0, 1] \times \mathfrak{K}$, after decreasing $\rho_{\mathfrak{K}}$.

For $r \geq 1$, $\chi_r(x; 0) = (-x)^r b_r(p)$ with $p = 1/(1 + y)$, and for $r = 0$ the determination above is $\alpha \log x + \log(1 + y)$ on $x \geq 1$. Use the Euler operator $x \partial_x$. Then $(x \partial_x)y = -\alpha y$, so for a function G holomorphic on a neighbourhood of $\{|1 + y| \geq \rho_{\mathfrak{K}}, |y| \leq 1\}$, $(x \partial_x)^i G(y) = (-\alpha)^i (y \partial_y)^i G(y)$, and $\partial_x^m = x^{-m} \sum_{i=1}^m \sigma_{m,i} (x \partial_x)^i$ for $m \geq 1$, with integer coefficients $\sigma_{m,i}$ (signed Stirling numbers of the first kind; there is no $i = 0$ term). Every function $(y \partial_y)^i G$ with $i \geq 1$ is y times a function bounded on that region, so it is $O(|y|) = O(x^{-\sigma_{\mathfrak{K}}})$. By Leibniz’s rule applied to $x^r \cdot b_r(1/(1 + y))$, every term of $\partial_x^{m_E} \chi_r(x; 0)$ is either x^{r-m_E} times some $(y \partial_y)^i G$ with $i \geq 1$, or, from the terms in which all derivatives fall on x^r , a multiple of $x^{r-m_E} b_r(p)$ with $m_E \leq r$, which does not occur because $m_E > r + 1$. Hence

$$|\partial_x^{m_E} \chi_r(x; 0)| \leq C_{\mathfrak{K}} x^{r-m_E-\sigma_{\mathfrak{K}}}, \quad x \geq 1, \quad \alpha \in \mathfrak{K},$$

and for $r = 0$ the extra term $\partial_x^{m_E}(\alpha \log x) = \alpha(-1)^{m_E-1}(m_E - 1)! x^{-m_E}$ is integrable because $m_E \geq 2$. The integrand of the last term of (21.7) is therefore dominated, uniformly on $\mathfrak{K}$, by an integrable function, and it is holomorphic in α for each x ; by dominated convergence and Morera’s theorem the integral is holomorphic. The boundary terms at $x = 1$ are finitely many derivatives of functions holomorphic in (x, α) near $\{1\} \times \mathfrak{K}$. On $(0, 1]$ the integrand $\chi_r(x; 0)$ is continuous, bounded on $(0, 1] \times \mathfrak{K}$ (since $|x^\alpha| \leq 1$ and $|1 + x^\alpha| \geq \rho_{\mathfrak{K}}$; for $r = 0$ the principal logarithm of $1 + x^\alpha$ is holomorphic there), and holomorphic in α , so $\int_0^1$ is holomorphic as well. For real α both determinations are the real logarithm, so the holomorphic function restricts to $r! \xi_r(\alpha)$, which is therefore real analytic. $\square$

Closed forms (source 100-89). For $\alpha > 1$, separating $\alpha \log x$ from $\log(1 + x^{-\alpha})$, applying Stirling’s formula and $\int_0^\infty \log(1 + x^{-\alpha}) dx = \pi \csc(\pi/\alpha)$ (integration by parts) gives

$$\xi_0(\alpha) = \frac{\alpha}{2} \log(2\pi) + \sum_{k \geq 1} \log(1 + k^{-\alpha}) - \pi \csc(\pi/\alpha). \quad (21.12)$$

At $\alpha = 2$, Euler’s product for $\sinh$ gives $\xi_0(2) = \log(1 - e^{-2\pi})$; and $\xi_0(1) = \frac{1}{2} \log(2\pi) - 1$, the continued value at the removable resonance, or directly from Stirling’s formula for $\log(1 + x)$. For $\alpha > 2$, $\chi_1(x; 0) = -x + x/(1 + x^\alpha)$ and $\mathfrak{L}[-x] = \frac{1}{12}$, so

$$\xi_1(\alpha) = \frac{1}{12} + \sum_{k \geq 1} \frac{k}{1 + k^\alpha} - \frac{\pi}{\alpha} \csc(2\pi/\alpha), \quad \xi_1(1) = \frac{7}{12} - \gamma_E, \quad \xi_1(2) = \frac{1}{12} - \Re \psi(1+i), \quad (21.13)$$

with Euler's constant γ_E and $\psi = \Gamma'/\Gamma$; the two separately divergent pieces of (21.13) are not evaluated individually at $\alpha = 2$. For $r \geq 2$ and $\alpha > r + 1$,

$$\xi_r(\alpha) = \frac{1}{r!} \left\{ \sum_{k \geq 1} (-k)^r b_r \left(\frac{k^\alpha}{1 + k^\alpha} \right) - \frac{\pi}{\alpha} \left(\frac{r+1}{\alpha} \right)^{r-1} \csc \left(\frac{\pi(r+1)}{\alpha} \right) \right\} : \quad (21.14)$$

with $v = (r+1)/\alpha \in (0, 1)$ and $y = x^\alpha$, integrating by parts $r - 2$ times reduces b_r to $b_2 = y/(1+y)^2$ and gives $\int_0^\infty x^r b_r(x^\alpha/(1+x^\alpha)) dx = \frac{(-v)^{r-2}}{\alpha} \frac{\pi v}{\sin \pi v}$; multiplication by $(-1)^r$ gives (21.14). Formula (21.11) continues it through all apparent positive- α resonances, so these poles create no fractional or logarithmic powers in $\Xi_\alpha(t)$.

A finite algorithm for integer powers (source 100-89). For $\alpha = p \in \mathbb{N}$ and $r \geq 1$, $\chi_r(x; 0) = (-x)^r b_r(x^p/(1+x^p))$ is rational, with poles among $z^p = -1$. Polynomial division and partial fractions, followed by

$$\begin{aligned} \mathfrak{L}[x^\ell] &= \zeta(-\ell) \quad (\ell \geq 0), & \mathfrak{L}[(x-z)^{-1}] &= \log(-z) - \psi(1-z), \\ \mathfrak{L}[(x-z)^{-j}] &= \zeta(j, 1-z) - \frac{(-z)^{1-j}}{j-1} \quad (j \geq 2), \end{aligned} \quad (21.15)$$

compute $r! \xi_r(p)$ exactly. The first is Faulhaber's formula inside the Euler–Maclaurin functional; the others follow from finite digamma sums and convergent Hurwitz zeta sums. Principal logarithms are used and conjugate roots are paired; no root is positive real, and $\Re(1-z) > 0$. A finite constant formula, valid also for $p = 1$, is

$$\xi_0(p) = \frac{p}{2} \log(2\pi) - \sum_{z^p=-1} \log \Gamma(1-z) + \sum_{z^p=-1} z \{1 - \log(-z)\}, \quad (21.16)$$

from Stirling's formula for each $\log(x-z)$ in $\log(1+x^p)$, whose functional is $\frac{1}{2} \log(2\pi) - \log \Gamma(1-z) + z - z \log(-z)$. For example,

$$\xi_2(1) = -\frac{1}{24} + \frac{3\gamma_E}{2} - \frac{\pi^2}{12}, \quad \xi_3(1) = -\frac{19}{36} - \frac{8\gamma_E}{3} + \frac{\pi^2}{4} - \frac{\zeta(3)}{3}.$$

The delivered symbolic checkers implement this functional through order six for $p = 1$ and reconstruct four square-weight rational cumulants at the poles $\pm i$.

[Merge note, batch 100.] The intake confirmed the six printed constants independently, by fitting a polynomial in t to $\Xi_\alpha(t)$ computed directly (40-digit mpmath sums and quadrature) at $t = 0.0075, \dots, 0.04$: $\xi_0(1), \xi_1(1), \xi_0(2), \xi_1(2)$ agree to 10–12 significant digits, $\xi_2(1)$ to 9 and $\xi_3(1)$ to 6 (the fit's own truncation error). [write] The write repeated this and also compared $\Xi_\alpha(t)$ at $t = 0.0025$ and 0.00125 with (21.12)–(21.13) at $\alpha = 1.5, 2.5, 3$, evaluating the slowly convergent sums with Hurwitz zeta tails: $(\Xi_\alpha(t) - \xi_0)/t$ settles (at $\alpha = 1.5$ near -0.00477) and $(\Xi_\alpha(t) - \xi_0 - \xi_1 t)/t^2$ settles at $\alpha = 2.5, 3$. Plain `mpmath.nsum` is not accurate enough for these sums near $\alpha = 2$ (it was off by about 0.01 in $\xi_1(2.5)$ and $\xi_0(1.5)$); the delivered programs use it only at $\alpha = 3$, where it is accurate. These are numerical confirmations, not certificates.

21.3 Relative expansions at the integral saddle

This subsection is source 100-89's Theorem 1.2, Corollary 1.3 and Sections 4–5, with source 100-82's Theorem 1.1 as the case $\alpha = 1$. Fix $J \geq 1$ and put $D = 2J + 2$ (source 100-89 writes $M = J - 1$ and $D = 2M + 4$). Let

$$F_D(t) = I_\alpha(t) + \sum_{r=0}^D \xi_r(\alpha) t^r, \quad (21.17)$$

so $F(t) - F_D(t) = O(t^{D+1})$ with every fixed derivative. For all large n let t_D be the unique small positive solution of $-F'_D(t_D) = n$, and put $B_D = F''_D(t_D)$, $\kappa_{j,D} = (-1)^j F_D^{(j)}(t_D)$, and $\mathcal{E}_\ell[D]$ for $\mathcal{E}_\ell$ of (6.2) with κ_j, B replaced by $\kappa_{j,D}, B_D$.

Theorem 21.4 (All-orders relative expansion with the exact integral; source 100-89). *Uniformly for $\alpha \in \mathfrak{A}$, as $n \rightarrow \infty$,*

$$a_\alpha(n) = \frac{\exp\{F_D(t_D) + nt_D\}}{\sqrt{2\pi B_D}} \left\{ 1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell[D] + O_{\mathfrak{A},J}(t_D^J) \right\}, \quad t_D \sim \frac{\alpha \log \sqrt{2n}}{\sqrt{2n}}. \quad (21.18)$$

The exact integral stays in the exponent; only finitely many boundary coefficients enter. A fixed expansion in inverse powers of $\log n$ would not give the same relative accuracy (source 100-89).

Proof (source 100-89). At the lattice saddle, that is, with F , t_n , B and κ_j in place of F_D , t_D , B_D and $\kappa_{j,D}$, (21.18) is Theorem 2.2 in the cumulative form (6.4), since $t = \alpha\varepsilon$; source 100-89's Section 4 re-proves it (Table 8). Replace the lattice by the model: Theorem 21.2 with derivatives gives $F^{(j)}(t) - F_D^{(j)}(t) = O(t^{D+1-j})$ for $0 \leq j \leq D$. The integral variance is positive and of the size (4.4), so the small saddle of F_D is unique, and its displacement from the lattice saddle is $O(t_D^D/B_D)$. The phase difference is $O(t_D^{D+1})$ plus a smaller quadratic displacement term. Substituting the F_D derivatives in each of the finitely many normalized cumulant terms changes them by $o(t_D^J)$. $\square$

Remark 21.5 (The remainder scale; [write]). The remainder $O(t^J) = O((\alpha s/M)^J)$ is the conservative scale of Theorem 2.2, not the sharp $O(\varepsilon_\#^J)$, $\varepsilon_\# = 1/(\alpha M s)$, of Theorem 16.3; source 100-89 claims nothing sharper, and its expansion is not sharper than Section 16. The replacement step transfers the sharp scale: with $D = 2J + 2$,

$$a_\alpha(n) = \frac{\exp\{F_D(t_D) + nt_D\}}{\sqrt{2\pi B_D}} \left\{ 1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell[D] + O_{\alpha,J}(\varepsilon_\#^J) \right\}.$$

Indeed the changes in the proof above are: the phase, by relative $O(t^{D+1})$; the saddle, displaced by $\delta = O(t^D/B)$, which changes the variance by relative $O(|\kappa_3|\delta/B) = O(|\kappa_3|t^D/B^2) = O(t^D s/M^2)$; and each normalized cumulant with $j \leq 2J$, by $O(t^{D+1-j} B^{-j/2})$ from the substitution and $O(|\kappa_{j+1}|\delta B^{-j/2})$ from the displacement. Since $t^{D+1}/\varepsilon_\#^J = \alpha^{D+1+J} s^{D+1+J} M^{J-D-1} \rightarrow 0$ and $t^D s/(M^2 \varepsilon_\#^J) = \alpha^{D+J} s^{D+J+1} M^{J-D-2} \rightarrow 0$, all of them are $o(\varepsilon_\#^J)$, so Theorem 16.3 (with its $J-1$) at the lattice saddle gives the display. This is a bookkeeping consequence of two proved theorems, not a new estimate. (Dated note, 5 October 2026: the first version listed the saddle change as relative $O(t^D/B)$ and omitted the displacement terms; an independent check made by the intake after the write supplied them and found the conclusion unchanged.)

For direct use, let t^{int} solve $-I'_\alpha(t^{\text{int}}) = n$ and put $B^{\text{int}} = I''_\alpha(t^{\text{int}})$, $I_\alpha^{(j)} = I_\alpha^{(j)}(t^{\text{int}})$.

Corollary 21.6 (First order at the integral saddle; source 100-89). *Uniformly for $\alpha \in \mathfrak{A}$,*

$$a_\alpha(n) = \frac{\exp\{I_\alpha(t^{\text{int}}) + nt^{\text{int}} + \xi_0(\alpha)\}}{\sqrt{2\pi B^{\text{int}}}} \left\{ 1 + \xi_1(\alpha)t^{\text{int}} + \frac{I_\alpha^{(4)}}{8(B^{\text{int}})^2} - \frac{5(I_\alpha^{(3)})^2}{24(B^{\text{int}})^3} + O_{\mathfrak{A}}((t^{\text{int}})^2) \right\}. \quad (21.19)$$

In particular, at $\alpha = 1$, where $\xi_0(1) = \frac{1}{2} \log(2\pi) - 1$,

$$a_1(n) \sim \frac{\exp\{I_1(t^{\text{int}}) + nt^{\text{int}} - 1\}}{\sqrt{B^{\text{int}}}} \quad (\text{source 100-82's Theorem 1.1}). \quad (21.20)$$

The integral cumulant block in (21.19) is kept exactly, rather than further truncated on a different asymptotic scale (source 100-89). The two factors $\sqrt{2\pi}$ in (21.20) cancel (source 100-82).

Proof (source 100-89). Start at t^{int} . Smoothness gives $F' = I'_\alpha + \xi_1 + O(t)$, hence $t_n - t^{\text{int}} = -\xi_1/B^{\text{int}} + O(t^{\text{int}}/B^{\text{int}}) + O(|I_\alpha^{(3)}|/(B^{\text{int}})^3)$. The change in the stationary phase is $O(1/B^{\text{int}})$; the relative variance change is $O(1/B^{\text{int}}) + O(|I_\alpha^{(3)}|/(B^{\text{int}})^2) = O(s/M^2) = O(t^2)$, and $1/B^{\text{int}} = O(s/M^3) = o(t^2)$. The lattice-integral differences in fixed cumulants are bounded, hence smaller still after normalization. Expanding $\exp(\xi_1 t^{\text{int}} + O(t^2))$ and using $1 + \mathcal{E}_1$ gives (21.19), with all saddle-shift errors at the claimed order. Source 100-82's proof of (21.20) is different; it is the second route of Section 21.5. $\square$

[Merge note, batch 100.] Comparison with the sharp first correction: here $\xi_1 t$ is of order s/M , larger than $\mathcal{E}_1 \sim -3/(8\alpha Ms)$ (Theorem 16.3) by a factor of order s^2 ; at the integral saddle the leading correction is the boundary term, not the Edgeworth term. The $O(t^2)$ remainder is again the conservative scale.

21.4 Exact-range inversion

This subsection is source 100-89's Section 6. Fix $J \geq 1$, $D = 2J + 2$, continue t_D , B_D and $\mathcal{E}_\ell[D]$ smoothly to large real x in place of n , and define

$$\Psi_J^{\text{int}}(x) = F_D(t_D) + xt_D - \frac{1}{2} \log(2\pi B_D) + \log\left(1 + \sum_{\ell=1}^{J-1} \mathcal{E}_\ell[D]\right), \quad (21.21)$$

the logarithm being defined eventually because the sum is $O(t_D)$. For integer n , Theorem 21.4 gives

$$\log a_\alpha(n) = \Psi_J^{\text{int}}(n) + O(t_D^J). \quad (21.22)$$

Write $P(t)$ for $1 + \sum_{\ell < J} \mathcal{E}_\ell[D]$ as a function of the saddle variable. Implicit differentiation gives $t'_D(x) = -1/B_D$ and

$$(\Psi_J^{\text{int}})'(x) = t_D + \frac{\partial_t B_D}{2B_D^2} - \frac{\partial_t P}{P B_D}, \quad (21.23)$$

with ∂_t the derivative in the saddle variable.

Proposition 21.7 (Slope of the inverse model; source 100-89, completed [write]). $(\Psi_J^{\text{int}})'(x) = t_D + O(s/M^2) + O(s^2/M^3) \sim t_D > 0$, uniformly on $\mathfrak{A}$.

Proof. Source 100-89 states the two bounds $\partial_t B_D = O(M^4/s)$ and $\partial_t P = O(M\varepsilon) = O(s)$, the latter from “fixed-order moment bounds one order higher” and the fact that every nonconstant term of P has weight at least two. [write] The details: the integral analogue of (4.5), $\int_0^\infty x^j p(1-p) dx = O(M^{j+1}/s)$, holds by the same proof (or by the sum-integral comparison of Section 7.3), and the boundary polynomial in (21.17) changes each derivative of order at most D by $O(1)$; so $|\kappa_{j,D}| = O(M^{j+1}/s)$ for $2 \leq j \leq D$, and $B_D \asymp M^3/s$. Then $\partial_t B_D = -\kappa_{3,D}$ gives the first bound, and $\partial_t B_D/(2B_D^2) = O((M^4/s)(s/M^3)^2) = O(s/M^2)$. Next, $\partial_t \kappa_{j,D} = -\kappa_{j+1,D}$, so for the normalized cumulants $x_j = \kappa_{j,D} B_D^{-j/2}$,

$$\partial_t x_j = -\frac{\kappa_{j+1,D}}{B_D^{j/2}} + \frac{j}{2} \frac{\kappa_{j,D} \kappa_{3,D}}{B_D^{j/2+1}} = O\left(\frac{M^{j+2}}{s} \left(\frac{s}{M^3}\right)^{j/2}\right) = O(M\varepsilon^{(j-2)/2}),$$

both terms having this size; only $j + 1 \leq 2J + 1 \leq D$ is used, the “one order higher”. Each monomial $\prod_j x_j^{m_j}$ of P has weight $w = \sum_j (j-2)m_j \geq 2$, because the only weight-one monomial, x_3 , has odd degree and does not occur; by the product rule its derivative is $O(M\varepsilon^{w/2}) = O(M\varepsilon)$. Hence $\partial_t P = O(M\varepsilon) = O(s)$, while $P = 1 + O(\varepsilon)$, so $\partial_t P / (PB_D) = O(s \cdot s/M^3) = O(s^2/M^3)$. Both error terms are $o(t_D)$ since $t_D \asymp s/M$. $\square$

Let $n_{*,J}^{\text{int}}(Y)$ denote the eventual real inverse of Ψ_J^{int} , and, as in Theorem 2.3, $N_\alpha(Y)$ the least n with $\log a_\alpha(n) \geq Y$.

Theorem 21.8 (Exact-range inverse; source 100-89). *For every fixed $J \geq 1$,*

$$n_{*,J}^{\text{int}}(\log a_\alpha(n)) = n + O(t_D(n)^{J-1}). \quad (21.24)$$

For $J \geq 2$, sequence indices are therefore recovered eventually by $n = \lfloor n_{,J}^{\text{int}}(\log a_\alpha(n)) + \frac{1}{2} \rfloor$. For arbitrary large Y , put $x = n_{*,J}^{\text{int}}(Y)$ and $t_x = t_D(x)$. There is a constant C such that*

$$\lfloor x - Ct_x^{J-1} \rfloor \leq N_\alpha(Y) \leq \lceil x + Ct_x^{J-1} \rceil. \quad (21.25)$$

If $J \geq 2$ and $\text{dist}(x, \mathbb{Z}) > Ct_x^{J-1}$, then $N_\alpha(Y) = \lceil x \rceil$.

Proof (source 100-89). Proposition 21.7 and (21.22) give (21.24) by the mean value theorem; t_D varies by a factor $1 + o(1)$ on bounded x -intervals. For thresholds, the coefficients increase strictly for $n \geq 1$ by the injection of Section 9, also for nonintegral α . By (21.22) and the positive slope, integers at least Ct_x^{J-1} below x lie below the threshold and those at least that far above lie above it, which is (21.25). The nearest-integer claim uses $t_D^{J-1} \rightarrow 0$ for $J \geq 2$. $\square$

There is no unconditional ceiling rule for arbitrary thresholds, and no assertion about how often thresholds fall into the exceptional neighbourhoods of integers. The exact-range error is $O((\log n/\sqrt{n})^{J-1})$ for every fixed J (source 100-89).

[Merge note, batch 100.] Relation to Theorem 16.4 (source 75-01). Exact index recovery at sequence values already follows from (16.9): the displacement $-3/(8\alpha^2 s^2)$ of the Gaussian inverse tends to zero, and with Ψ_1 it is $O(M^{-1}s^{-3})$. Source 100-89’s theorem gives the same conclusion for every truncation order, at the conservative scale: for $J = 2$ its error $O(t) = O(\alpha s/M)$ is weaker than the Ψ_1 bound by a factor of order s^4 , and for large J it is the only all-orders statement here. By Remark 21.5 the error in (21.24) improves to $O(\varepsilon_\#^J/t)$. Its conditional ceiling rule is consistent with the sentence after Theorem 16.4: no universal rounding rule exists for arbitrary thresholds; source 100-89 says the same. The threshold step is the staircase theorem `p0:thm:staircase` of [25].

Remark 21.9 (A resummed inverse at $\alpha = 1$; source 100-82). Source 100-82 inverts $\Psi^{\text{int}}(x) = I_1(t^{\text{int}}) + xt^{\text{int}} - 1 - \frac{1}{2} \log B^{\text{int}}(t^{\text{int}})$, the logarithm of (21.20). Exact differentiation gives $(\Psi^{\text{int}})'(x) = t^{\text{int}} + \partial_t B^{\text{int}} / (2(B^{\text{int}})^2)$, and $|\partial_t B^{\text{int}}| \leq \int x^3 p(1-p) dx = O(M^4/s)$, so $(\Psi^{\text{int}})' = t^{\text{int}} + O(s/M^2) \sim t^{\text{int}}$. If $\nu(y)$ is the eventual inverse of $e^{\Psi^{\text{int}}}$, monotone bracketing gives $N_1(\log y) = \nu(y) + o(1/t_{\nu(y)}^{\text{int}}) + O(1)$; the saddle scale varies by $1 + o(1)$ on this window since $(t^{\text{int}})' = -1/B^{\text{int}}$. “A relative forward equivalent does not justify an exact rounding rule.” This is weaker than both Theorem 16.4 and Theorem 21.8.

21.5 Second routes and forms that differ

The constant $\frac{1}{2} \log(2\pi) - 1$ by an L^1 limit (source 100-82). At $\alpha = 1$ write $f_t(x) = \log(1 + xe^{-tx})$, $t = se^{-s}$, $M = e^s$. Then $f_t(0) = 0$, $f_t'(0) = 1$ and

$$f_t'' = -\frac{p^2}{x^2} - \frac{2tp(1-p)}{x} + t^2 p(1-p), \quad p = p(x), \quad (21.26)$$

with continuous limits at zero. The first term converges in $L^1(0, \infty)$ to $-(1+x)^{-2}$, since $p(x) \leq x/(1+x)$; and $\int_0^\infty p x^{-1} dx = O(s)$, $\int_0^\infty p dx = O(M)$, so the other two terms have L^1 norms $O(ts)$ and $O(t^2 M)$, which tend to zero. Second-order Euler–Maclaurin summation gives $F(t) - I_1(t) = -\frac{1}{12} - \frac{1}{2} \int_0^\infty \tilde{B}_2(x) f_t''(x) dx$, and the L^1 convergence allows the limit inside the remainder. The limiting functional is evaluated by the finite formula for $\log(1+x)$ on $[0, X]$: by Stirling’s formula $\sum_{k \leq X} \log(1+k) - \int_0^X \log(1+x) dx - \frac{1}{2} \log(X+1) \rightarrow \frac{1}{2} \log(2\pi) - 1$, with the same Bernoulli remainder. Hence $F(t) = I_1(t) + \frac{1}{2} \log(2\pi) - 1 + o(1)$: the small-part boundary is essential for this constant. This is the case $r = 0$, $\alpha = 1$ of Theorem 21.2, proved by a second-order expansion only.

The local limit at the integral saddle (source 100-82). For $h = xp$ and $h = x^2 p(1-p)$ the bounded-variation inequality $|\sum_k h(k) - \int_0^\infty h| \leq \int_0^\infty |h'|$ (as in Section 7.3) gives, at $\alpha = 1$, $\mathbb{E}S = -I_1'(t) + O(sM)$ and $\text{Var } S = I_1''(t) + O(sM^2) = I_1''(t)(1 + o(1))$; the third absolute centred moment sum is $O(M^4/s) + O(sM^3) = o((I_1'')^{3/2})$. At $t = t^{\text{int}}$ the target is therefore within $o(\sqrt{\text{Var } S})$ of the mean, and the block bounds give $\mathbb{P}(S = n) \sim (2\pi B^{\text{int}})^{-1/2}$, the uniform local limit (5.7) evaluated at the integral saddle. With the constant above this proves (21.20) without Theorem 21.2.

A third proof of the block inequality (source 100-82 and 100-89). For a block of ℓ consecutive integers, the sum in Lemma 5.1 is at least $(\ell - |Z_\ell|)/2$ with $Z_\ell = \sum_{j < \ell} e^{ij\theta}$. If $|\theta| \geq 2\pi/\ell$, then $|Z_\ell| \leq \pi/|\theta| \leq \ell/2$. At smaller angles both sources take a sub-block of length $m = \lfloor \ell/8 \rfloor$ and use the identity

$$m^2 - |Z_m|^2 = \sum_{i,j=0}^{m-1} (1 - \cos((i-j)\theta)) \geq c m^4 \theta^2,$$

from $1 - \cos u \geq cu^2$ on a fixed small interval; hence $m - |Z_m| \geq c' m^3 \theta^2$, independently of the starting point. This replaces the uniform-convergence step of the first two proofs.

The Sommerfeld expansion with the operator $\mathcal{T}$ (source 100-82 and 100-89). Source 100-89’s (31) is, at $t = \alpha s e^{-s}$,

$$I_\alpha(t) = \alpha M \left\{ \frac{s}{2} - 1 + \sum_{j=1}^J \frac{2\eta(2j)}{\alpha^{2j}} \frac{\mathcal{T}^{2j-1}s}{s} \right\} + O_{\mathfrak{A},J}(M s^{-2J-2}),$$

with $\mathcal{T} = \frac{s}{s-1} \frac{d}{ds}$, and source 100-82’s Lemma 4.1 is its case $\alpha = 1$. This is Lemma 7.1: $s^{-1} \mathcal{T}^{2j-1} s = U_{2j-1}(s)$ (for instance $s^{-1} \mathcal{T} s = 1/(s-1) = U_1$ and $s^{-1} \mathcal{T}^3 s = (2s+1)/(s-1)^5 = U_3$), and, as $\mathcal{T} s = f(s)$, it is also source 48’s (7.14). The kernel moments are written $\int_{\mathbb{R}} u^{2j-2} \log(1 + e^{-|u|}) du = 2(2j-2)! \eta(2j)$ (sources 100-82 and 100-89 write $2(1-2^{1-2j})\zeta(2j)$ for $2\eta(2j)$, and D_b for $\mathcal{T}$). Both prove it by inverting $u = tx - \alpha \log x$ near M on $|u| \leq cs$ and Taylor-expanding the Jacobian, with $\mathcal{T}^r$ of the Jacobian $O_r(s^{-r-1})$ by induction from $s/(s-1) = 1 + 1/(s-1)$. Source 100-82 adds the lower edge explicitly: $\int_0^\infty \max(\log x - tx, 0) dx = M(s/2 - 1) + x_- - tx_-^2/2$, with the lower zero $x_- \rightarrow 1$.

Eliminating the saddle as a minimized phase (source 100-82 and 100-89). At $\alpha = 1$ write $\mathcal{A}_J(s)$ for the braces above, so that $I_1(se^{-s}) = e^s \mathcal{A}_J(s) + O(e^s s^{-2J-2})$. Both sources solve formally, for $\delta = s - r = O(r^{-2})$,

$$e^{2\delta} \frac{2[\mathcal{A}_J(r+\delta) + \mathcal{A}_J'(r+\delta)]}{r+\delta-1} = 1,$$

and expand $e^\delta \mathcal{A}_J(r + \delta) + \frac{1}{2}(r + \delta)e^{-\delta}$, whose coefficients through r^{-R} are $r - 1 + \sum_{j \leq R} P_j(\pi^2/6)r^{-j}$ for every sufficiently large J ; source 100-89 multiplies by α for general α . With $\mathcal{A}_J = s/2 - 1 + d_1(s)$ this is (8.2) and (8.3): $2(\mathcal{A}_J + \mathcal{A}'_J)/(s - 1) = Q_1(s)$, so the stationary equation is $r = s + \frac{1}{2} \log Q_1(s)$, and $e^\delta = Q_1^{-1/2}$ turns the expanded value into $(s(Q_1 + 1)/2 - 1 + d_1)/\sqrt{Q_1}$. The justification differs from the first two: the exact integral saddle minimizes $I_1(t) + nt$; on $|s - r| \leq 1$ the approximate phase $e^s \mathcal{A}_J(s) + nse^{-s}$ is uniformly within $O(Kr^{-2J-2})$ of it and strictly convex, with second derivative comparable to Kr ; so the two minima differ by the same amount, and the implicit function theorem in $z = 1/r$ gives the unique branch $\delta = O(z^2)$. The first terms are $\delta = -cr^{-2} - cr^{-3} + c^2r^{-4} + O(r^{-5})$, which is (8.8), and $s_4 = c - c^2/2 + 2 \cdot \frac{7\pi^4}{360} = P_4(\pi^2/6)$ (source 100-82).

The inverse equation (source 100-82). With $X_0 = Y/w = K_0$ and $C = w + 1 = R_0$, source 100-82's equation $e^\varepsilon(C + \varepsilon - 1 + \sum_j s_j(C + \varepsilon)^{-j}) = C - 1$ for the inverse correction is (9.1) with $E = e^\varepsilon$, as for source 75-06; its reversion $\varepsilon = -cC^{-2} - cC^{-3} + (c^2 - c)C^{-4} + O(C^{-5})$ gives the coefficients $-c, -c, \frac{3}{2}c^2 - c$ of (9.2). The remainder is transported by the slope $\log X$ of the leading phase in $X = \sqrt{2n}$, as in Section 9.

21.6 Finite-size data

Source 100-82 compares (21.20) with exact coefficients (the product computed through $n = 10000$, its first 21 terms checked against the OEIS prefix); the last column subtracts $\log a_1(n)$ from the four-term expression of Theorem 2.1. Source 100-89's table gives the relative error, approximation divided by exact coefficient minus one, of the leading integral equivalent and of (21.19) without its remainder; its exact coefficients were checked by the recurrence (12.2), and two quadrature variables at 35 and 65 digits agree to more than 25 digits. Quadratures are numerical orientation, not certified bounds or proof inputs (both sources).

Source 100-82, $\alpha = 1$				Source 100-89		
n	ratio	log error	α	n	leading	corrected
100	1.0095049294	5.2419251	1	1000	$+1.890945 \cdot 10^{-3}$	$-4.561189 \cdot 10^{-6}$
1000	1.0018909450	4.1573353	2	1000	$+3.770255 \cdot 10^{-3}$	$+4.122684 \cdot 10^{-4}$
10000	1.0003719130	3.4161122	3	100	$+8.354550 \cdot 10^{-2}$	$+6.970208 \cdot 10^{-2}$
			3	1000	$+4.319863 \cdot 10^{-2}$	$+3.840891 \cdot 10^{-2}$
			3	3000	$-3.250594 \cdot 10^{-2}$	$-3.528618 \cdot 10^{-2}$

Table 9: Left: prediction (21.20) divided by $a_1(n)$, and the four-term logarithmic error (source 100-82). Right: relative errors of the leading integral equivalent and of (21.19) (source 100-89, whose caption reads “Verified numerical diagnostics for fixed powers. No effective onset is claimed”).

[Merge note, batch 100.] The two tables measure the same quantity at $\alpha = 1$, $n = 1000$ ($1.0018909450 - 1 = 1.890945 \cdot 10^{-3}$). The corrected integral-saddle error there, $-4.56 \cdot 10^{-6}$, is comparable in size with the corrected lattice-saddle error $+4.50 \cdot 10^{-6}$ of Table 3, a different approximant. The log errors continue source 48's (3.55 at $n = 6400$ for the same four terms). “Verified” in source 100-89's caption means that the decimals were reproduced by independent exact recurrences and quadratures; they are not interval-certified, and the intake reran both tables on copies with identical output. The $\alpha = 3$ rows show the finite-size residual of Section 16.3, about 4% with a sign change between $n = 1000$ and 3000, unchanged by the correction: “consistent with the need to control weighted-lattice minor arcs, but the table

does not identify their asymptotic form” (source 100-89). The batch-100 note in Section 16.3 records a numerical decomposition of that residual.

21.7 Literature, scope and non-claims of sources 100-82 and 100-89

Source 100-89 compares four general theories, none of which supplies these boundary coefficients: Granovsky–Stark [14] allow inside weights at most one, and their decreasing-weight examples differ from k^α ; Bridges, Brindle, Bringmann and Franke [29] study nonnegative multiplicities in products $\prod (1 - q^k)^{-f(k)}$; Ahmadi, Gómez-Aíza and Ward [30] place their weights outside the distinct-part factors; and the exponential Hayman criterion of Cantón, Fernández, Fernández and Maciá [31] assumes nonnegative coefficients of the logarithm, unavailable here. “General saddle methods remain central to the proof.” It reads Gikunda’s thesis [10] as Section 1.2 does: its Lemma 4.2.2 and Proposition 4.2.5 impose a shrinking tilt $a = O(t^\delta) \rightarrow 0$, which does not cover fixed positive powers. Source 100-82 cites Naranjo and Ramírez [27], “printed page 12”, who identify this sequence and product but “do not derive its asymptotics”, and Granovsky–Stark as above.

[Merge note, batch 100.] Source 100-89 cites the thesis at the ResearchGate address 402798310 (as source 77-51’s source check did); this report keeps the DOI of [10].

Non-claims. Source 100-89: the results are uniform only for α in a fixed compact positive interval; they do not address α growing with n , convergence of the boundary Taylor series, optimal truncation, or a complete exponentially small lattice expansion; no effective onset is known; exact-range rounding and arbitrary-threshold rounding have different, explicitly stated scopes; “The symbolic checks verify finite coefficient identities; they do not replace the analytic all-orders and minor-arc proofs”; and a uniform theory for growing α , “possibly involving a theta factor, is a separate research problem and is not a consequence claimed here”. Source 100-82: no convergent full transseries, no remainder uniform as the truncation order grows, no exact threshold rounding rule, no exhaustive novelty; “Finite logarithmic truncations are not confused with relative coefficient asymptotics”; quadratures are orientation, not proof premises; the lower boundary “is negligible at every inverse-logarithmic scale but cannot be omitted from the relative equivalent”. Neither claims Lean or Rocq verification, and neither uses a repository theorem.

21.8 Further questions and research (sources 100-82 and 100-89)

Every claim of the two sources is proved above or listed here (Vladimir’s standing rule of 4 October 2026). Source 100-89’s two sketched steps, the holomorphy of ξ_r and the slope bound of its inverse model, are completed by Lemma 21.3 and Proposition 21.7; no claim of either source was found to be wrong.

100.1 The size of the boundary coefficients. Source: source 100-89, after Theorem 1.1 and Section 7.2 (convergence and optimal truncation are “not addressed”). Open: the growth of $\xi_r(\alpha)$ in r ; whether $\sum_r \xi_r(\alpha) t^r$ converges near $t = 0$ or is Gevrey, and what optimal truncation of (21.6) gives. Sketch: none; the functional (21.11) and the Hurwitz rule (21.15) give every coefficient at integer α . Overlaps project 2 and question 75-06.2.

100.2 Effective onset. Source: both sources (“No effective onset is claimed”). Open: explicit constants in Theorems 21.2, 21.4 and 21.8, so that the threshold bracket (21.25) can be certified at a stated Y . Overlaps project 8, 48.7, 75-01.8 and 77-51.1.

- 100.3 An exponentially small lattice correction.** Source: both sources (no complete exponentially small expansion, no convergent transseries). Open, overlapping projects 1–2 and 48.1; the critical window of Section 22 is the only regime in which the noncentral contributions are computed.
- 100.4 Uniformity in the truncation order.** Source: source 100-82 (“uniform remainder as truncation order grows” not claimed). Open: a remainder in Theorem 2.1 uniform in R , or a proof that none exists.
- 100.5 Rounding at arbitrary thresholds.** Source: source 100-82 (“No exact ceiling rule is asserted”) and source 100-89 (“no assertion about the frequency of their near-integer exceptions”). Re-scoped: Theorem 21.8 gives the ceiling rule off an $O(t^{J-1})$ -neighbourhood of the integers. Open: how often $n_{*,J}^{\text{int}}(Y)$ falls into that neighbourhood as Y varies, for instance for $Y = \log 10^m$.
- 100.6 The finite-size residuals at $\alpha = 3$.** Source: source 100-89, Section 7.1 (“does not identify their asymptotic form”). Re-scoped: the numerical decomposition in the batch-100 note of Section 16.3 attributes the residual at $n = 5000$ to the Fourier cells around $\pm 2\pi/M$. Open: a *frozen-exact satellite formula at fixed α* , an analogue of Lemma 22.4 in which the satellite phase and damping use the exact Bernoulli probabilities of the lower bulk, which is not frozen at fixed α , with a proof that it gives the cell contributions to relative error $o(1)$. Missing: everything beyond Theorem 11.1, which gives the modulus at one point.
- 100.7 Growing α .** Source: source 100-89, Section 7.1. Answered in the critical window $\alpha \asymp M^{1/3}/\log M$ by Section 22; the adjacent regimes are its questions 100-93.4 and 100-93.5.

22 The critical theta window for growing powers

This section is source 100-93 (Section 21), translated: its $a, K, b, W, \lambda, L, R, \mu, U, C, V, Q, T_r$ and z are written $\alpha, M, s, H, \Lambda, F(t), R_{\geq 2}, \mu_R, B_{RR}, B_{RS}, B, Q_{\perp}, T_r$ and K (fourth table of Section 1.5). Every other section of this report keeps α fixed; here α grows with n . Source 100-93’s own scope paragraph is kept: it is “an unrefereed mathematical research note, not a formal proof-assistant development”; its critical-window theorem is self-contained; source 100-89 is “its companion and background dependency”, but “None of those compact- a error estimates is extended to growing a here”, and “The present results have a different uniformity domain and a weaker inverse-error conclusion”; its numerical diagnostics are illustrations independent of the proof; and “The source comparison is bounded and makes no exhaustive priority claim”. Its pinned snapshot 6591a56b3 contains neither this report nor the lattice-peak report of Section 22.8.

22.1 The critical relative equivalent

For real $\alpha > 0$ and $t > 0$ let X_k, p_k and $S = \sum_k kX_k$ be as in (4.1), and let

$$R_{\geq 2} = \sum_{k \geq 2} X_k \tag{22.1}$$

be the number of selected parts of size at least two. The saddle $t = t_n$ is defined exactly by $\mathbb{E}S = n$, for every real $n > 0$. Put

$$\begin{aligned} \mu_R &= \mathbb{E}R_{\geq 2}, & B_{RR} &= \text{Var } R_{\geq 2}, & B_{RS} &= \text{Cov}(R_{\geq 2}, S), \\ B &= \text{Var } S, & Q_{\perp} &= B_{RR} - \frac{B_{RS}^2}{B}. \end{aligned} \quad (22.2)$$

Thus $Q_{\perp}$ is the variance of $R_{\geq 2}$ after linear regression on S . With M the larger root of $M^{\alpha}e^{-tM} = 1$ (the transition site of (2.7)), $s = \log M$ and $H = M/(\alpha(s-1))$ as there, set

$$\Lambda = \frac{\alpha(s-1)}{M^{1/3}}, \quad \text{so} \quad H = \frac{M^{2/3}}{\Lambda}, \quad \alpha = \frac{\Lambda M^{1/3}}{s-1}, \quad t = \frac{\Lambda s}{s-1} M^{-2/3}. \quad (22.3)$$

The *critical regime* is: $M \rightarrow \infty$ with Λ in an arbitrary fixed compact interval $\mathfrak{J} \subset (0, \infty)$; all constants and limits below are uniform on $\mathfrak{J}$ unless stated otherwise. In it $\alpha \rightarrow \infty$ and $\alpha/\log M \rightarrow \infty$. Define the positive 1-periodic function

$$\Theta(\mu, Q) = \sum_{j \in \mathbb{Z}} \exp(-2\pi^2 Q j^2 - 2\pi i j \mu) = \frac{1}{\sqrt{2\pi Q}} \sum_{l \in \mathbb{Z}} \exp\left(-\frac{(l - \mu)^2}{2Q}\right), \quad Q > 0, \quad (22.4)$$

the equality being Poisson summation for a Gaussian.

Theorem 22.1 (Critical theta equivalent; source 100-93). *For positive integers n in the critical regime,*

$$a_{\alpha}(n) = \frac{\exp(F(t) + nt)}{\sqrt{2\pi B}} \{\Theta(\mu_R, Q_{\perp}) + o(1)\}, \quad (22.5)$$

and

$$n \sim \frac{M^2}{2}, \quad B \sim M^2 H, \quad Q_{\perp} = \frac{\pi^2}{3\Lambda^3} + o(1). \quad (22.6)$$

$\Theta(\mu_R, Q_{\perp})$ is bounded above and away from zero, uniformly in the phase μ_R , so (22.5) is a uniform relative equivalent.

The exact phase μ_R cannot in general be replaced by a coarse leading approximation. In particular $R_{\geq 2}$ excludes X_1 : although the lower bulk becomes frozen, $p_1 = 1/(1 + e^t) \rightarrow \frac{1}{2}$, and Section 22.3.1 explains why including it in a Gaussian count gives a wrong variance and phase (source 100-93).

Corollary 22.2 (A direct criterion; source 100-93). *Let $\alpha = \alpha(n) > 0$. If*

$$\frac{\alpha \log \sqrt{2n}}{(2n)^{1/6}} \longrightarrow \Lambda_{\infty} \in (0, \infty), \quad (22.7)$$

then the exact saddle lies in the critical regime, $M/\sqrt{2n} \rightarrow 1$ and $\Lambda \rightarrow \Lambda_{\infty}$, so Theorem 22.1 applies.

22.2 Uniform edge estimates

Let $\varpi(x) = \alpha \log x - tx$ be the log odds, as in Section 21.2, and $v_k = p_k(1 - p_k)$. At the upper root, $\varpi(M) = 0$ and $\varpi'(M) = -1/H$, and uniformly on bounded u -intervals

$$\varpi(M + Hu) = -u + O\left(\frac{M^{-1/3}u^2}{\log M}\right). \quad (22.8)$$

For large M , $\varpi' \leq -1/(2H)$ on $[M/2, M]$ and $\varpi' \leq -1/H$ on $[M, \infty)$, so both v_k and the defect probability $\min(p_k, 1 - p_k)$ are at most $\exp(-|k - M|/(2H))$ on $[M/2, \infty)$. On $[2, M/2]$ concavity bounds ϖ below by its smaller endpoint value; $\varpi(2) = \alpha(\log 2 - 2 \log M/M)$ and $\varpi(M/2) = \alpha(\frac{\log M}{2} - \log 2)$, so $\varpi \geq c_3 \alpha$ there, with $c_3 > 0$. Since $\alpha/\log M \rightarrow \infty$, every $M^{A_0} e^{-c_3 \alpha}$ with fixed A_0 tends to zero. This controls the frozen lower bulk without a small-phase approximation (source 100-93).

Lemma 22.3 (Edge moments; source 100-93). *For every fixed integer $r \geq 0$,*

$$\mathsf{T}_r := \sum_{k \geq 2} (k - M)^r v_k = H^{r+1} \left\{ \int_{\mathbb{R}} u^r \frac{e^u}{(1 + e^u)^2} du + o(1) \right\}, \quad (22.9)$$

and $\sum_{k \geq 2} |k - M|^r v_k = O_r(H^{r+1})$. In particular $B_{RR} = \mathsf{T}_0 \sim H$, $\mathsf{T}_1 = o(H^2)$ and $\mathsf{T}_2 \sim \frac{\pi^2}{3} H^3$.

Proof (source 100-93). On bounded u -intervals, (22.8) and the mesh $1/H \rightarrow 0$ give uniform Riemann-sum convergence; the exponential edge bound gives a uniform integrable majorant for every absolute moment; the far lower interval contributes at most $M^{r+1} e^{-c_3 \alpha}$. The limit density is the standard logistic density, whose bilateral exponential moment is $\pi z / \sin(\pi z)$ for $|z| < 1$, with second derivative $\pi^2/3$ at 0. $\square$

Compare Lemma 4.1 and (4.6), the same logistic profile at fixed α , where the lower bulk is not frozen.

Put $v_1 = p_1(1 - p_1)$. Directly from the definitions,

$$B_{RS} = MB_{RR} + \mathsf{T}_1, \quad B = M^2 B_{RR} + 2M\mathsf{T}_1 + \mathsf{T}_2 + v_1, \quad (22.10)$$

$$Q_{\perp} = \frac{B_{RR}\mathsf{T}_2 - \mathsf{T}_1^2 + B_{RR}v_1}{B}. \quad (22.11)$$

Consequently $B \sim M^2 H$ and

$$Q_{\perp} = \frac{\pi^2 H^3}{3M^2} + o(1) = \frac{\pi^2}{3\Lambda^3} + o(1); \quad (22.12)$$

the contribution of part one to $Q_{\perp}$ is $O(M^{-2})$. The leading terms $M^2 B_{RR}^2$ of $B_{RR}B$ and of B_{RS}^2 cancel in (22.11): conditioning on the size removes the variance $B_{RR} \sim H$ of the count down to order H^3/M^2 . The same defect estimates compare $\mathbb{E}S$ with $\sum_{k \leq M} k$, with difference

$$O(MH + H^2) + O(M^2 e^{-c_3 \alpha}) = O(MH), \quad (22.13)$$

so $n = M^2/2 + O(MH)$, the mean estimate in (22.6). The exact regression-centre identity

$$\frac{B_{RS}}{B} - \frac{1}{M} = -\frac{M\mathsf{T}_1 + \mathsf{T}_2 + v_1}{MB} \quad (22.14)$$

implies

$$\frac{B_{RS}}{B} = \frac{1}{M} + o\left(\frac{H}{M^2}\right), \quad \sqrt{B}\left(\frac{B_{RS}}{B} - \frac{1}{M}\right) = o(1), \quad (22.15)$$

because the three terms $-\mathsf{T}_1/\sqrt{B}$, $-\mathsf{T}_2/(M\sqrt{B})$ and $-v_1/(M\sqrt{B})$ tend to zero: $H^{3/2}/M$ is bounded and $H^{5/2}/M^2 = O(M^{-1/3})$ (source 100-93).

Proof of Corollary 22.2 (source 100-93). Fix $0 < \varepsilon < 1$ and put $M_{\pm} = (1 \pm \varepsilon)\sqrt{2n}$, with trial saddles $t_{\pm} = \alpha \log M_{\pm}/M_{\pm}$. Their critical parameters tend to $\Lambda_{\infty}(1 \pm \varepsilon)^{-1/3}$, so the uniform mean estimate applies at both, and their means lie above and below n for large n . On $M > e$, $\alpha \log M/M$ decreases in M while the mean decreases in t , so the exact root lies between M_- and M_+ . Letting $\varepsilon \downarrow 0$ gives $M/\sqrt{2n} \rightarrow 1$ and then $\Lambda \rightarrow \Lambda_{\infty}$. This uses the edge mean estimate rather than presupposing the location of the saddle. $\square$

22.3 The local Fourier satellites

Let $\phi(\theta) = \mathbb{E} \exp(i\theta(S - n))$; at the saddle this is ϕ_t of Section 5. For every integer j , since $R_{\geq 2}$ is integer valued,

$$\phi(\theta) = e^{-2\pi i j \mu_R} \mathbb{E} \exp(i[\theta(S - n) - 2\pi j(R_{\geq 2} - \mu_R)]), \quad (22.16)$$

with no further n -dependent phase because the saddle mean is exactly n . Set

$$\theta_j = 2\pi j \frac{B_{RS}}{B}. \quad (22.17)$$

The variance inside the expectation in (22.16) is exactly

$$B(\theta - \theta_j)^2 + 4\pi^2 j^2 Q_{\perp}. \quad (22.18)$$

Lemma 22.4 (Satellites; source 100-93). *For fixed $j \in \mathbb{Z}$ and $y \in \mathbb{R}$,*

$$\phi\left(\theta_j + \frac{y}{\sqrt{B}}\right) = \exp(-2\pi i j \mu_R - 2\pi^2 j^2 Q_{\perp} - y^2/2) \{1 + o(1)\}, \quad (22.19)$$

uniformly on bounded sets of (j, y) and on $\mathfrak{J}$.

Proof (source 100-93). At $\theta = \theta_j + y/\sqrt{B}$ the centred Bernoulli phases in (22.16) are $\omega_1 = \theta$ and $\omega_k = \theta k - 2\pi j$ for $k \geq 2$. By (22.15), $\theta = O(1/M)$ and

$$|\omega_k| \leq C \left(\frac{|k - M|}{M} + \frac{H}{M} + H^{-1/2} \right), \quad (22.20)$$

so the absolute edge moments give

$$\sum_{k \geq 2} v_k |\omega_k|^3 = O(H^4/M^3 + H^{-1/2}) = O(M^{-1/3}). \quad (22.21)$$

This weighted estimate alone does not permit a small-angle logarithm on the whole tail. Fix a large constant C_1 . In the active band $|k - M| \leq C_1 H \log M$ the largest $|\omega_k|$ tends to zero uniformly, and for a Bernoulli variable with parameter p and small real ω the branch of the logarithm continuous from zero satisfies $\log \mathbb{E} e^{i\omega(X-p)} = -\frac{1}{2}p(1-p)\omega^2 + O(p(1-p)|\omega|^3)$, uniformly in $0 \leq p \leq 1$; summing over the band has error $o(1)$ by (22.21). For a frozen lower-tail variable, $p = 1 - \epsilon$, the exact factor is $e^{i\epsilon\omega} \{1 + \epsilon(e^{-i\omega} - 1)\}$, and for a frozen upper-tail variable, $p = \epsilon$, it is $e^{-i\epsilon\omega} \{1 + \epsilon(e^{i\omega} - 1)\}$; in both cases

$$|\mathbb{E} e^{i\omega(X-p)} - 1| \leq \epsilon(|\omega| + 2). \quad (22.22)$$

For every prescribed $A > 0$, a large C_1 makes the sum of the right side outside the band $O(M^{-A})$: between $M/2$ and $M - C_1 H \log M$ use $e^{-(M-k)/(2H)}$, below $M/2$ use $e^{-c_3 \alpha}$, above $M + C_1 H \log M$ use $e^{-(k-M)/H}$; the linear weight in (22.20), or the coarser $O(1 + k/M)$, changes only polynomial factors. The same bounds with ω_k^2 show that the discarded tail variance is $o(1)$. Every characteristic factor has modulus at most one, so a telescoping product bound and (22.22) make the whole frozen-tail product $1 + o(1)$, without a logarithm across a large phase. Part one has $\omega_1 = O(1/M)$ and contributes $1 + o(1)$; its variance may also be kept in the exact quadratic expression. Now (22.18) gives (22.19). $\square$

22.3.1 The exceptional part of size one

If $R_{\geq 2}$ were replaced by $R_{\geq 2} + X_1$, the part-one satellite phase would be $\theta - 2\pi j$, not small for fixed $j \neq 0$. Its exact centred characteristic factor tends to $\exp(2\pi i j p_1)$, which cancels the added mean phase; it has no Gaussian attenuation. A Gaussian replacement would instead introduce a spurious limiting conditional variance $\frac{1}{4}$ and a half-unit mean shift. Excluding X_1 from $R_{\geq 2}$ while keeping it in S and B is therefore necessary for the satellite calculation (source 100-93).

22.4 Uniform control of the Fourier complement

Choose the odd consecutive block $\mathbf{B} = \{k_{\mathbf{B}} - m_{\mathbf{B}}, \dots, k_{\mathbf{B}} + m_{\mathbf{B}}\}$, with $k_{\mathbf{B}}$ an integer nearest M , $m_{\mathbf{B}} = \lfloor H/2 \rfloor$ and $N_{\mathbf{B}} = 2m_{\mathbf{B}} + 1$. Then $N_{\mathbf{B}}/H \rightarrow 1$, $|k_{\mathbf{B}} - M| \leq \frac{1}{2}$, and all v_k on $\mathbf{B}$ are bounded below by a positive constant.

Lemma 22.5 (A two-scale block inequality; source 100-93). *For all sufficiently large critical parameters, with an absolute $c' > 0$,*

$$\sum_{k \in \mathbf{B}} \sin^2(k\theta/2) \geq c' \{H \operatorname{dist}(M\theta, 2\pi\mathbb{Z})^2 + H^3\theta^2\}, \quad |\theta| \leq \frac{1}{2H}, \quad (22.23)$$

$$\sum_{k \in \mathbf{B}} \sin^2(k\theta/2) \geq c'H, \quad \frac{1}{2H} \leq |\theta| \leq \pi. \quad (22.24)$$

Proof (source 100-93). With the normalized Dirichlet kernel $D_N(\theta) = \sin(N\theta/2)/(N \sin(\theta/2))$ (equal to 1 at 0), symmetry gives

$$\sum_{k \in \mathbf{B}} \sin^2(k\theta/2) = \frac{N_{\mathbf{B}}}{2} \{1 - D_{N_{\mathbf{B}}}(\theta) \cos(k_{\mathbf{B}}\theta)\}. \quad (22.25)$$

For $|\theta| \leq 1/(2H)$, $D_{N_{\mathbf{B}}}$ is bounded below by a positive constant and $1 - D_{N_{\mathbf{B}}}(\theta) = N_{\mathbf{B}}^{-1} \sum_{|i| \leq m_{\mathbf{B}}} (1 - \cos(i\theta)) \geq c'' N_{\mathbf{B}}^2 \theta^2$. Since $1 - \cos(k_{\mathbf{B}}\theta) \geq c'' \operatorname{dist}(k_{\mathbf{B}}\theta, 2\pi\mathbb{Z})^2$, the decomposition $1 - D \cos = (1 - D) + D(1 - \cos)$ gives the bound with centre $k_{\mathbf{B}}$; the distance function is Lipschitz, and $\operatorname{dist}(M\theta, 2\pi\mathbb{Z})^2 \leq 2 \operatorname{dist}(k_{\mathbf{B}}\theta, 2\pi\mathbb{Z})^2 + \theta^2/2$, whose last term is absorbed by $N_{\mathbf{B}}^3 \theta^2$. For (22.24) take $A = 4\pi$. If $|\theta| \geq A/N_{\mathbf{B}}$, $\sin(|\theta|/2) \geq |\theta|/\pi$ gives $|D_{N_{\mathbf{B}}}(\theta)| \leq \frac{1}{4}$. On the remaining interval $x = N_{\mathbf{B}}|\theta|/2$ ranges over a compact subset of $(0, 2\pi]$ bounded away from zero, where $|D_{N_{\mathbf{B}}}(\theta)| = |\sin x/x| \{1 + O(N_{\mathbf{B}}^{-2})\}$ and $\sup |\sin x/x| < 1$. Hence $|D_{N_{\mathbf{B}}}(\theta)| \leq 1 - \eta'$ for a fixed $\eta' > 0$, and (22.25) proves (22.24). This includes the transition just above a small multiple of $1/H$, where a crude geometric-sum estimate alone is insufficient. $\square$

[Merge note, batch 100.] Lemma 5.1 gives the second bound and the term $H^3\theta^2$ of the first. The new term $H \operatorname{dist}(M\theta, 2\pi\mathbb{Z})^2$ of (22.23) is what resolves the cells around the resonances $2\pi j/M$; it is the block-level form of the action in Theorem 11.1.

A Bernoulli factor satisfies $|1 - p_k + p_k e^{ik\theta}| = \sqrt{1 - 4v_k \sin^2(k\theta/2)} \leq \exp(-2v_k \sin^2(k\theta/2))$. Dropping every factor outside $\mathbf{B}$, Lemma 22.5 gives

$$|\phi(\theta)| \leq \exp\{-c_3 [H \operatorname{dist}(M\theta, 2\pi\mathbb{Z})^2 + H^3\theta^2]\}, \quad |\theta| \leq 1/(2H), \quad (22.26)$$

$$|\phi(\theta)| \leq e^{-c_3 H}, \quad 1/(2H) \leq |\theta| \leq \pi. \quad (22.27)$$

Lemma 22.6 (Cell bound; source 100-93). *Let $\mathcal{I}_j = \{\theta : |M\theta - 2\pi j| \leq \pi\}$. Uniformly in all integers j ,*

$$\int_{\mathcal{I}_j \cap \{|\theta| \leq 1/(2H)\}} |\phi(\theta)| d\theta \leq \frac{C_3}{M\sqrt{H}} e^{-c_3 j^2}. \quad (22.28)$$

Proof (source 100-93). Write $M\theta = 2\pi j + x$, $|x| \leq \pi$. The exponent in (22.26) is $Hx^2 + \Lambda^{-3}(2\pi j + x)^2$, because $H^3/M^2 = \Lambda^{-3}$. Since Λ^{-3} stays bounded above and below, for large M , $Hx^2 + \Lambda^{-3}(2\pi j + x)^2 \geq (H/2)x^2 + 2\pi^2 \Lambda^{-3} j^2$, by $(u+v)^2 \geq u^2/2 - v^2$ and $H \geq 2\Lambda^{-3}$. Integrating the Gaussian in x with $d\theta = dx/M$ gives the claim, also for truncated cells. $\square$

Proof of Theorem 22.1 (source 100-93). For fixed j , (22.15) gives $\sqrt{B}(\theta_j - 2\pi j/M) = o(1)$. In the coordinate $y = \sqrt{B}(\theta - \theta_j)$ the cell endpoints tend to $\mp\infty$, and (22.26) gives an integrable majorant $C_j e^{-c_3 y^2}$. Lemma 22.4 and dominated convergence give

$$\int_{\mathcal{I}_j} \phi(\theta) d\theta = \sqrt{\frac{2\pi}{B}} e^{-2\pi i j \mu_R - 2\pi^2 j^2 Q_\perp} + o(B^{-1/2}) \quad (22.29)$$

for every fixed j , uniformly in the parameters, with no assumption that μ_R converges: its exact unit-modulus phase is kept. The outer range contributes $O(e^{-c_3 H}) = o(B^{-1/2})$, and Lemma 22.6 bounds the cells with $|j| > j_0$ by $C_3 B^{-1/2} \sum_{|j| > j_0} e^{-c_3 j^2}$. Summing (22.29) over finitely many j and then letting $j_0 \rightarrow \infty$, Fourier inversion at the integer n gives $\mathbb{P}_t(S = n) = (2\pi B)^{-1/2} \{\Theta(\mu_R, Q_\perp) + o(1)\}$, and (4.2) is exact. Finally, the Fourier series in (22.4) is uniformly bounded on compact Q -intervals, and the Gaussian series with l an integer nearest μ gives

$$\Theta(\mu, Q) \geq \frac{1}{\sqrt{2\pi Q}} \exp\left(-\frac{1}{8Q}\right), \quad (22.30)$$

positive on compact Q -intervals, which turns the additive error into a uniform relative one. $\square$

22.5 Integer powers obstruct a uniform single Gaussian

Let $G_\alpha(n) = \exp(F(t) + nt)/\sqrt{2\pi B}$, the single Gaussian G of Section 12 with exact saddle and exact variance, so no continuum approximation enters the comparison.

Corollary 22.7 (Two limits of the single-Gaussian ratio; source 100-93). *For every fixed $\Lambda_0 > 0$ and each sufficiently large integer power $\alpha = p$ there are integer indices $n_p^{(0)}$ and $n_p^{(1/2)}$ with $\Lambda \rightarrow \Lambda_0$ along which*

$$\frac{a_p(n_p^{(0)})}{G_p(n_p^{(0)})} \rightarrow \Theta\left(0, \frac{\pi^2}{3\Lambda_0^3}\right) > 1, \quad (22.31)$$

$$\frac{a_p(n_p^{(1/2)})}{G_p(n_p^{(1/2)})} \rightarrow \Theta\left(\frac{1}{2}, \frac{\pi^2}{3\Lambda_0^3}\right) < 1. \quad (22.32)$$

Thus no single-Gaussian relative equivalent holds uniformly over the integer-power critical array.

Proof (source 100-93). For each large integer p let $M_p > e^4$ be the unique root of $p(\log M_p - 1)/M_p^{1/3} = \Lambda_0$ on that branch (unique by differentiation; it exists because the maximum at $M = e^4$ grows linearly with p), and let n_p^0 be the corresponding real mean. At fixed $\alpha = p$, exactly,

$$\frac{d\mu_R}{dn} = \frac{B_{RS}}{B} \sim \frac{1}{M_p}, \quad \frac{dn}{dM} = \frac{B}{MH} \sim M_p. \quad (22.33)$$

On an n -interval of length DM_p with fixed $D > 2$, M changes by $O(1)$ and $\Lambda = \Lambda_0 + o(1)$ uniformly (first on a bounded M -interval by the second identity, then on its image), and by the first identity μ_R traverses $D + o(1)$ units. So the interval contains real indices where μ_R is an integer and where it is a half-integer; rounding them to integers moves μ_R by $O(1/M_p)$ and keeps $\Lambda \rightarrow \Lambda_0$. Theorem 22.1 gives the limits. At phase 0 the nonconstant Fourier terms are positive. At phase $\frac{1}{2}$, pairing terms gives $\Theta(\frac{1}{2}, Q) - 1 = 2 \sum_{m \geq 1} \{-e^{-2\pi^2 Q(2m-1)^2} + e^{-2\pi^2 Q(2m)^2}\} < 0$, and $\Theta(\frac{1}{2}, Q) > 0$ by the Gaussian form. $\square$

This is a qualitative nonuniformity statement. It does not contradict the compact-fixed-power expansions (Theorem 2.2, Section 21.3), in which $\Lambda \rightarrow 0$, and it supplies no effective power or index at which either limit becomes numerically accurate (source 100-93).

22.6 Theta-adjusted inversion

For fixed $\alpha > 0$, continue the exact saddle quantities to real $x > 0$ (the saddle $t(x)$ of Section 16.4) and define

$$\Psi_{\Theta}(x) = F(t(x)) + xt(x) - \frac{1}{2} \log(2\pi B) + \log \Theta(\mu_R, Q_{\perp}) = \Psi_0(x) + \log \Theta(\mu_R, Q_{\perp}), \quad (22.34)$$

that is, the Gaussian inverse Ψ_0 of (16.7) plus the logarithm of the theta factor. The quantifiers are joint: α is fixed for differentiation or inversion and then tends to infinity through critical locations. Nothing here is a fixed- α , $n \rightarrow \infty$ statement (source 100-93).

Proposition 22.8 (Derivative control; source 100-93). *Uniformly on $\mathfrak{J}$, $\Psi'_{\Theta}(x) = t(x) + O(M^{-1})$ and $\Psi'_{\Theta}(x)/t(x) = 1 + O(M^{-1/3})$. In particular the real interpolation is eventually strictly increasing on each such large- M interval.*

Proof (source 100-93). All sums are locally differentiable in t by exponential domination. With $\rho_{RS} = B_{RS}/B$, $r_1 = 0$ and $r_k = 1$ for $k \geq 2$, exponential tilting gives

$$\frac{dx}{dt} = -B, \quad \frac{dt}{dx} = -\frac{1}{B}, \quad \frac{d\mu_R}{dx} = \rho_{RS}, \quad \frac{dB}{dt} = -\sum_{k \geq 1} k^3 v_k (1 - 2p_k). \quad (22.35)$$

Since $Q_{\perp} = \text{Var}(R_{\geq 2} - \rho_{RS}S)$ and $\text{Cov}(R_{\geq 2} - \rho_{RS}S, S) = 0$, the derivative of ρ_{RS} cancels at the point of differentiation, and

$$\frac{dQ_{\perp}}{dt} = -\sum_{k \geq 1} (r_k - \rho_{RS}k)^2 k v_k (1 - 2p_k), \quad (22.36)$$

which can also be checked by differentiating $B_{RR} - B_{RS}^2/B$. The edge moments give $\sum k^3 v_k = O(M^3 H)$, so $|(\log B)'(x)| = O((MH)^{-1})$. By (22.14), $|1 - \rho_{RS}M| = O(H/M)$; with $d = k - M$, $(1 - \rho_{RS}k)^2 \leq C[(H/M)^2 + (d/M)^2]$, so $\sum_{k \geq 2} k(1 - \rho_{RS}k)^2 v_k \leq C[\frac{H^2}{M^2}(MH + H^2) + \frac{1}{M^2}(MH^3 + H^4)] = O(H^3/M) = O(M)$. Part one contributes $O(t/M^2)$ to the absolute $Q_{\perp}$ -sum and $O(t)$ to the absolute B -sum; the range $2 \leq k \leq M/2$ contributes $O(M^2 e^{-c_3 \alpha})$ and $O(M^4 e^{-c_3 \alpha})$. Hence $|dQ_{\perp}/dt| = O(M)$ and $|dQ_{\perp}/dx| = O((MH)^{-1}) = O(M^{-5/3})$. The theta series and its first derivatives in μ and Q converge uniformly on compact positive Q -intervals, and (22.30) bounds the derivatives of $\log \Theta$. Also $(F(t(x)) + xt(x))' = t(x)$ exactly. Since $d\mu_R/dx \sim 1/M$, these bounds give the first assertion, and $t \sim 1/H$ gives the second. No derivative of the unspecified forward $o(1)$ error is taken. $\square$

At fixed α ,

$$\frac{dt}{dM} = -\frac{1}{MH}, \quad \frac{dx}{dM} = \frac{B}{MH} \sim M, \quad \frac{d \log \Lambda}{dM} = \frac{1/(\log M - 1) - 1/3}{M}, \quad (22.37)$$

so if x_0 has saddle parameters M_0, t_0 , then for $|h| = o(M_0^{2/3})$

$$M(x_0 + h) - M_0 = O(|h|/M_0) = o(M_0^{-1/3}), \quad \frac{\Lambda(x_0 + h)}{\Lambda(x_0)}, \quad \frac{t(x_0 + h)}{t_0} = 1 + O(|h|/M_0^2). \quad (22.38)$$

These hold first on a slightly enlarged compact Λ -interval, and by continuity on bounded multiples of $1/t_0 \asymp M_0^{2/3}$; hence $\Psi'_{\Theta} = t_0(1 + o(1))$ there. The coefficients increase strictly for $n \geq 1$ by the injection of Section 9 (source 100-93's Lemma 6.2, also for nonintegral α).

Theorem 22.9 (Theta-adjusted inverse; source 100-93). *Suppose the real critical-window solution $n_\Theta(Y)$ of $\Psi_\Theta(n_\Theta) = Y$ has $M_0 \rightarrow \infty$ and Λ_0 in a fixed compact positive interval, and let $N_\alpha(Y)$ be as in Theorem 2.3. Then*

$$N_\alpha(Y) - n_\Theta(Y) = o(1/t_0) = o(M_0^{2/3}). \quad (22.39)$$

More precisely, there is $\varrho \rightarrow \infty$ with $\varrho = o(1/t_0)$ such that

$$\log a_\alpha(\lfloor n_\Theta - \varrho \rfloor) < Y < \log a_\alpha(\lceil n_\Theta + \varrho \rceil), \quad (22.40)$$

so $\lfloor n_\Theta - \varrho \rfloor < N_\alpha(Y) \leq \lceil n_\Theta + \varrho \rceil$. At sequence values,

$$\Psi_\Theta^{-1}(\log a_\alpha(n)) = n + o(1/t(n)). \quad (22.41)$$

Proof (source 100-93). Theorem 22.1 and positivity give $\log a_\alpha(m) = \Psi_\Theta(m) + o(1)$ uniformly at critical integers; let $e_{\max} \rightarrow 0$ be the supremum of this error on a slightly enlarged compact interval, put $\varsigma = \sqrt{e_{\max} + t_0}$ and $\varrho = \varsigma/t_0$, so $\varsigma \rightarrow 0$, $e_{\max} = o(\varsigma)$, $t_0 = o(\varsigma)$, $\varrho \rightarrow \infty$ and $\varrho = o(1/t_0)$. By Proposition 22.8 and (22.38), $\Psi_\Theta(\lfloor n_\Theta - \varrho \rfloor) = Y - \varsigma(1 + o(1))$ and $\Psi_\Theta(\lceil n_\Theta + \varrho \rceil) = Y + \varsigma(1 + o(1))$; rounding costs $O(t_0) = o(\varsigma)$ and the forward error is $o(\varsigma)$, which proves (22.40). Strict monotonicity gives the rest; at a sequence value the threshold is the index itself. $\square$

The brackets are asymptotic, not certified at any finite parameter. If a certified forward error bound e and a slope lower bound $m_\bullet > 0$ were available on a neighbourhood, endpoints at distance d with $m_\bullet d > e$ would give certified inequalities by the same argument; no such bound is supplied. The allowed discrepancy $o(M^{2/3})$ may diverge, so nearest-integer recovery is not justified in this regime (source 100-93).

Remark 22.10 (The theta shift of the threshold; [write]). Let $n_*(Y)$ be the Gaussian inverse of Section 16.4, $\Psi_0(n_*) = Y$; in the critical regime $\Psi'_0 = t - \kappa_3/(2B^2) = t(1 + o(1))$, since $\kappa_3 = O(M^3H)$ and $B^2 \asymp M^4H^2$. Then

$$n_\Theta(Y) = n_*(Y) - \frac{\log \Theta(\mu_R, Q_\perp)}{t} + o\left(\frac{1}{t}\right), \quad (22.42)$$

with $\mu_R, Q_\perp, t$ evaluated at either point. Indeed $\Psi_0(n_\Theta) - \Psi_0(n_*) = -\log \Theta$ at n_Θ , which is $O(1)$; so $|n_\Theta - n_*| = O(1/t)$, on which window (22.38) gives $\Psi'_0 = t(1 + o(1))$, and μ_R moves by $O(1/(tM)) = o(1)$ and $Q_\perp$ by $o(1)$, so $\log \Theta$ changes by $o(1)$. With Theorem 22.9, the threshold $N_\alpha(Y)$ is displaced from the single-Gaussian inverse by $-\log \Theta/t$, oscillating with the phase μ_R . This displacement is of order at most $1/t \asymp M^{2/3}$, and of exact order $1/t$ at phases bounded away from the zeros of $\log \Theta(\cdot, Q_\perp)$. These occur twice per phase period: with $q = e^{-2\pi^2 Q}$ one has $\Theta(\mu, Q) = \vartheta_3(\pi\mu, q)$, whose Jacobi product $\prod_{m \geq 1} (1 - q^{2m})(1 + 2q^{2m-1} \cos 2\pi\mu + q^{4m-2})$ is even and strictly decreasing in $\mu \in [0, \frac{1}{2}]$, and $\Theta(0, Q) > 1 > \Theta(\frac{1}{2}, Q)$, so $\log \Theta(\cdot, Q)$ vanishes exactly at $\mu \equiv \pm\mu_0(Q) \pmod{1}$ for one $\mu_0(Q) \in (0, \frac{1}{2})$; there the displacement is $o(1/t)$. In particular it is of exact order $1/t$ near $\mu_R \in \mathbb{Z}$ and near $\mu_R \in \mathbb{Z} + \frac{1}{2}$. (Dated note, 5 October 2026: the first version of this sentence read “displaced from the single-Gaussian inverse by $-\log \Theta/t$, of exact order $1/t \asymp M^{2/3}$ and oscillating with the phase μ_R ”; an independent check made by the intake after the write found that wording overstated at the two zeros of $\log \Theta$ per period, and the rest of the remark valid.) By the proof of Corollary 22.7 the displacement is near $-\log \Theta(0, \cdot)/t < 0$ and near $-\log \Theta(\frac{1}{2}, \cdot)/t > 0$ on every index interval of length a fixed multiple of M , so no constant correction of order $1/t$ is valid.

Proposition 22.11 (Leading logarithm; source 100-93). *Uniformly in the critical regime, with $K = \sqrt{2n}$,*

$$\log a_\alpha(n) = \alpha K(\log K - 1) + O(M) \sim \alpha K(\log K - 1), \quad (22.43)$$

and the same holds for $\Psi_\Theta(n)$.

Proof (source 100-93). Let $m = \lfloor M \rfloor$. For large critical parameters ϖ is nonnegative on $2 \leq k \leq m$ and negative for $k > M$. Since $0 \leq \log(1 + e^\varpi) - \max(\varpi, 0) \leq e^{-|\varpi|}$, the edge and frozen estimates give $F(t) = \alpha \log(m!) - t\{m(m+1)/2 - 1\} + O(H)$, the part-one term being $O(1)$. By (22.13), $n - m(m+1)/2 = O(MH)$, while $t = O(1/H)$, and Stirling's formula gives $F(t) + nt = \alpha M(\log M - 1) + O(M)$. The variance logarithm is $O(\log M)$ and $\log \Theta = O(1)$. Also $K - M = O(H)$, and the mean value theorem gives $\alpha\{K(\log K - 1) - M(\log M - 1)\} = O(\alpha H \log M) = O(M)$. The main term is of order $M^{4/3}$, which gives the relative equivalence. $\square$

Solving the leading equation $Y = \alpha K(\log K - 1)$ gives exactly the Lambert data of (2.10): $K_0 = Y/(\alpha W_0(Y/(\alpha e)))$ and

$$n_0 = \frac{K_0^2}{2} = \frac{Y^2}{2\alpha^2 W_0(Y/(\alpha e))^2}, \quad (22.44)$$

with the principal branch W_0 (source 100-93 writes z_L, n_L ; its W is the width H). For $Y = \log a_\alpha(n)$ with critical n , Proposition 22.11 gives $n_0/n \rightarrow 1$. Conversely, for a joint sequence (α, Y) with $K_0 \rightarrow \infty$ and $\Lambda_0 = \alpha(R_0 - 1)/K_0^{1/3}$ in a fixed compact subset of $(0, \infty)$, trial-root bracketing as in Corollary 22.2 shows that the exact saddle at n_0 has $M/K_0 \rightarrow 1$; the saddles at $(1 \pm \varepsilon)n_0$ stay in a slightly enlarged compact critical interval, and their leading logarithms divided by Y tend to $\sqrt{1 \pm \varepsilon}$. Letting $\varepsilon \downarrow 0$,

$$N_\alpha(Y) \sim \frac{Y^2}{2\alpha^2 W_0(Y/(\alpha e))^2}, \quad (22.45)$$

and a nearby critical solution of $\Psi_\Theta(n) = Y$ exists, to which Theorem 22.9 then applies. Formula (22.44) by itself is only a leading relative inverse (source 100-93).

[Merge note, batch 100.] This is the leading term of Theorem 2.3, now in a joint limit in which α grows; Theorem 2.3 and its corrections are fixed- α statements and are not extended here.

22.7 Relation to the fixed-power sections

The satellites are the resonances of Section 11. The damping $2\pi^2 Q_\perp j^2$ of the j th satellite is, by (22.12), $\frac{2\pi^4 j^2}{3} \frac{H^3}{M^2} + o(1)$: exactly the action of Theorem 11.1 at $\ell = j$. At fixed α that action is of order $M/s^3 \rightarrow \infty$, so the noncentral arcs are exponentially small; in the critical window $H^3/M^2 = \Lambda^{-3}$ is bounded and every satellite survives. Section 11 says that its theorem “does not yet determine the shifted secondary saddles, their phases, or the contribution of their full neighborhoods”. In the critical window Theorem 22.1 determines all three: the centres $\theta_j = 2\pi j B_{RS}/B = 2\pi j/M + o(B^{-1/2})$, the phases $-2\pi j \mu_R$, and the full-cell contributions (22.29), with relative error $o(1)$ and no effective constant.

Questions of Section 13. Project 1 (an effective multi-saddle expansion) is answered at leading order in the critical window only; at fixed α it stays open. Project 4 (growing moments) gains one sharp range: for $\Lambda \asymp 1$, that is $\alpha \asymp M^{1/3}/\log M$, single-saddle dominance fails (Corollary 22.7) and the theta law holds; whether single-saddle dominance holds for every $\alpha = o(M^{1/3}/\log M)$, and the localization near maximal-norm partitions as $\Lambda \rightarrow \infty$, stay open (questions 100-93.4–5). Project 6 (conditioned fluctuations of the number of parts) is touched: $Q_\perp$ is the regression variance of the number of parts at least two given the size, and (22.11) exhibits the cancellation that project anticipates ($B_{RR} \sim H$ against $Q_\perp \sim \pi^2 H^3/(3M^2)$); this is proved in the window only. Dated notes at those projects record this.

Not an explanation of the fixed- α finite-size errors. Source 100-93's theorem does not explain the residuals of Section 12 at $\alpha = 3, 4, 5$, $n = 5000$. It needs the lower bulk to be frozen ($\alpha/\log M \rightarrow \infty$), which fails at fixed α , and at fixed α one has $\Lambda \rightarrow 0$, outside its window. Numerically, its formula evaluated with the exact moments at this report's exact saddles gives $\Theta(\mu_R, Q_\perp) = 1.0000055, 0.98057$ and 1.19429 at $\alpha = 3, 4, 5$, $n = 5000$ (exact $Q_\perp = 0.535, 0.229, 0.114$, larger than $\pi^2/(3\Lambda^3)$ because the lower bulk still contributes), against the observed ratios $0.96163, 1.23052$ and 1.58110 ; its own diagnostics at $\alpha = 8$ (Table 10) still differ from Θ by 3% and 16%. ([write]; 30-digit sums, uncertified.) A direct decomposition of those residuals into Fourier cells is recorded, as numerical evidence only, in the batch-100 note of Section 16.3; the formula that would explain them is question 100.6.

The phase average is the single Gaussian ([write]). For every $Q > 0$,

$$\int_0^1 \Theta(\mu, Q) d\mu = 1, \quad (22.46)$$

since only the term $j = 0$ of the uniformly convergent Fourier series in (22.4) survives integration. So the single Gaussian G is the phase average of the theta law, and Corollary 22.7's two limits lie on either side of it. Source 100-93 does not state this.

22.8 Relation to the lattice-peak transfer theorem

Part II of `oeis-sequence-asymptotics/a357825-theta-ballot-power-sums` (written in 2b257ba94, labels `tbs:lpt:`) proves a general lattice-peak transfer theorem [32]: for a positive row $C_{n,k}$ on a moving lattice of span d_n , raised to a power r_n , its Theorem `tbs:lpt:thm:transfer` gives $S_n(r_n)/\Omega_n^{r_n} = \vartheta(q_n, \delta_n) + o(1)$ with $\vartheta(q, \delta) = \sum_{j \in \mathbb{Z}} e^{-q(j-\delta)^2/2}$, under a local quadratic law (H1), Gaussian or exponential domination (H2) or (H2e), and interior completeness (H3). Source 100-93 is not a power sum, and its theorem is not a corollary of that one. It is the same mechanism in Fourier-dual form. By (22.4),

$$\Theta(\mu, Q) = (2\pi Q)^{-1/2} \vartheta(1/Q, \mu),$$

and $\sqrt{2\pi B} \mathbb{P}_t(S = n) = \sqrt{2\pi B} \sum_l \mathbb{P}_t(S = n, R_{\geq 2} = l)$. Read with the row $l \mapsto C_{n,l} = \mathbb{P}_t(S = n, R_{\geq 2} = l)$, $r_n = 1$, $d_n = 1$, peak $x_n = \mu_R$ (so $\delta_n \equiv \mu_R \pmod{1}$), $q_n = 1/Q_\perp \rightarrow 3\Lambda^3/\pi^2$ and $\Omega_n = (2\pi\sqrt{BQ_\perp})^{-1}$ (the bivariate Gaussian density of $(S, R_{\geq 2})$ at its mean, since $BQ_\perp = BB_{RR} - B_{RS}^2$), the conclusion of the transfer theorem is exactly Theorem 22.1. But:

- hypothesis (H1) for this row is a local limit theorem for the pair $(S, R_{\geq 2})$ on the fibre $S = n$, which source 100-93 does not prove: it integrates the one-dimensional characteristic function over cells, and its Gaussian satellites are the Fourier modes of ϑ , not lattice sites;
- (H2e) would follow from log-concavity of $l \mapsto \mathbb{P}_t(S = n, R_{\geq 2} = l)$ (Part II's Proposition `tbs:lpt:prop:logconcave`), which is open;
- in Part II's normalization the critical narrowing comes from a growing power r_n with curvature scale $\sigma_n \rightarrow \infty$; here $r_n = 1$ and the growing α narrows the conditional law of $R_{\geq 2}$ itself to width $\sqrt{Q_\perp} = O(1)$, so σ_n stays bounded (the transfer theorem's proof uses only (H1)–(H3)).

Proving (H1) and (H2e) for this row is question 100-93.7. The further correspondences are of the same kind. Corollary 22.7 is the two-cluster-value conclusion of Part II's Corollary

`tbs:lpt:cor:clusters`, derived here from Theorem 22.1. Theorem 22.9 and Remark 22.10 have the form of Part II's one-step inverse transfer (Theorem `tbs:lpt:thm:inverse`) with $G = \Psi_0$ and the theta term $\log \Theta$, whose regularity ratios hold here ($G''/G'^2 \asymp -1/(Bt^2) = O(H/M^2)$, $(\log \Theta)' / G' = O(H/M)$); but that theorem assumes $G' \rightarrow \infty$ or G' bounded below, whereas here $G' = t \rightarrow 0$, so Remark 22.10 proves the first-order statement directly, and its last sentence is the conclusion of Part II's Corollary `tbs:lpt:cor:no-constant` in this case. The phase average (22.46) is Part II's `tbs:lpt:eq:theta-average` after the factor $(2\pi Q)^{-1/2}$. Not covered by Part II: the exclusion of part one (Section 22.3.1), the frozen-tail product, and the transition range of Lemma 22.5. A reciprocal note in Part II is proposed with this batch.

22.9 Numerical diagnostics (source 100-93)

Table 10 estimates the normalized multiplier $\sqrt{2\pi B} \mathbb{P}_t(S = n) = a_\alpha(n)/G_\alpha(n)$. These are Fourier diagnostics, not exact integer coefficients or interval-certified enclosures; the theorems do not depend on them. For each integer power and target Λ , the program selects a real saddle with μ_R integral or half integral near a reference large-root saddle, rounds the mean index to the nearest integer and re-solves the discrete saddle equation there. The data files record the integer n , the actual M and Λ , and the final phase discrepancy; every displayed Θ uses the finite discrete moments and the final exact-saddle phase. Across the 18 cases the phase discrepancy satisfies $M|\mu_R - \mu_{\text{target}}| < 0.470$.

Target Λ	α	Target phase	Estimated $a_\alpha(n)/G_\alpha(n)$	$\Theta(\mu_R, Q_\perp)$	ratio -1
4	8	0	1.7428029039	1.6909021187	$3.069 \cdot 10^{-2}$
4	8	$\frac{1}{2}$	0.2999985664	0.3583997652	$-1.629 \cdot 10^{-1}$
4	12	0	1.7566327608	1.7554693502	$6.627 \cdot 10^{-4}$
4	12	$\frac{1}{2}$	0.3063539563	0.3122316641	$-1.882 \cdot 10^{-2}$
4	20	0	1.7591230654	1.7597783471	$-3.724 \cdot 10^{-4}$
4	20	$\frac{1}{2}$	0.3085945296	0.3092996789	$-2.280 \cdot 10^{-3}$
4	32	0	1.7596083198	1.7597997181	$-1.088 \cdot 10^{-4}$
4	32	$\frac{1}{2}$	0.3090890657	0.3092866384	$-6.388 \cdot 10^{-4}$
4	48	0	1.7597309948	1.7597987101	$-3.848 \cdot 10^{-5}$
4	48	$\frac{1}{2}$	0.3092172775	0.3092871616	$-2.260 \cdot 10^{-4}$
5	12	0	2.4477171475	2.4470400564	$2.767 \cdot 10^{-4}$
5	12	$\frac{1}{2}$	0.0422729991	0.0443723681	$-4.731 \cdot 10^{-2}$
5	20	0	2.4567852457	2.4590372192	$-9.158 \cdot 10^{-4}$
5	20	$\frac{1}{2}$	0.0424850721	0.0425829467	$-2.298 \cdot 10^{-3}$
5	32	0	2.4584706021	2.4591006134	$-2.562 \cdot 10^{-4}$
5	32	$\frac{1}{2}$	0.0425481285	0.0425747323	$-6.249 \cdot 10^{-4}$
5	48	0	2.4588810005	2.4590996451	$-8.891 \cdot 10^{-5}$
5	48	$\frac{1}{2}$	0.0425653301	0.0425746945	$-2.200 \cdot 10^{-4}$

Table 10: Fourier diagnostics at integer indices (source 100-93's Table 1; the last column is the estimate divided by Θ , minus one). The displayed precision is conservative relative to the recorded sensitivity checks; target parameters differ slightly from the resolved discrete-saddle parameters recorded in the data files. The first two rows deliberately include an earlier, visibly preasymptotic power.

The limiting multipliers are $\Theta(0, \pi^2/(3\Lambda^3)) = 1.759797211458$ and $\Theta(\frac{1}{2}, \cdot) = 0.309288136393$ at $\Lambda = 4$, and 2.459097973253 and 0.042574972058 at $\Lambda = 5$. The cases approach these strongly non-unit values. At $\alpha = 8$, and less so at $\alpha = 12$, rare omissions of low parts have not yet become negligible in the count $R_{\geq 2}$, which explains the appreciable mismatch; the proof asserts no effective onset. Including part one in the count would predict values close to 1: at

$\alpha = 32$ the wrong phase-0 and half-phase values are about 0.9948 and 1.0052 for $\Lambda = 4$, and 0.9914 and 1.0086 for $\Lambda = 5$.

Method and accuracy limits (source 100-93). The program keeps the frozen baselines $\lfloor M \rfloor - 1$ (for $R_{\geq 2}$) and $\lfloor M \rfloor (\lfloor M \rfloor + 1)/2 - 1$ (for S) as exact integers: below the upper edge deviations are rare holes, above it rare additions; means are accumulated as corrections, v_k is formed from the rare probability, and $Q_{\perp}$ is evaluated through (22.11), avoiding the subtraction of nearly equal variances; part one is kept separately in S . The centred characteristic function is integrated around θ_j in the coordinate $y = \sqrt{B}(\theta - \theta_j)$, with 64-bit-mantissa extended precision, 160-point Gauss–Legendre quadrature on $[-11, 11]$, degree-20 or degree-30 discrete-cumulant acceleration with exact rational Bernoulli cumulant coefficients, and an independent check of the full rare-variable product at five points per satellite; satellites $|j| \leq 8$ are kept for $\Lambda = 4$ and $|j| \leq 10$ for $\Lambda = 5$. Omitted arcs are assessed by pointwise algebraic envelopes built from consecutive edge blocks, not by the limiting theta tail; their integrated normalized estimates are below $3.4 \cdot 10^{-14}$, but the integration uses ordinary adaptive quadrature, not interval arithmetic. The discarded-variable coupling bounds are below $6.4 \cdot 10^{-22}$; cutoff, window, node-count, Taylor-degree and satellite-count changes support about 12 significant digits. An independent C++ binary128 computation evaluates the full product at every node without acceleration: for $\alpha = 8$, target $\Lambda = 4$, phase 0, $n = 3320242$ its retained-arc multiplier is 1.742802903877193692 (96 and 128 nodes agree through 33 digits; the accelerated value differs by $6.2 \cdot 10^{-15}$), and for $\alpha = 12$, target $\Lambda = 5$, half phase, $n = 21224105$, it is 0.042272999079952589 (31 digits; difference $1.6 \cdot 10^{-16}$). These digits concern retained arcs only.

[Merge note, batch 100.] The intake reran source 100-93's exact identity checks (eight algebraic identities, among them (22.11), (22.14), (22.18) and (22.36)) and its consistency checks of the 18 saved rows on copies, and compiled and ran the binary128 program, reproducing all four of its outputs; the main extended-precision runs were not repeated.

22.10 Further questions and research (source 100-93)

Source 100-93's six questions are kept with its numbers; the seventh is added by this write. No claim of source 100-93 was found to be wrong; every step of its main chain was checked at placement.

100-93.1 Effective errors. Explicit constants for the edge approximation, the satellite remainders and the integrated complement, to make finite threshold brackets certifiable (overlaps project 8 and 100.2).

100-93.2 Higher satellite corrections. Uniform cumulant corrections, summed as theta derivatives; the cubic bound (22.21) need not give the optimal rate. (Part II of the lattice-peak report has a theta-derivative calculus for its all-orders transfer theorem.)

100-93.3 Sharper inversion. Enough uniform correction orders for a logarithmic forward error $o(t)$ and an inverse error $o(1)$; exact-range rounding would then need its own error-below-one-half argument.

100-93.4 Small critical parameter. Uniformity as $\Lambda \downarrow 0$, where the first Fourier correction has size $\exp[-2\pi^4/(3\Lambda^3)]$, matching the single-Gaussian regime; at fixed α this is the scale of Theorem 11.1 (overlaps project 4).

100-93.5 Large critical parameter. $\Lambda \rightarrow \infty$, where the conditional variance decreases and the multiplier concentrates on the count phase; the lower bound (22.30) is not uniform there. The diagonal A292190 ($\alpha = n$) has $\Lambda \rightarrow \infty$ and lies here (overlaps project 4, 75-01.4, 75-06.4).

100-93.6 Other inside weights. $w_k = \exp(\alpha \log k + g(k))$ with a comparable upper-edge width: which lower parts stay non-frozen, and how their finite factors change the theta law (overlaps project 7).

100-93.7 Fibre lattice peaks ([write]). Prove (H1) and (H2e) of the lattice-peak transfer theorem for the row $l \mapsto \mathbb{P}_t(S = n, R_{\geq 2} = l)$ in the critical window: a local limit theorem for $(S, R_{\geq 2})$ on the fibre $S = n$, uniform for $|l - \mu_R| \leq C$, and a summable majorant, for instance from log-concavity in l . This would make Theorem 22.1 an instance of Theorem tbs:lpt:thm:transfer. Sketch: the two-dimensional Fourier integral over $(\theta, \omega) \in [-\pi, \pi]^2$ with the methods of Lemmas 22.4–22.6; missing: the two-dimensional complement bound, and log-concavity.

Source 100-89’s question of a uniform theory for growing α (question 100.7) is answered in the window by this section.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the five results this write supplied in Sections 21 and 22, made by the intake after the write, found Lemma 21.3, Proposition 21.7, Remark 21.5 and the phase average (22.46) valid, and Remark 22.10 valid with one overstatement (“of exact order $1/t$ ”, false at the two zeros of $\log \Theta$ per period); that wording and an understated displacement term in Remark 21.5 are corrected where they stand, with dated notes. It found no counterexample and no gap in any proof chain. It reread every step and tested them numerically against exact coefficients of $\prod_k (1 + k^\alpha q^k)$, computed by a big-integer recursion to $n = 4000$ for $\alpha = 1, 2, 3$ (the first two are A022629 and A092484) and by tilted binary64 recursions for $\alpha = 1.5, 6, 8, 10$. For Lemma 21.3 it evaluated the boundary functional at complex α by Euler–Maclaurin: it reproduces the printed $\xi_0(1), \dots, \xi_3(1), \xi_0(2), \xi_1(2)$ to 19–25 digits, satisfies the Cauchy–Riemann equations to 10^{-11} or better at four points (two with $\Re \alpha < 1$), and its circle means about $\alpha = 1, 2$ reproduce the real constants to $4 \cdot 10^{-10}$. For Proposition 21.7 the numerical derivative of the inverse model matches (21.23) to $5 \cdot 10^{-11}$ or better ($\alpha = 1, 1.5, 2, 3, n = 250\text{--}4000, J = 1, 2, 3$), and both error terms stay bounded on their scales. For Remark 21.5 the lattice–integral replacement alone is a stable multiple of t^{D+1} ($\alpha = 0.5, 2, 3, n \leq 8000$), and at $\alpha = 1$ the full residual over $\varepsilon_{\sharp}^J$ stays bounded, tending to $-3/8$ for $J = 1$ (the sharp $\mathcal{E}_1$); at $\alpha = 2, 3$ and $n \leq 4000$ the residual is dominated by the noncentral Fourier cells of Section 16.3, exponentially small only asymptotically. For (22.46) both forms of (22.4) integrate to 1 to 16 digits for $0.02 \leq Q \leq 3$. For Remark 22.10, at exact lattice saddles ($\alpha = 8\text{--}32, \Lambda = 4, 5$, about three phase periods) $|(n_\Theta - n_*)t + \log \Theta| \leq 6 \cdot 10^{-6}$ while $(n_\Theta - n_*)t$ ranges over $[-0.88, 3.09]$ with both signs; at exact thresholds ($\alpha = 6, 8, 10, n \approx 2\text{--}3 \cdot 10^4$) the displacement changes sign with the phase and correlates with $-\log \Theta/t$ at 0.90–0.99, with magnitudes still preasymptotic; this is an illustration only. This was a careful reading with numerical tests, not a formal verification or an external review. A numerical-method note: the closed form (21.14), evaluated with `mpmath.nsum` at $(r, \alpha) = (2, 4.5)$ and $(3, 5.5)$, differs from the Euler–Maclaurin functional by about $2 \cdot 10^{-7}$, the accuracy of `nsum` on these slowly convergent sums (compare the merge note after (21.12)); it is not a defect of the formula.

A Reproduction guide and file manifest

Besides this source, its compiled PDF and the README, the report directory contains the two sources' programs and recorded outputs under prefixed names.

File	Purpose
code/40-norm-moments-verify.py	Exact integer sequences, independent recurrence, and numerical saddle/Edgeworth diagnostics (source 40).
code/40-norm-moments-derive_series.py	Symbolic generation of the boundary derivatives and logarithmic coefficients.
code/40-norm-moments-derive_inverse.py	Symbolic reversion through six inverse-logarithmic orders.
code/40-norm-moments-check_continuum.py	Independent continuum quadrature diagnostics.
code/40-norm-moments-check_resonances.py	Direct numerical product moduli at the first resonance.
data/40-norm-moments-*	Computed integer lists, CSV diagnostics, symbolic output, execution logs, and the requirements file.
code/48-a022629-coefficients.py	Source 48's operator construction of six forward and six inverse coefficients.
code/48-a022629-verify.py	Source 48's exact table through $n = 6400$ and its saddle diagnostics.
code/48-a022629-Makefile	Source 48's build and verify targets (delivery layout).
data/48-a022629-*	Source 48's exact table, recorded checks, symbolic output, table fragment, and requirements.
48-a022629-PROVENANCE.md	Source 48's sources and verification record.
code/75-01-jensen-verify.py	Source 75-01's exact tables, log-concavity audits and saddle comparisons.
code/75-01-jensen-derive_series.py	Source 75-01's exact check of five forward and five inverse coefficients.
code/75-01-jensen-extra_checks.py	Source 75-01's recurrence, Jensen root counts, precision comparison and Gaussian inverse.
data/75-01-jensen-*	Source 75-01's exact table to $n = 10000$, audits, saddle tables, symbolic output and requirements.
75-01-jensen-BUILD_AND_VALIDATION.txt, 75-01-jensen-OEIS_PROPOSED_NOTES.md	Source 75-01's build record and draft OEIS notes.
code/75-06-powers-formal.py	Source 75-06's exact formal algebra: eight forward, seven inverse coefficients.
code/75-06-powers-numerics.py, code/75-06-powers-checks.py	Source 75-06's exact coefficients and saddle estimates; recurrence and convolution checks.
data/75-06-powers-*	Source 75-06's coefficient, numerical, verification and source-audit records and requirements.
75-06-powers-proposed_oeis_updates.txt	Source 75-06's draft OEIS notes.
code/77-51-fp-*	Source 77-51's producer checks and symbolic generator, three independent checkers, root driver and build script.
data/77-51-fp-*, 77-51-fp-*.md	Source 77-51's receipts, approvals and requirements; proof snapshots, audits and reviews.

File (batch 100)	Purpose
<code>code/100-82-integral-*</code>	Source 100-82's exact formal series ($s_1, \dots, s_{10}, v_2, \dots, v_{10}$), independent low-order check, exact coefficients to $n = 10000$ with integral-saddle comparisons, endpoint-constant illustration, exact-check driver.
<code>code/100-89-boundary-*</code>	Source 100-89's exact boundary coefficients and Gaussian blocks, reconstructions, integer-root and Gamma-constant checks, boundary illustration, exact-coefficient and quadrature comparisons, exact-check driver.
<code>code/100-93-theta-*</code>	Source 100-93's eight exact identity checks and check driver; (<code>-diag-</code>) its Fourier diagnostics, envelope checks, summary assembly, binary128 audit and saved-output check.
<code>data/100-*</code>	Their JSON records and provenance records; source 100-93's diagnostic inputs, nodes, binary128 outputs, results, sensitivity runs and summary.
<code>100-93-theta-diagnostics-README.md</code>	Source 100-93's computational README for its diagnostics.

The mathematical proofs do not require running the programs. The programs write their outputs under the names of the delivered packages, so they should be rerun on a copy of the delivered layout; the README gives the commands. The source uses standard LaTeX packages and no external bibliography processor. No Lean code or machine-checked proof is bundled or implied.

[**Merge note, batch 73O1.**] Source 40's manifest listed its files under their delivered names (`verify.py`, `derive_series.py`, `data/`, `README.md`, `requirements.txt`) with the commands `python verify.py --max-n 5000 --dps 50`, `python derive_series.py`, `python derive_inverse.py`, `python check_continuum.py`, `python check_resonances.py` and two `pdflatex` passes; source 48's were `python code/coefficients.py --order 6` and `python code/verify.py --max-n 6400 --dps 45`. The shipped names are those above; the delivered READMEs are replaced by the report README.

[**Merge note, batch 75.**] Source 75-01's appendix and source 75-06's Section 10.4 give these commands, in their delivered layouts:

```
python code/verify.py --exact --s 1 --nmax 10000 # 75-01; also --s 2, --s 3 with --
nmax 2500
python code/verify.py --s 1 --dps 40
python code/verify.py --s 1 --samples 10000 --dps 60
python code/verify.py --s 2 --samples 1000,2500 --dps 40 # and --s 3
python code/derive_series.py
python code/extra_checks.py
python formal.py --order 8 # 75-06
python checks.py
python numerics.py
bash build.sh
```

Their manuscripts, PDFs and READMEs, source 75-01's `exact_s2.json` and `exact_s3.json`, and source 75-06's `build.sh` and `SHA256SUMS.txt` are not shipped; the README says how to rerun the programs on a copy.

[Merge note, batch 77.] Source 77-51’s README gives `python verify.py`, run from the delivered root; it checks the delivered file manifest and PDF, so it runs only on an extraction of the arrival archive. Its five checkers take an `--output-dir` argument; the README gives the commands. Its manuscript, PDF, README and SHA256SUMS are not shipped.

[Merge note, batch 100.] Sources 100-82 and 100-89 give `bash run_checks.sh` (their exact checks) and `python -0 verify_package.py` (file hashes), run from their delivered roots; source 100-93 adds `python3 -0 checks/check_identities.py`, `python3 -0 checks/diagnostics/verify_summary.py` and the diagnostics commands of its README (shipped as 100-93-theta-diagnostics-README.md). Their manuscripts, PDFs, READMEs, `verify_package.py` and `build_local.sh`, source 100-89’s `check_logarithmic_reversion.py` and its JSON (byte copies of source 100-82’s `check_formal_series.py` and its JSON), source 100-93’s `diagnostics.tex`, `context.tex` and `references_extra.tex`, and its diagnostics checksum list are not shipped; the README says how to rerun the programs on a copy.

B Compact proposed OEIS formula updates

For the entry corresponding to $\alpha = 1, \dots, 5$, write $K = \sqrt{2n}$ and $r = \log K$. The following common formula is proved in Theorem 2.1:

$$\log a_\alpha(n) = \alpha K(r - 1) + \frac{\pi^2 K}{6\alpha r} + \frac{\pi^2 K}{6\alpha r^2} + O_\alpha(K/r^3).$$

It is appropriate to link a reviewed proof before changing a database’s conjectural status. This report has not submitted or edited any OEIS entry. For the array [A292189](#), the result holds down every fixed positive column; it makes no uniform assertion about the main diagonal or columns whose index grows with the row.

[Batch 100, 5 October 2026.] Columns whose index grows like $\alpha \asymp (2n)^{1/6}/\log n$ are now covered, with a theta-corrected equivalent and the leading logarithm $\alpha K(r - 1)$ (Section 22); the main diagonal is not.

Source 48’s suggested A022629 addition

Source 48 suggests the following text for independent editorial review; it has not been submitted.

The logarithmic conjecture follows with the stronger estimate $\log a(n) = \sqrt{2n}(\log \sqrt{2n} - 1) + O(\sqrt{n}/\log n)$. More precisely, with $K = \sqrt{2n}$ and $r = \log K$,

$$\log a(n) = K \left(r - 1 + \frac{\pi^2}{6r} + \frac{\pi^2}{6r^2} + \frac{\pi^2/6 - \pi^4/72}{r^3} + O(r^{-4}) \right).$$

A proof, an all-orders coefficient rule, and a separate relative saddle-point formula are supplied in the accompanying research report.

A particularly short proof of the first assertion uses the partition $(1, 2, \dots, m - 1, m + d)$ with $n = m(m + 1)/2 + d$ for the lower bound, and $a(n) \leq e^{tn} \prod_k (1 + ke^{-tk})$ with $t = \log \sqrt{2n}/\sqrt{2n}$ for the upper bound. Splitting the logarithm into its positive-part contribution and a smoothing remainder of width $O(\sqrt{n}/\log n)$ finishes the argument; see Section 3.1.

[Merge note, batch 7301.] The “accompanying research report” is now this report.

Sources 75-01 and 75-06's suggested notes

Both sources ship draft OEIS notes, not submitted or approved: 75-01-jensen-OEIS_PROPOS ED_NOTES.md and 75-06-powers-proposed_oeis_updates.txt. Beyond the expansion above (source 75-01 through r^{-5} , source 75-06 through r^{-4} , both in the form π^{2j} or u^j), source 75-01 proposes for A022629 the log-concavity gap $\log(a(n)^2/(a(n-1)a(n+1))) \sim \log \sqrt{2n}/(2n)^{3/2}$, eventual hyperbolicity of fixed-degree Jensen polynomials (by the Griffin–Ono–Rolen–Zagier mechanism, which is not new), and the finite observation that $n = 77, \dots, 9999$ are strictly log-concave, explicitly not a proof that 77 is the threshold. Source 75-06 proposes the same formula for the positive columns of A292189 and, with μK_μ for αK and $u = \pi^2$, for the positive columns of A297321. Both insist that a fixed truncation of a logarithmic expansion is not a relative asymptotic, and both ask for review and a broader priority search before use.

C Verification boundaries and source audit of source 48

Item	Boundary of the claim
OEIS status	A022629 was retrieved with the 2018 formula still labeled a conjecture. This is evidence about that entry, not a guarantee of absence of a proof elsewhere.
Repository inspection	The repository README and the Transseries and Inversion README were read; a targeted A022629 code search returned no match. No complete audit of every research artifact is claimed.
Mathematical results	Conventional proofs are given in the body of this article. The proof of the displayed OEIS conjecture is independent of the more elaborate local limit and formal-expansion arguments.
Symbolic calculations	Six forward and six inverse coefficients are generated by exact rational-polynomial arithmetic in c using the stated operator recurrence.
Integer computations	The product recurrence is evaluated through $n = 6400$; the independent logarithmic-derivative recurrence is checked through $n = 100$.
Floating-point diagnostics	The saddle comparisons use 45 decimal digits and explicit analytic truncation bounds. They are not interval-certified evaluations.
Formal verification	No Lean or Rocq/Coq proof of the new analytic theorems is included or claimed. Repository formalization material is a possible integration path, not a certificate of this article.
Novelty	No worldwide priority or “breakthrough” status is certified. The contribution is the supplied proof and its stated extensions.

D Provenance and merge record

Source	Archive (arrival commit)	Main file, PDF	Pin	Author line
40 (base)	OEIS_Distinct_Partition_Norms.zip (f8c3a392a)	article.tex, 22 pp.	1f1981f68	Research report prepared for Vladimir Reshetnikov
48	A022629_Research_Package.zip (3a9518c52)	article.tex, 20 pp.	dc1a7242d	Research report prepared for Vladimir Reshetnikov; developed with AI assistance; proofs and computations supplied for review
75-01	A022629_research_package.zip (4b874cea0)	article.tex, 20 pp.	29aca108e	Research report prepared for Vladimir Reshetnikov (no AI wording)
75-06	OEIS_Norm_Weighted_Partitions.zip (4b874cea0)	article.tex, 26 pp.	29aca108e	Prepared for Vladimir Reshetnikov (no AI wording)
77-51	oeis-fixed-power-asymptotics-result.zip (096ee7b87)	fixed-power-partition-asymptotics.tex, 11 pp.	none	Research report (no AI wording)
100-82	ProveIt_OEIS_A022629_Asymptotics_and_Inversion.zip (60f54ea06)	article.tex, 7 pp.	none	Research note prepared for Vladimir Reshetnikov with OpenAI
100-89	ProveIt_Power_Weighted_Distinct_Partitions_Relative_Expansions.zip (60f54ea06)	article.tex, 9 pp.	none	the same
100-93	ProveIt_Critical_Theata_Power_Weighted_Distinct_Partitions.zip (60f54ea06)	article.tex and three included files, 13 pp.	6591a56b3	the same

Both are manuscripts of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, numbered 40 and 48 there; the placement commit [9df4ba51a](#) staged their files and deleted the archives, which survive in their arrival commits. Source 40 arrived first. Neither cites the other, and their texts share almost nothing (8-word shingle containment 0.004 both ways; the longest common run is a `pdflatex` command line): they are independent. Both pins are ProveIt commits of 1 October 2026; source 48 calls its identifier a “tree response” and does not assert that it pins every subsequent read.

What each contributed. Source 40: the base text, in its order, and everything in it, including the five sequences, P_7 , P_8 , the resonance theorem and the finite-size analysis. Source 48: the second proofs of Proposition 3.2, Lemmas 4.1 and 5.1, Theorem 2.2 and the threshold transfer; Propositions 3.4 and 3.5; Lemma 3.6; the operator route; Corollaries 8.1 and 10.1; the A022629 forms; its computations, projects, OEIS wording and audit table.

Where the merge chose.

- Base: source 40, the more general manuscript (five sequences, one more coefficient, the resonance theorem).
- Notation: source 40's throughout; source 48's symbols renamed as in Section 1.5.
- Source 48's λ -centred Gumbel law is printed as a corollary of Theorem 2.4 with its own proof, since its centring is explicit where source 40's is the exact root $b(t_n)$.
- Source 48's statements that duplicate source 40's (its Theorem 1.1, Theorem 2.4, Lemma 3.1, Lemma 3.2, Theorem 3.3, Theorem 6.1, Proposition 7.2) are printed once, credited, with only their different proofs or forms kept. Its abstract, title page and package-contents paragraph are replaced by this text and the README.
- Both data tables of A022629 are shipped (they agree on $n \leq 5000$).
- The two bibliographies are merged; source 48's OEIS item is source 40's [1].

Batch 75. Sources 75-01 and 75-06 are manuscripts 01 and 06 of batch 75 of the intake, both in the arrival commit 4b874cea0; the placement commit 6ea60e367 staged their files under the prefixes 75-01-jensen- and 75-06-powers- and deleted the archives, which survive in the arrival commit. The prefixes continue this report's source-numbered scheme, with the batch named so that 01- and 06- cannot be read as batch-73 manuscript numbers. Both pin 29aca108ed25d714f31b1316512777fbbdc8e006 (1 October 2026), which already held the archives of sources 40 and 48 unopened. Their texts share 0.55–1.41 % of their word 8-grams with sources 40 and 48 and with each other (preamble, OEIS references and standard displays): they are independent.

What each contributed. Source 75-01: Lemma 16.1 and Remark 16.2, Theorems 16.3, 16.4, 17.1 and 17.2, the smooth approximant of Section 17, the finite data of Tables 3 and 4, the analytic-inverse route of Section 15.2, ten research questions and its OEIS notes. Source 75-06: the second proof of Lemma 16.1, Section 18 (the μ -layer theorems, Δ_7 , Δ_8 , the OEIS identifications), the exact b_8 , the coefficient-structure remark, the finite-size readings, twelve research questions and its OEIS notes.

Where the batch-75 merge chose.

- Placement: both sources were added to this report rather than filed separately; a separate report would have repeated Sections 2–10. The convolution powers were not split off either: their coefficients are the same P_j , and every proof is this report's at n/μ .
- Duplicates: every result already proved here is recorded in Table 2 and not reprinted; second-route text is kept only where the argument differs (Section 15.2, the second proof of Lemma 16.1, and the μ -layer modifications of Section 18).
- Placement in the text: the new sections follow the conclusion, so that every earlier section, theorem, equation and table keeps its number; unnumbered merge notes in the earlier sections point forward.
- Base for the shared new results: source 75-01 for the sharp cumulant scales and the remainder theorem (it proves the full remainder), source 75-06's Proposition 5.4 as the second proof.
- Notation: this report's throughout; both sources' symbols are renamed as in the second table of Section 1.5. New symbols: μ , K_μ , r_μ , $\varepsilon_\sharp$, n_* , Δ_j , $\mathcal{J}$, H_d , τ_n , γ_k , $\mathcal{R}(n)$.

- Source 75-01's finite-size residuals at $\alpha = 2, 3$, which it reads as slow convergence, are kept with its own words and identified by a merge note as the noncentral contribution of Section 11.
- Both sources' novelty paragraphs and abstracts are replaced by Section 15; their priority caveats are kept there.
- Source 75-01's tables for $\alpha = 2, 3$ duplicate source 40's and are not shipped; its $\alpha = 1$ table, which extends the range to $n = 10000$, is.

Batch 77. Source 77-51 is manuscript 51 of batch 77 of the intake (cluster P2), in the arrival commit 096ee7b87; the placement commit aa7345800 staged its 33 support files under the prefix 77-51-fp- and deleted the archive, which survives in the arrival commit. The prefix continues this report's source-numbered scheme, with the batch named as for batch 75. The archive has no wrapper directory; its three `*-optimization-negative-control.txt` files were shortened to `*-negative-control.txt` to keep paths short. Source 77-51 names no ProveIt commit (its approval records name only the producer's working directory) and records no repository search. Its text shares 0.69 % of its word 8-grams with this report and 0.11 % in the other direction: it is independent. Its checksum ledger SHA256SUMS was verified (36/36) and retired.

What it contributed. The fifth proofs recorded in Table 6; Proposition 20.1 (the resummed forms); the cutoff-sum route, the coefficient generation in its variables and the exact-saddle inverse of Section 20.2; Theorem 20.2 with its monotonicity, uniform estimates and constants; Corollary 20.3; the citation of Naranjo and Ramírez; Table 7; four research questions.

Where the batch-77 merge chose.

- Placement: added to this report, not filed separately, because its core theorems are this report's and its family contains this report's at $\beta = 0$.
- Duplicates: recorded in Table 6 and not reprinted; the resummed forms, the cutoff-sum route, the generation in its variables and the exact-saddle inverse are kept because they differ in form or route.
- Placement in the text: Section 20 follows the batch-75 sections and precedes the appendices, so every earlier number is unchanged.
- Notation: this report's throughout, with the family notation $A_\alpha^{[\beta]}$ new; source 77-51's symbols are renamed as in the third table of Section 1.5.
- Bibliography: source 77-51's OEIS items for A022629 and A092484 are [1] and [2], its Granovsky–Stark item is [14], and its Gikunda item is [10] (with the DOI rather than either of its two ResearchGate addresses); [27] and [28] are new.
- Its abstract, title, novelty sentences and package description are replaced by Section 20 and the README; its priority caveats are kept there.

Batch 100. Sources 100-82, 100-89 and 100-93 are Research Reports 82, 89 and 93 of the session bundle of Reports 1–243, which arrived in commit 60f54ea06 (5 October 2026); they formed one cluster of batch 100, whose placement commit 36571ae0e staged their files under the prefixes 100-82-integral-, 100-89-boundary- and 100-93-theta- and retired the archives, which survive in the arrival commit. The prefixes carry the bundle report numbers, because batch 100's cluster-local numbers 01–03 are used by other clusters. Their provenance

manifests were verified at placement (13/13, 19/19 and 39/39 files), as was source 100-93's diagnostics checksum list (5/5, not shipped). Source 100-93's README pins source 100-89's archive by SHA-256 (225b2070...); it matches. The three are one author series: source 100-89 calls source 100-82 "the earlier A022629 report", and source 100-93 calls source 100-89 its fixed-power companion. None cites sources 40 to 77-51. Their texts share 1.12 %, 0.63 % and 0.15 % of their word 8-grams with this report as it stood before the merge, and 0.6–2.8 % with each other (status lines, preamble, standard displays). Sources 100-82 and 100-89 name no ProveIt commit; source 100-93 pins 6591a56b3 [33]. All three were delivered with `verify_package.py`, `build_local.sh`, a README and a PDF, none of which is shipped; no "prepared for private review" line, personal path or third-party data occurs in them.

What each contributed. Source 100-89: Theorem 21.2 and the closed forms and integer-power rule of Section 21.2; Theorem 21.4 and Corollary 21.6; Theorem 21.8 with the slope bound of Proposition 21.7; the sub-block route and the Sommerfeld and minimized-phase forms of Section 21.5 (with source 100-82); its table in Table 9; three citations; its non-claims. Source 100-82: P_9 , P_{10} , Δ_9 , Δ_{10} (Proposition 21.1, from its exact record); the case (21.20) with its L^1 and local-limit proofs; the resummed inverse of Remark 21.9; its table; its non-claims. Source 100-93: all of Section 22 except the parts marked [write].

Where the batch-100 merge chose.

- Placement: all three were added to this report; sources 100-82 and 100-89 re-prove its main theorems, and source 100-93 answers parts of its projects 1, 4 and 6. The report has no Parts, so, as in batches 75 and 77, they became Sections 21 and 22, after Section 20 and before the appendices; every earlier number is unchanged.
- Base of Section 21: source 100-89, the more general (every real α , uniform on compacts); source 100-82 is its case $\alpha = 1$, printed as a corollary, with its different proofs as second routes. Source 100-93 has a regime of its own and a section of its own.
- Duplicates: recorded in Table 8, not reprinted.
- Renamed: source 100-89's Edgeworth order M is written $J - 1$, so that its theorems read with this report's J ; every other renaming is in the fourth table of Section 1.5.
- Source 100-82's P_9 , P_{10} , Δ_9 , Δ_{10} exist only in its JSON record; they are printed with that provenance and with the write's confirmation by source 75-06's program.
- Source 100-89's two sketched steps are completed in the text (Lemma 21.3, Proposition 21.7); the sharp-scale form of its expansion (Remark 21.5), the theta shift of the threshold (Remark 22.10) and the phase average (22.46) are the write's. An independent check after the write found all five valid; one overstatement in Remark 22.10 and one understated term in Remark 21.5 are corrected (dated note at the end of Section 22.10).
- Source 100-93 is not presented as explaining this report's fixed- α finite-size residuals: its hypotheses fail there, and its formula does not reproduce them (Section 22.7). The intake's cell decomposition of those residuals is recorded as numerical evidence in Section 16.3.
- Its relation to the lattice-peak transfer theorem of the neighbouring report is stated in Section 22.8; a reciprocal note there is proposed separately.
- Abstracts, title pages, status paragraphs and package descriptions of the three sources are replaced by Section 21, Section 22 and the README; their priority caveats and non-claims are kept. Source 100-89's Gikunda item is [10] with the DOI; its other items are [14], [29], [30], [31] and the OEIS entries already cited; source 100-82's are [1], [27], [14]; source 100-93's are [6], source 100-89 and [33].

References

- [1] OEIS Foundation Inc., *A022629: Expansion of $\prod_{m \geq 1} (1 + mq^m)$* . Conjectural formula attributed to V. Kotešovec, May 8, 2018. Internal entry consulted October 1, 2026; sequence revision #54, March 13, 2026. <https://oeis.org/A022629/internal>.
- [2] OEIS Foundation Inc., *A092484: Expansion of $\prod_{m \geq 1} (1 + m^2 q^m)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A092484>.
- [3] OEIS Foundation Inc., *A265840: Expansion of $\prod_{k \geq 1} (1 + k^3 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265840>.
- [4] OEIS Foundation Inc., *A265841: Expansion of $\prod_{k \geq 1} (1 + k^4 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265841>.
- [5] OEIS Foundation Inc., *A265842: Expansion of $\prod_{k \geq 1} (1 + k^5 q^k)$* . Conjectural formula attributed to V. Kotešovec, December 27, 2020. Consulted October 1, 2026. <https://oeis.org/A265842>.
- [6] OEIS Foundation Inc., *A292189: Square array of coefficients of $\prod_{j \geq 1} (1 + j^k q^j)$* . Consulted October 1, 2026. <https://oeis.org/A292189>.
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- [8] W. Bridges and W. Craig, *On the distribution of the norm of partitions*, *The Ramanujan Journal* **66** (2025), Article 40. DOI: [10.1007/s11139-024-00969-5](https://doi.org/10.1007/s11139-024-00969-5). Author version: <https://arxiv.org/html/2308.00123v1>.
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- [10] D. K. Gikunda, *The Distribution of the Product of Parts in Integer Partitions*, Ph.D. dissertation, Stellenbosch University, March 2026, Chapter 4, especially Section 4.2 and Proposition 4.2.5. Author-uploaded text consulted October 1, 2026. DOI: [10.13140/RG.2.2.22314.07368](https://doi.org/10.13140/RG.2.2.22314.07368). Repository record: <https://scholar.sun.ac.za/handle/10019.1/136012>.
- [11] V. Reshetnikov, *ProveIt*, repository snapshot `1f1981f682b2878bde51a6ad40c22777f362fc05`, *Analysis/Transseries/README.md*, consulted October 1, 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
- [12] V. Reshetnikov, *ProveIt*, repository and the documentation at *Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md*. <https://github.com/VladimirReshetnikov/ProveIt>. Repository tree response inspected during preparation of source 48: `dc1a7242d2ab4a4496b7dfee63256ee767b21fc9` (a commit); the documentation was read from the default branch, so that identifier is not asserted to pin every subsequent read. Consulted October 1, 2026.
- [13] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. In particular, the saddle-point framework of Chapter VIII. Official companion site: <https://ac.cs.princeton.edu/home/>.
- [14] B. L. Granovsky and D. Stark, *Developments in the Khintchine–Meinardus probabilistic method for asymptotic enumeration*, *Electronic Journal of Combinatorics* **22**(4) (2015), Paper P4.32, 26 pp. doi:[10.37236/4581](https://doi.org/10.37236/4581). <https://arxiv.org/abs/1311.2254>; version 2, November 12, 2015. The assumptions of Theorem 1 and the distinct-size products of Section 3 were checked by source 48 in the primary-source HTML version.
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- [21] OEIS Foundation Inc., *A022630: Expansion of $\prod_{m \geq 1} (1 + mq^m)^2$ and A022631: Expansion of $\prod_{m \geq 1} (1 + mq^m)^3$* . <https://oeis.org/A022630>, <https://oeis.org/A022631>. Consulted October 2, 2026 (intake).
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- [23] OEIS Foundation Inc., *A022661: Expansion of $\prod_{m \geq 1} (1 - mq^m)$* , including the zero question attributed to Robert Israel, June 2, 2015. <https://oeis.org/A022661>. Inspected by source 75-01, October 1, 2026.
- [24] V. Reshetnikov, *ProveIt*, repository map and the Lambert- W drafts directory `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/lambert-w/`, as inspected by source 75-01 on October 1, 2026 (tree 29aca108ed25d714f31b1316512777fbbdc8e006). <https://github.com/VladimirReshetnikov/ProveIt>.
- [25] [Added in the batch-75 merge.] ProveIt, *Transseries and Inversion* volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, Theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion`, `p0:thm:staircase` and `p0:thm:backward-error`.
- [26] [Added in the batch-75 merge.] ProveIt, Lean file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, declarations `Fabius.staircase_ceil` and `Fabius.staircase_separation`.
- [27] [Added in the batch-77 merge.] B. J. Naranjo and J. L. Ramírez, *On colored partitions and Euler-type identities*, INTEGERS **26** (2026), A20. <https://math.colgate.edu/~integers/aa20/aa20.pdf>. Cited by source 77-51.
- [28] [Added in the batch-77 merge.] OEIS Foundation Inc., *A266891: Expansion of $\prod_{k \geq 1} (1 + kx^k)^k$* . <https://oeis.org/A266891>. Asymptotic conjecture credited to V. Kotešovec, 8 May 2018; accessed by source 77-51, October 1, 2026.
- [29] [Added in the batch-100 merge.] W. Bridges, B. Brindle, K. Bringmann and J. Franke, *Asymptotic expansions for partitions generated by infinite products*, Mathematische Annalen **390** (2024), 2593–2632. <https://arxiv.org/abs/2303.11864>. Cited by source 100-89.
- [30] [Added in the batch-100 merge.] L. Ahmadi, R. Gómez-Aíza and M. D. Ward, *A unified treatment of families of partition functions*, La Matematica (2024). <https://doi.org/10.1007/s44007-024-00134-w>. Cited by source 100-89.
- [31] [Added in the batch-100 merge.] A. Cantón, J. L. Fernández, P. Fernández and V. J. Maciá, *Khinchin families and Hayman class*, Computational Methods and Function Theory **21** (2021), 851–904. <https://d-nb.info/1251615767/34>. Cited by source 100-89.

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SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a357825-theta-ballot-power-sums, Part II (written in commit 2b257ba94), Theorem `tbs:lpt:thm:transfer`, Corollaries `tbs:lpt:cor:clusters` and `tbs:lpt:cor:no-constant`, Theorem `tbs:lpt:thm:inverse`, Proposition `tbs:lpt:prop:logconcave`.
- [33] [Added in the batch-100 merge.] V. Reshetnikov, *ProveIt*, repository snapshot 6591a56b3551e08700aef333cb98849c8cdfe69e, examined by source 100-93 on 2 October 2026 for its bounded source comparison. <https://github.com/VladimirReshetnikov/ProveIt>.